

大模型基础：理论与技术的演进

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LangChain Developer

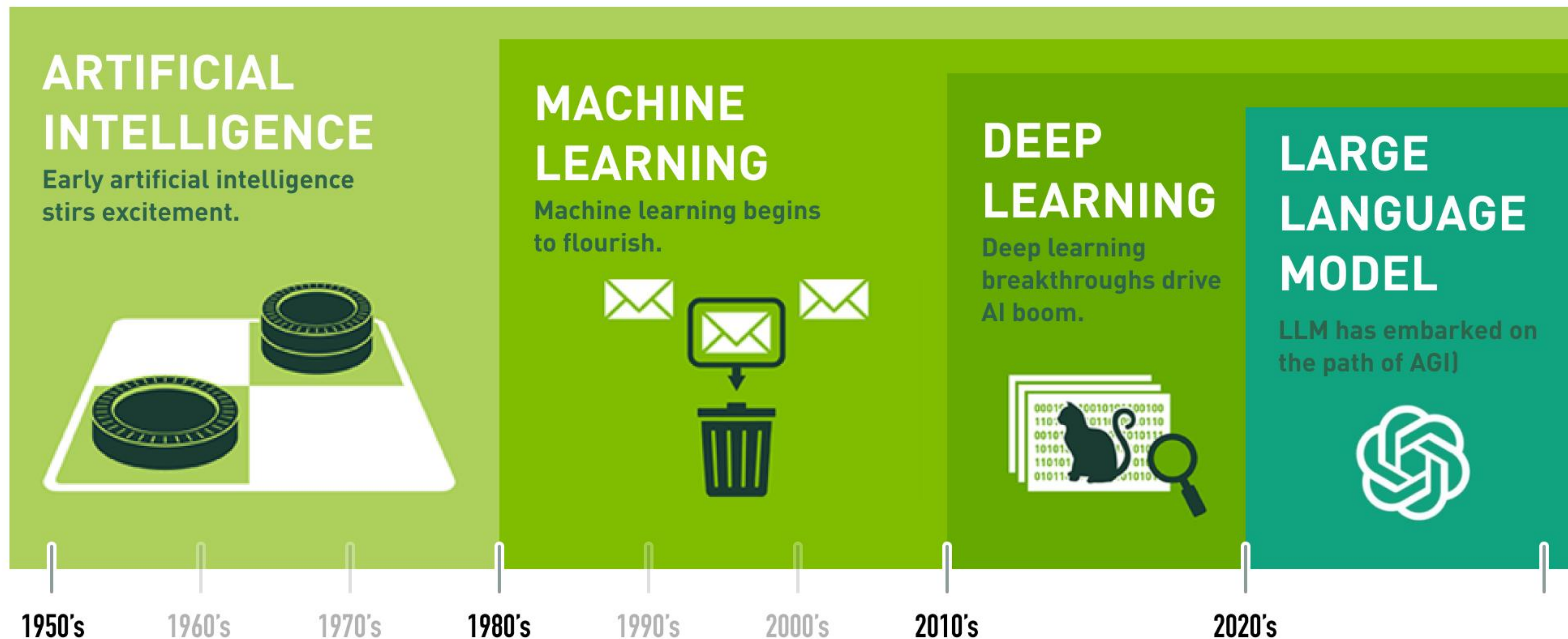


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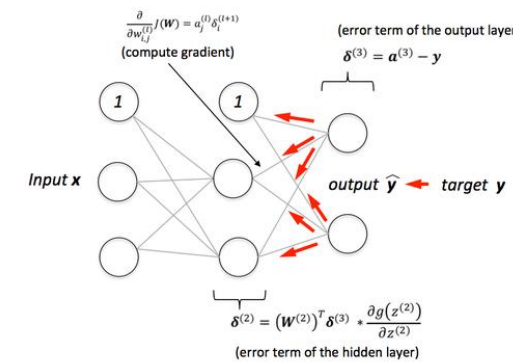
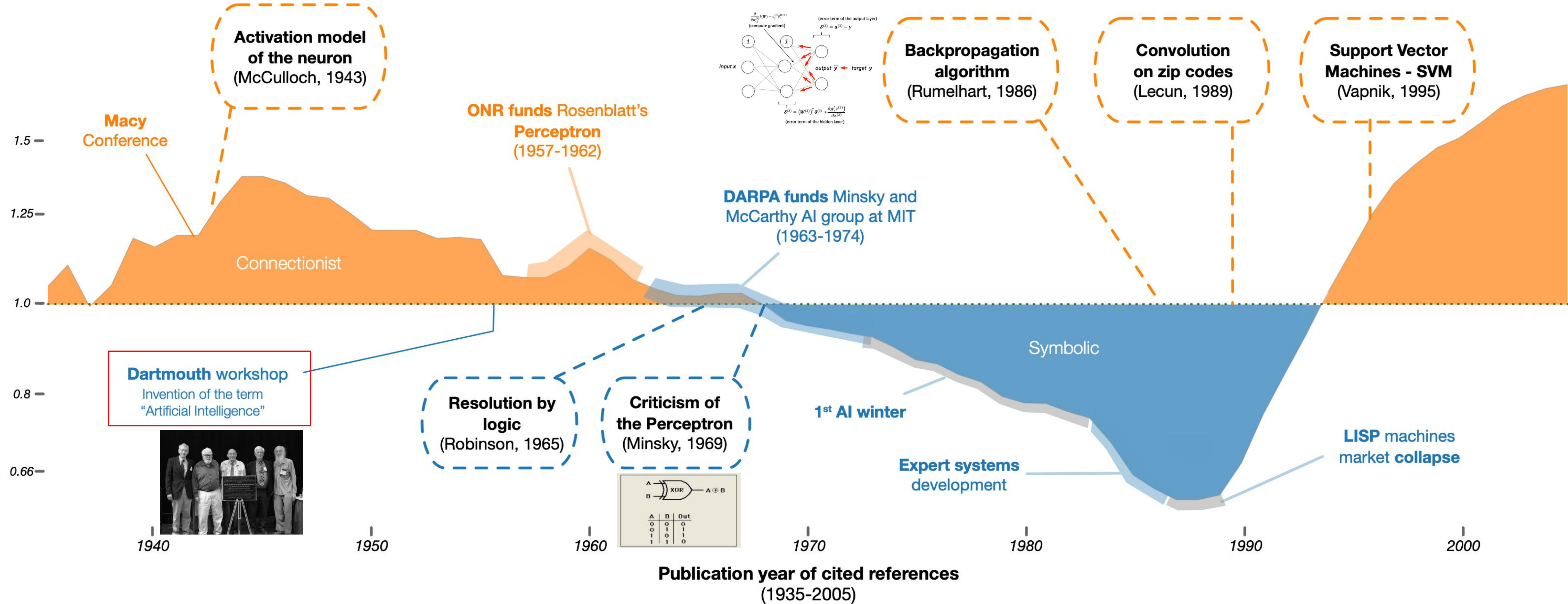
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 - GPT-4：一个新的开始

初探大模型：起源与发展

AI 历史上的四次大发展



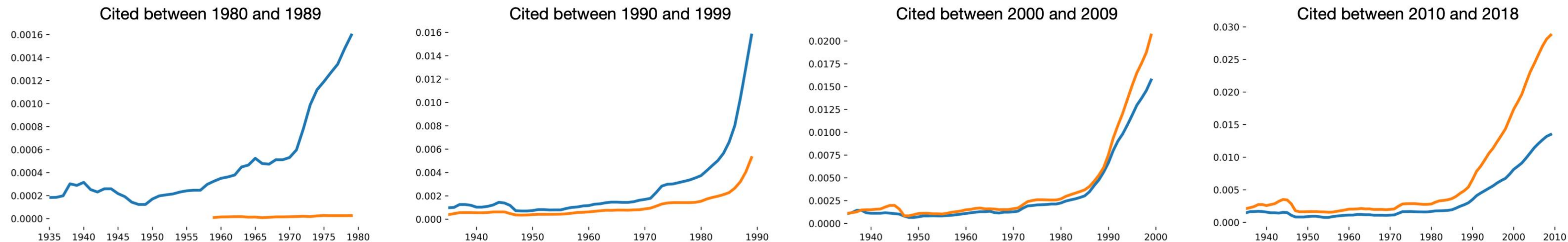
Ratio connectionist / symbolic (log)

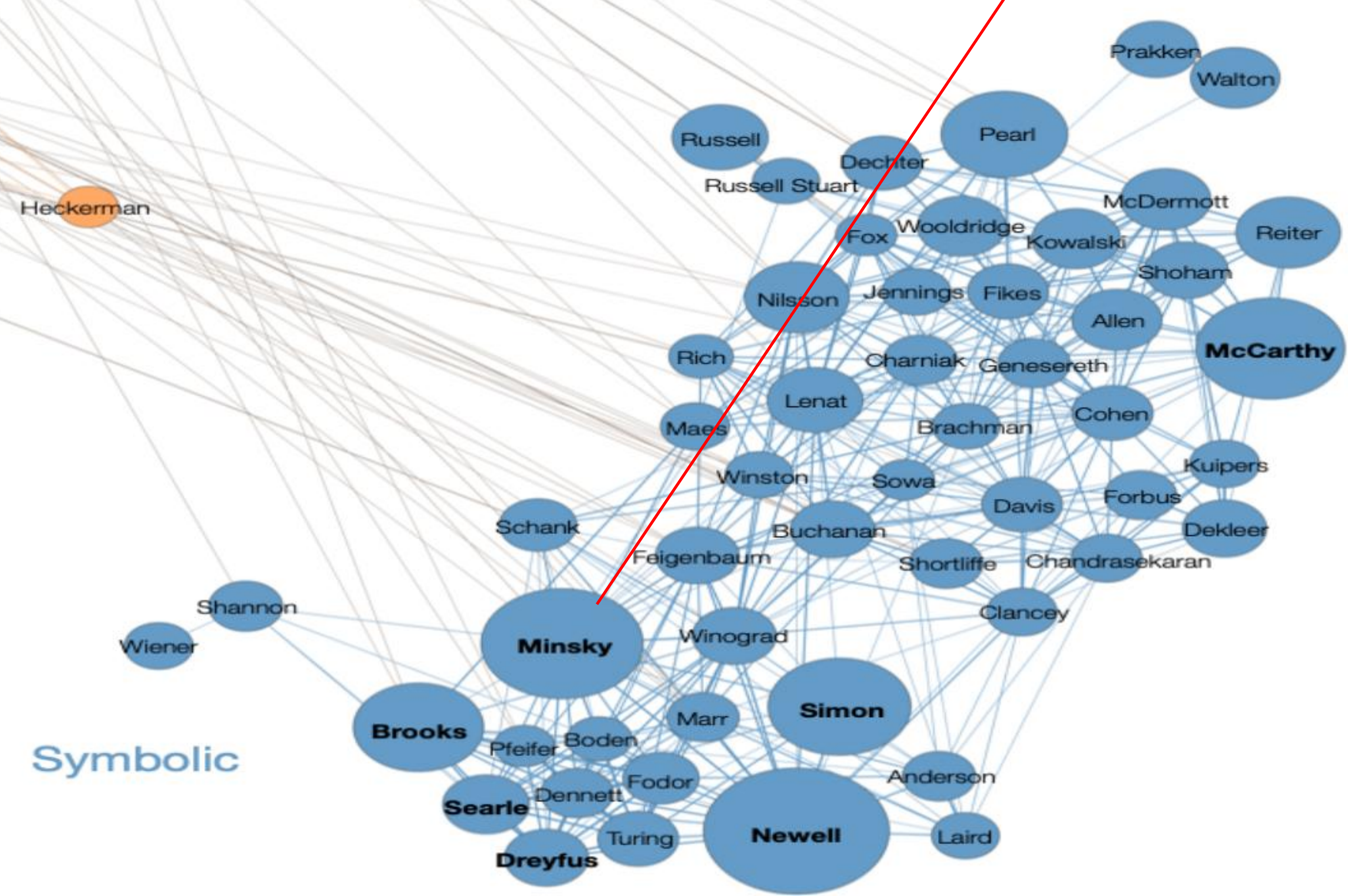
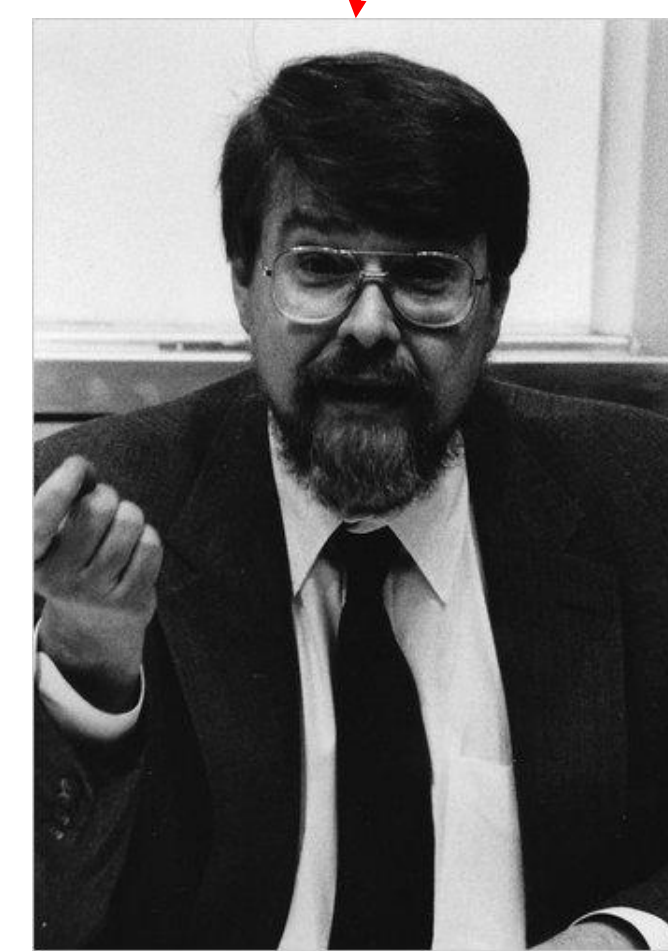
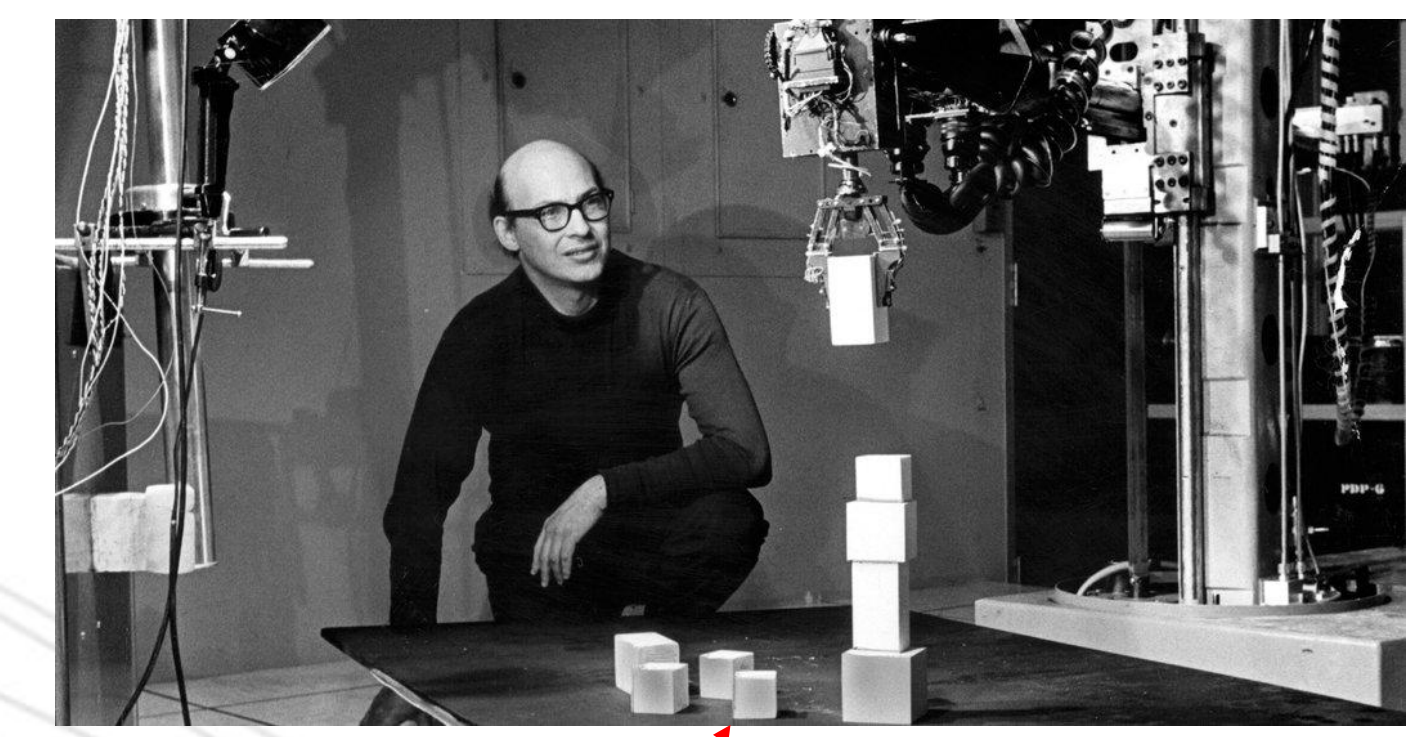
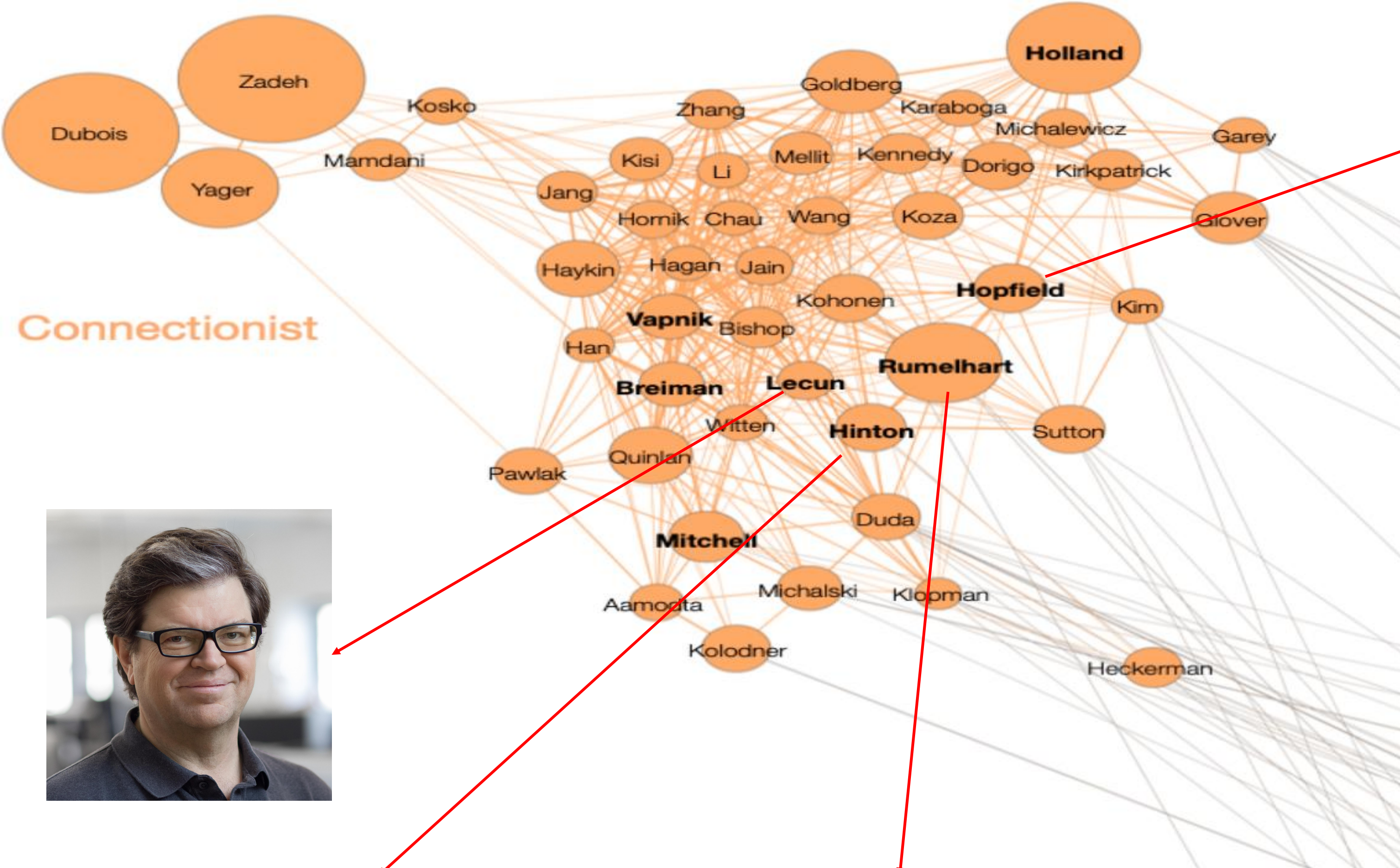


XOR

A	B	Out
0	0	0
0	1	1
1	0	1
1	1	0

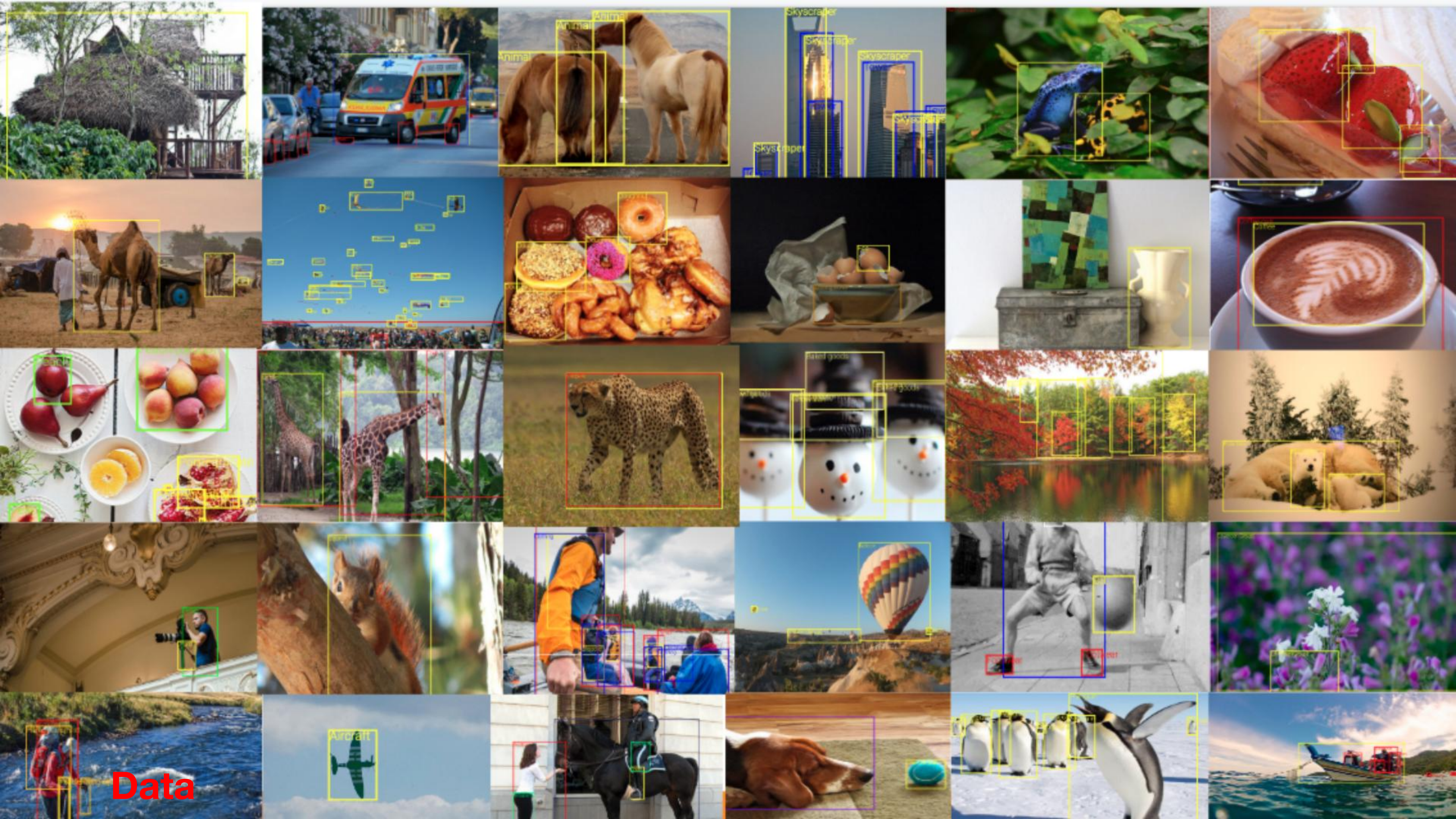
% of WoS





IMAGENET 数据集与竞赛



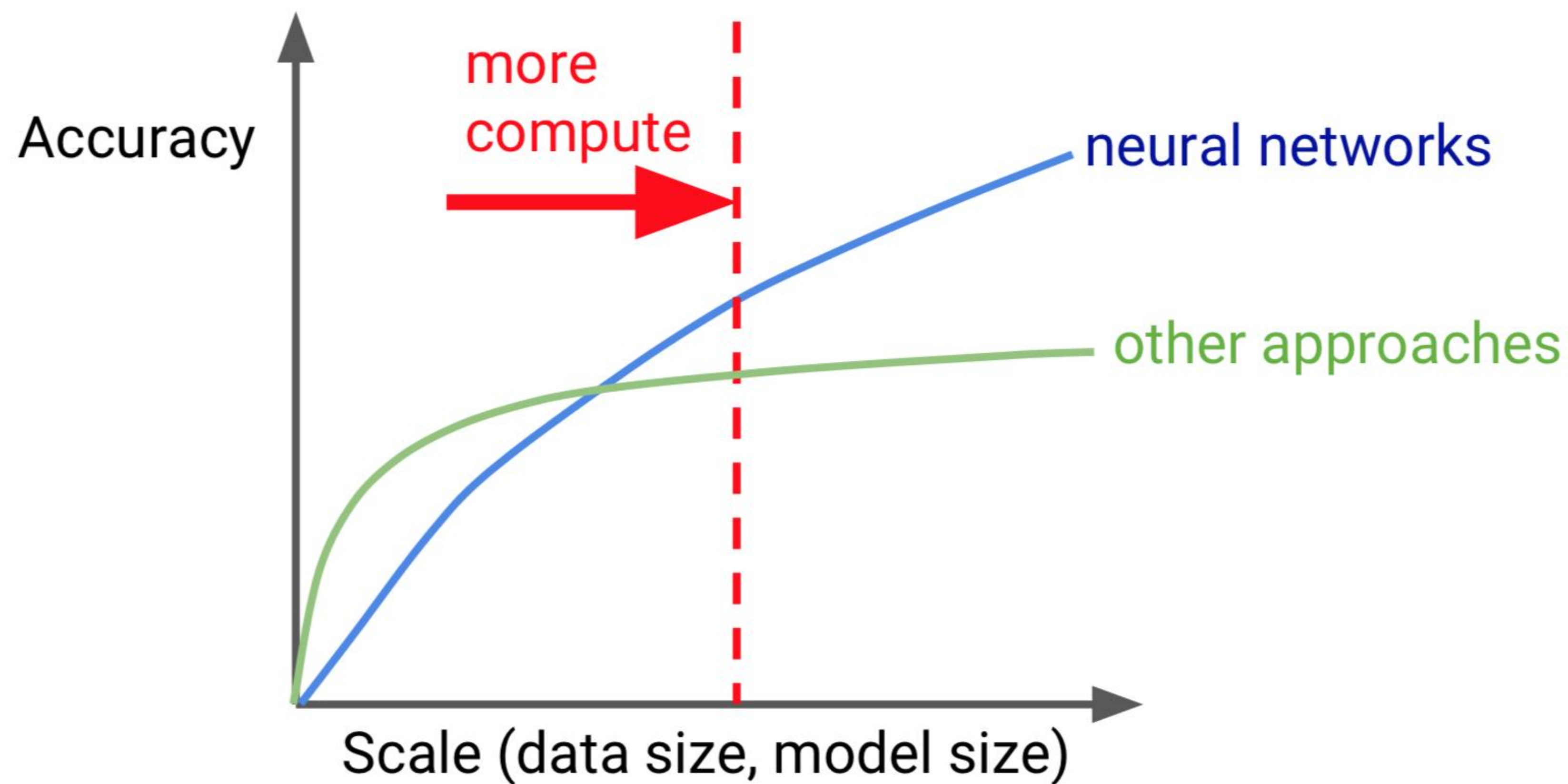


Data



Compute

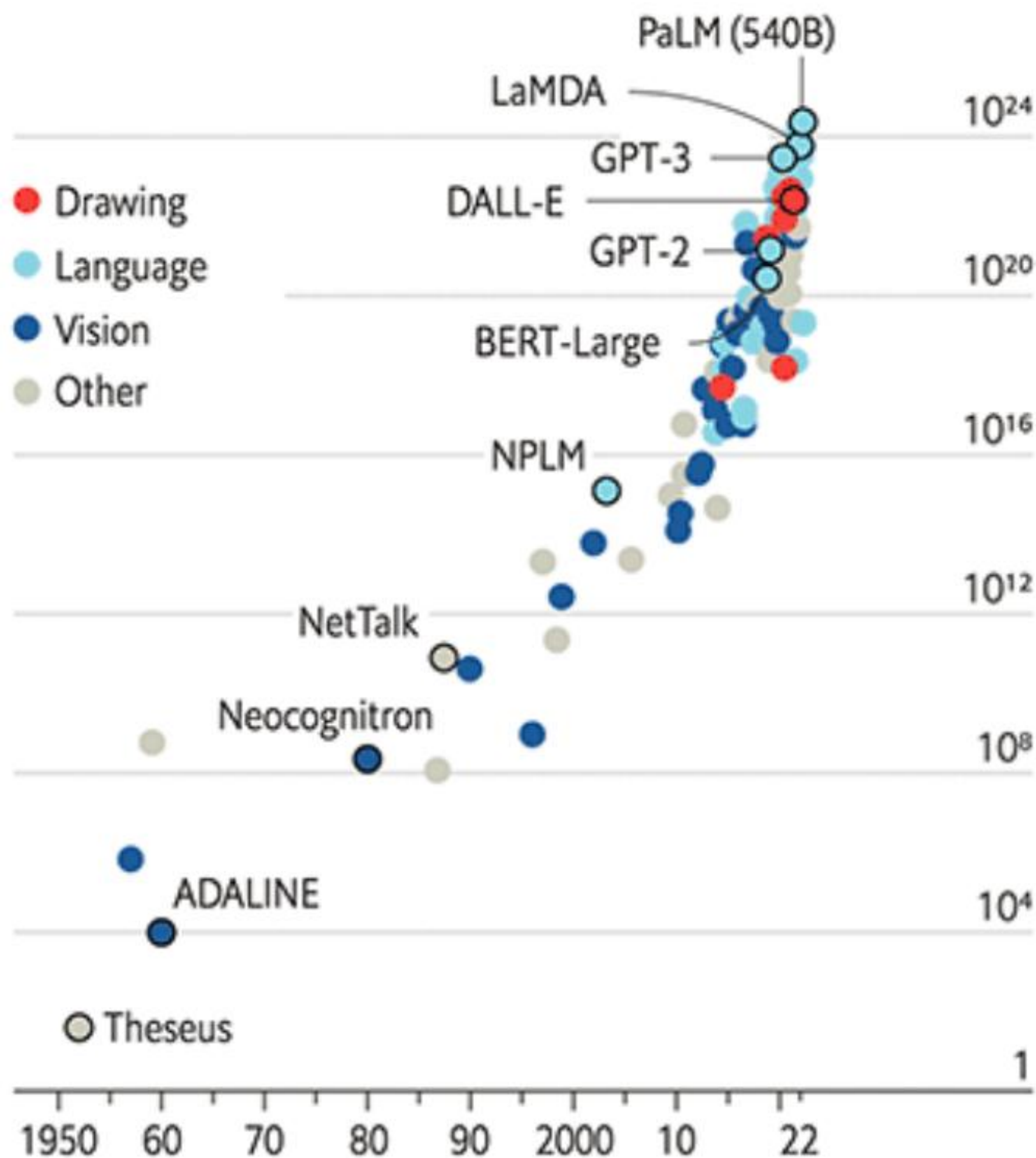
数据和算力增长推动深度学习的繁荣发展 (2010's)



The blessings of scale

AI training runs, estimated computing resources used

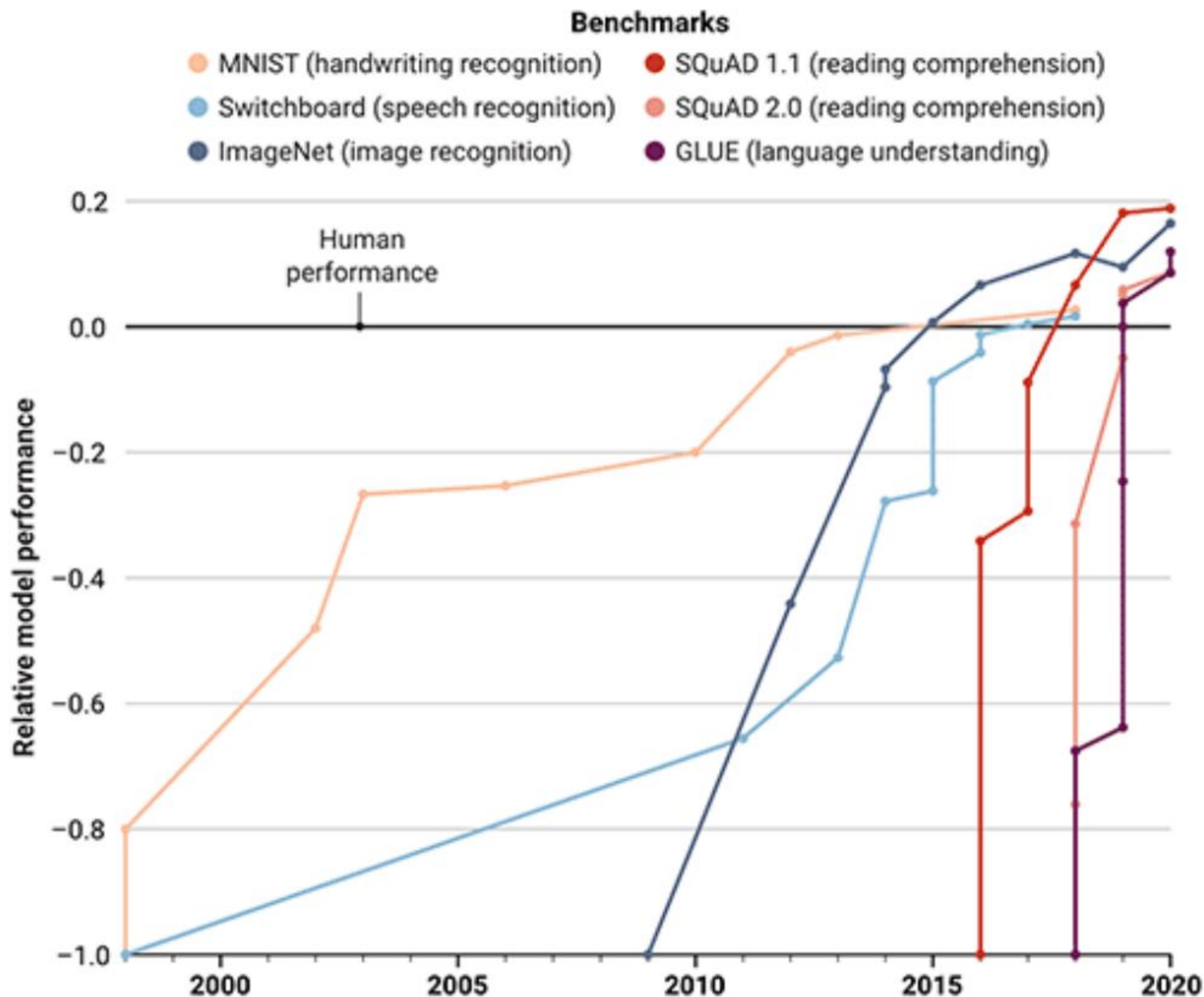
Floating-point operations, selected systems, by type, log scale



Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

Quick learners

The speed at which artificial intelligence models master benchmarks and surpass human baselines is accelerating. But they often fall short in the real world.



(GRAPHIC) K. FRANKLIN/SCIENCE; (DATA) D. KIELA ET AL., DYNABENCH: RETHINKING BENCHMARKING IN NLP, DOI:10.48550/ARXIV.2104.14337

预热篇：解码注意力机制 (**Attention Mechanism**)

What is Attention?



说重点！！！！

领导高水平发言
深刻理解



问：“我”去了几次咖啡店？

"昨天，我在一个繁忙的一天结束后，决定去我最喜欢的咖啡店放松一下。我走进咖啡店，点了一杯拿铁，然后找了一个靠窗的位置坐下。我喝着咖啡，看着窗外的人们匆匆忙忙，感觉非常惬意。然后，我从咖啡店出来，回到了家中。"

问：“我”去了几次咖啡店？

"昨天，我在一个繁忙的一天结束后，决定去我最喜欢的咖啡店放松一下。我走进咖啡店，点了一杯拿铁，然后找了一个靠窗的位置坐下。我喝着咖啡，看着窗外的人们匆匆忙忙，感觉非常惬意。然后，我从咖啡店出来，回到了家中。"

高频 ≠ 重点

"昨天，我在一个繁忙的一天结束后，决定去我最喜欢的咖啡店放松一下。我走进咖啡店，点了一杯拿铁，然后找了一个靠窗的位置坐下。我喝着咖啡，看着窗外的人们匆匆忙忙，感觉非常惬意。然后，我从咖啡店出来，回到了家中。"

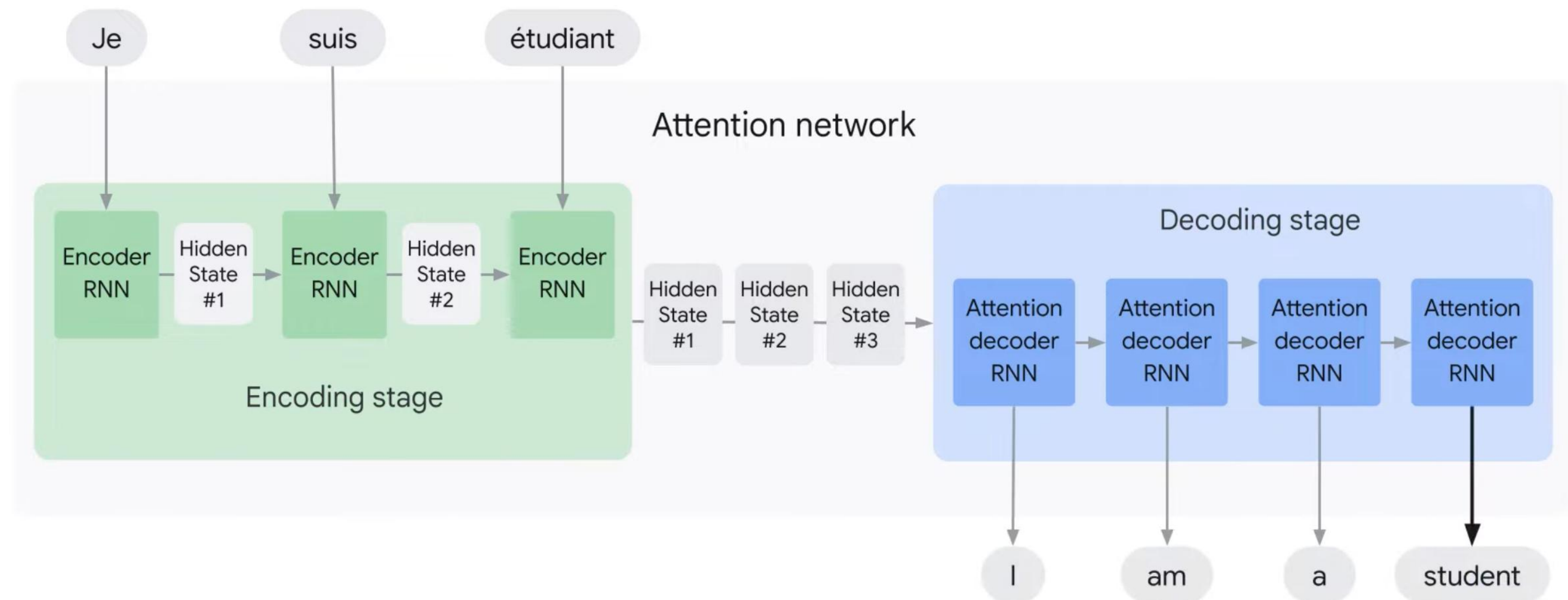
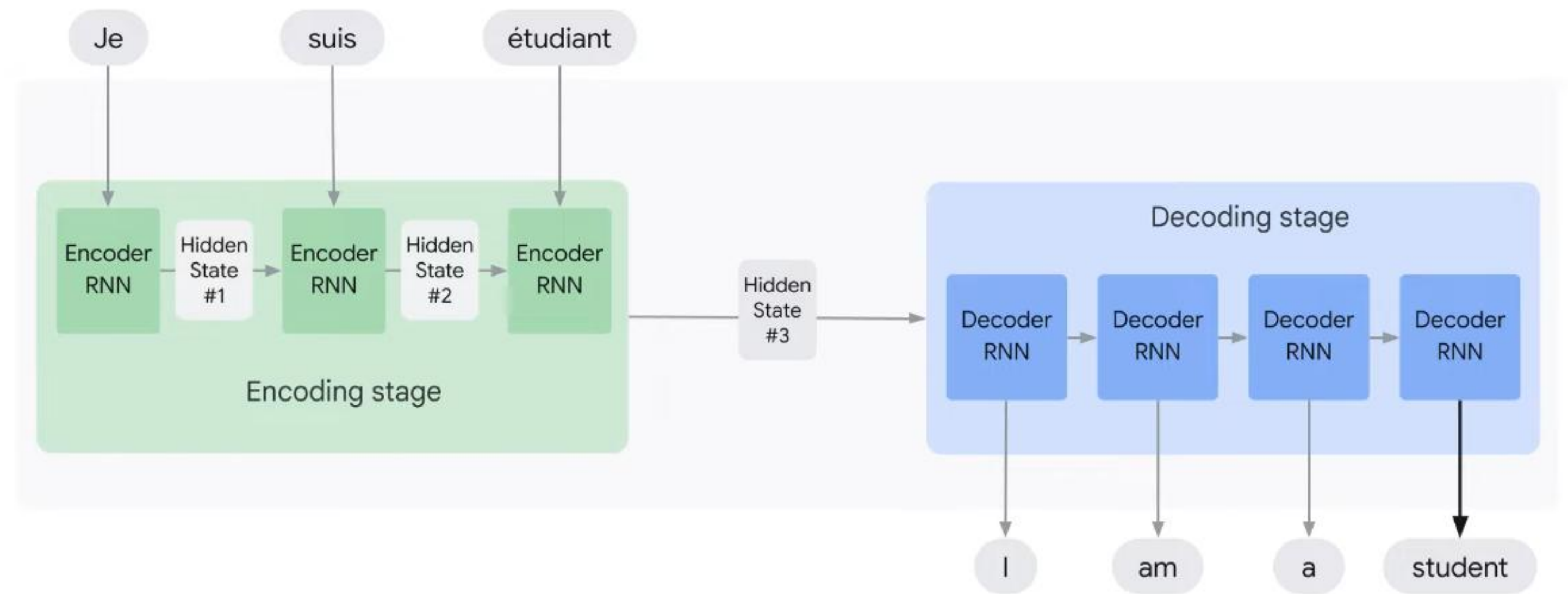
在这个例子中，"咖啡店"出现了三次，但并非每次都是关键信息。

- 当我们理解"我走进咖啡店，点了一杯拿铁"时，"走进"和"点了一杯拿铁"是关键动作，而"咖啡店"的信息已经被我们知道了。
- 同样，当理解"我从咖啡店出来，回到了家中"时，"从咖啡店出来"和"回到了家中"是关键信息，而"咖啡店"的信息仍然是冗余的。

注意力机制就是帮助模型在处理这样的句子时，能够更好地**关注到关键的信息**，而忽略冗余的信息。

But How?

Encoder-decoder architecture



Encoder-decoder Architecture with Attention Model

Function	Traditional Encoder-Decoder	Encoder-Decoder with Attention
Encode	$h_i = f(x_i, h_{i-1})$	$h_i = f(x_i, h_{i-1})$
Context	$c = h_T$	$c_j = \sum_{i=1}^T \alpha_{ij} h_i$ $\alpha_{ij} = p(e_{ij})$ $e_{ij} = a(s_{j-1}, h_i)$
Decode	$s_j = f(s_{j-1}, y_{j-1}, c)$	$s_j = f(s_{j-1}, y_{j-1}, c_j)$
Generate	$y_j = g(y_{j-1}, s_j, c)$	$y_j = g(y_{j-1}, s_j, c_j)$

$x = (x_1, \dots, x_T)$: input sequence, T : length of input sequence, h_i : hidden states of encoder, c : context vector, α_{ij} : attention weights over input, s_j : decoder hidden state, y_j : output token, f, g : non-linear functions, a : alignment function, p : distribution function

Table 1. Encoder-decoder architecture: traditional and with attention model

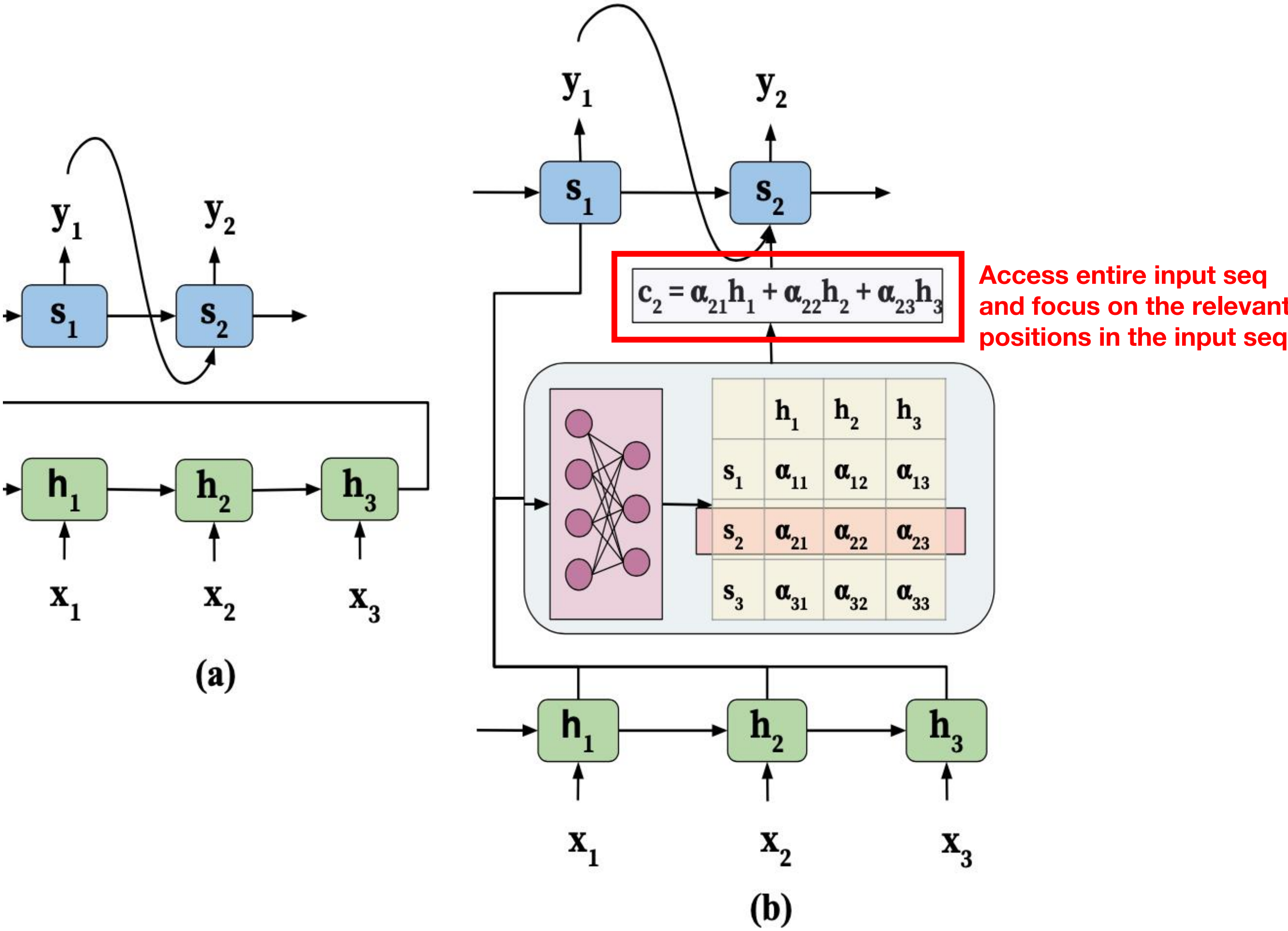


Fig. 2. Encoder-decoder architecture: (a) traditional (b) with attention model

Encoder-decoder Architecture with Attention Model

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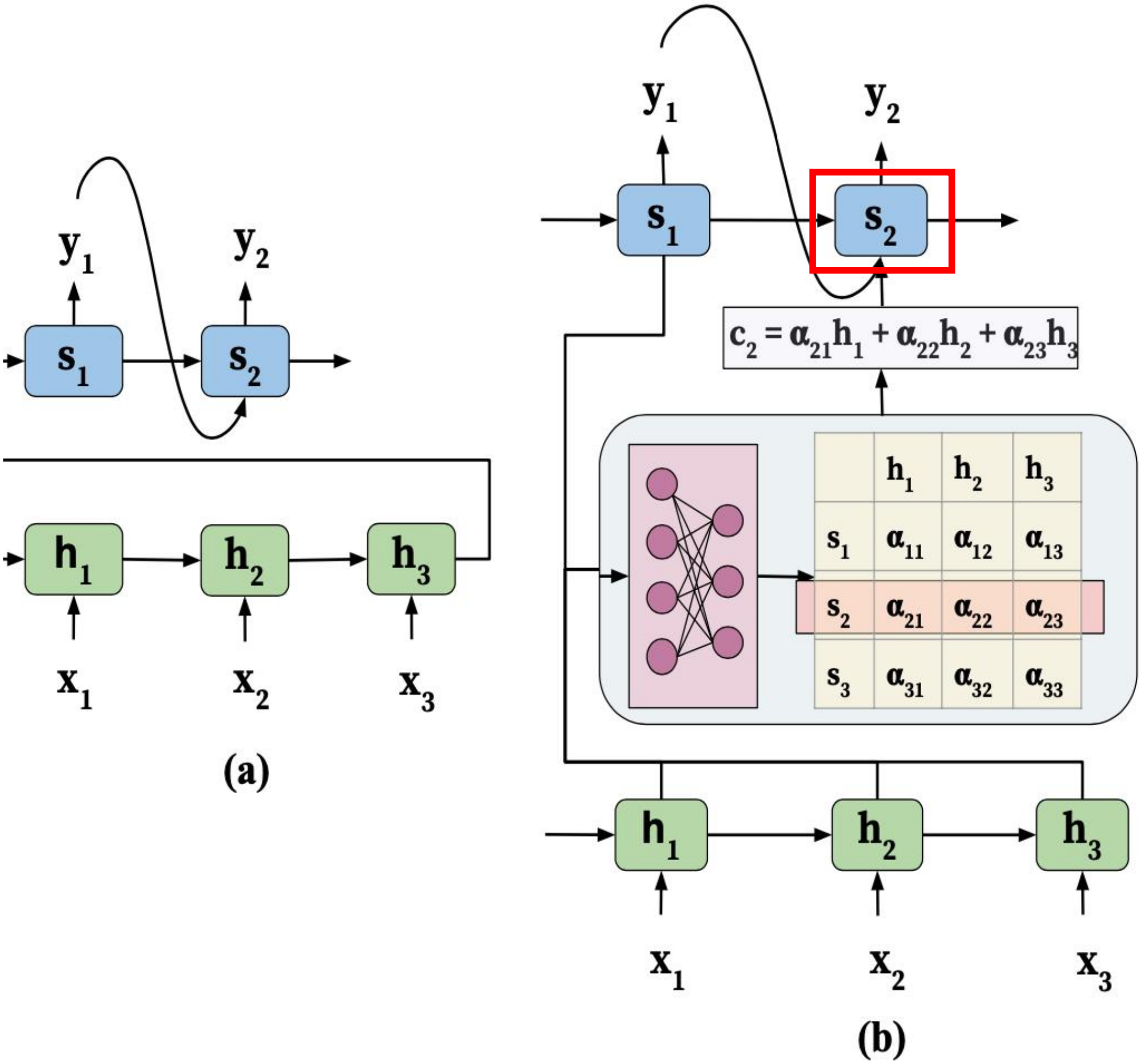
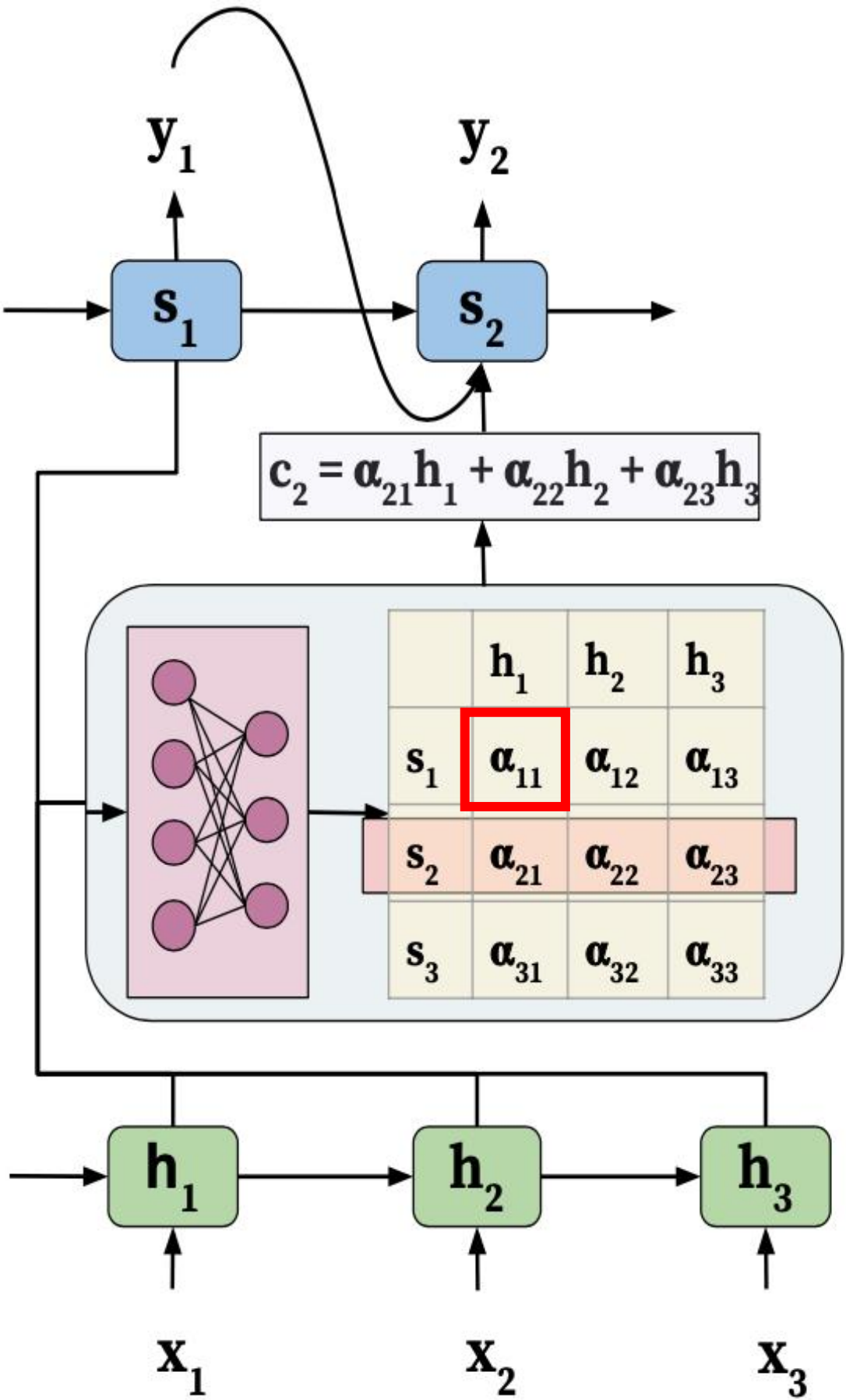


Fig. 2. Encoder-decoder architecture: (a) traditional (b) with attention model

Function	Traditional Encoder-Decoder	Encoder-Decoder with Attention
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Attention weights α
Capture the relevance
between h and s

Table 1. Encoder-decoder architecture: traditional and with attention model

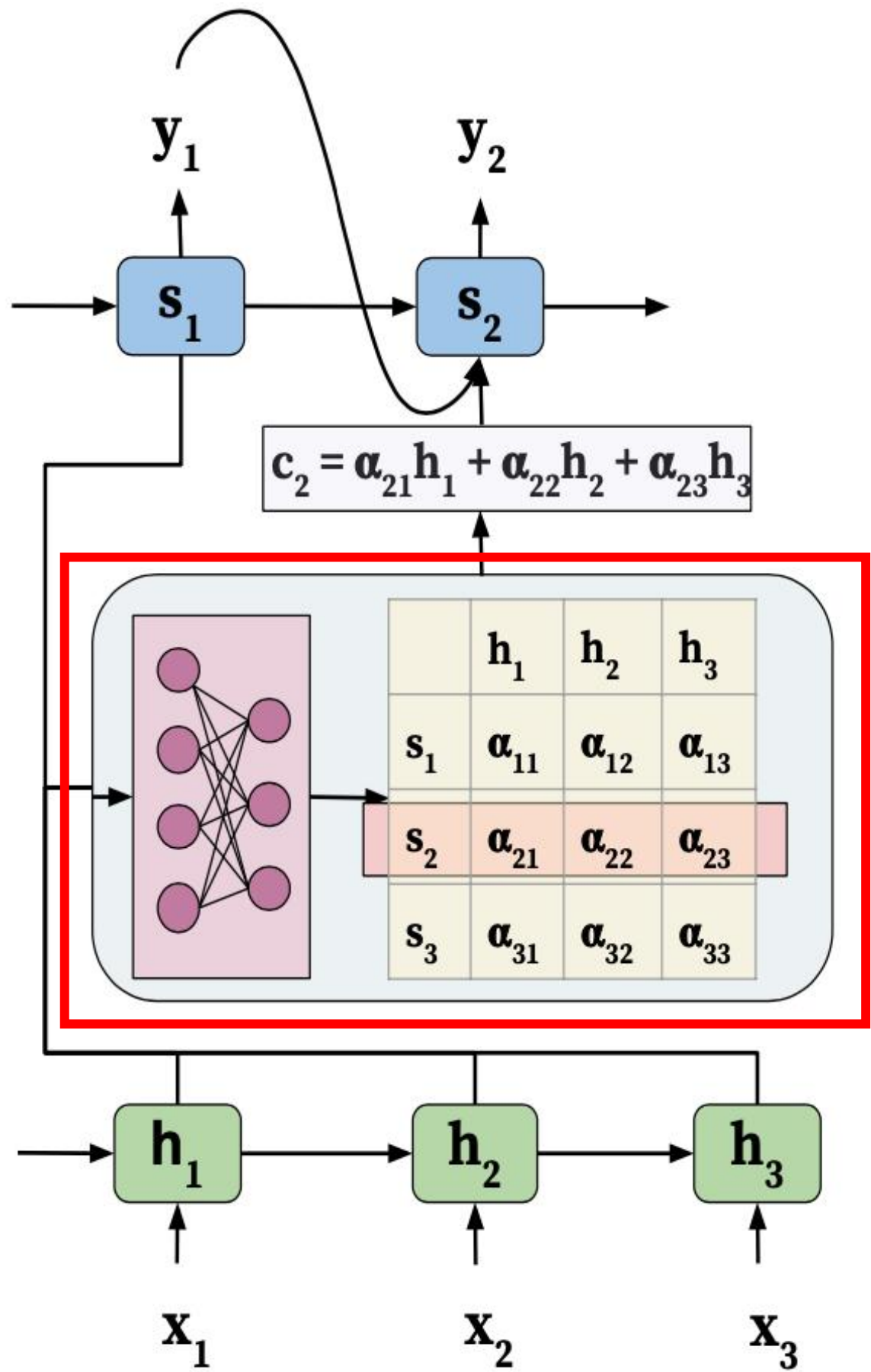
↓
sigmoid & softmax

Encoder-decoder Architecture with Attention Model

Function	Traditional Encoder-Decoder	Encoder-Decoder with Attention
Encode	$h_i = f(x_i, h_{i-1})$	$h_i = f(x_i, h_{i-1})$
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Table 1. Encoder-decoder architecture: traditional and with attention model



Learning attention weights

a: alignment function

based on a notion of comparing **query** representations with **key** representations

$$e = a(K, q) \text{ and } \alpha = p(e)$$

e.g. softmax

$$A(q, K, V) = \sum_i p(a(k_i, q)) * v_i$$

$k_i = v_i = h_i$ In some cases, there is also **additional input of values V**

Attention Mechanism

$$A(q, K, V) = \sum_i p(a(k_i, q)) * v_i$$



$$A(q, K, V) = \sum_i p(a(k_i, q)) * v_i$$

Function	Equation	References
similarity	$a(k_i, q) = sim(k_i, q)$	[Graves et al. 2014a]
dot product	$a(k_i, q) = q^T k_i$	[Luong et al. 2015a]
scaled dot product	$a(k_i, q) = \frac{q^T k_i}{\sqrt{d_k}}$	[Vaswani et al. 2017]
general	$a(k_i, q) = q^T W k_i$	[Luong et al. 2015a]
biased general	$a(k_i, q) = k_i(W q + b)$	[Sordoni et al. 2016]
activated general	$a(k_i, q) = act(q^T W k_i + b)$	[Ma et al. 2017b]
generalized kernel	$a(k_i, q) = \phi(q)^T \phi(k_i)$	[Choromanski et al. 2021]
concat	$a(k_i, q) = w_{imp}^T act(W[q; k_i] + b)$	[Luong et al. 2015a]
additive	$a(k_i, q) = w_{imp}^T act(W_1 q + W_2 k_i + b)$	[Bahdanau et al. 2015]
deep	$a(k_i, q) = w_{imp}^T E^{(L-1)} + b^L$ $E^{(l)} = act(W_l E^{(l-1)} + b^l$ $E^{(1)} = act(W_1 k_i + W_0 q) + b^l$	[Pavlopoulos et al. 2017]
location-based	$a(k_i, q) = a(q)$	[Luong et al. 2015a]
feature-based	$a(k_i, q) = w_{imp}^T act(W_1 \phi_1(K) + W_2 \phi_2(K) + b)$	[Li et al. 2019a]

$a(k_i, q)$: alignment function for query q and key k_i , sim : similarity functions such as cosine, d_k : length of input, $(W, w_{imp}, W_0, W_1, W_2)$: trainable parameters, b : trainable bias term, act : activation function.

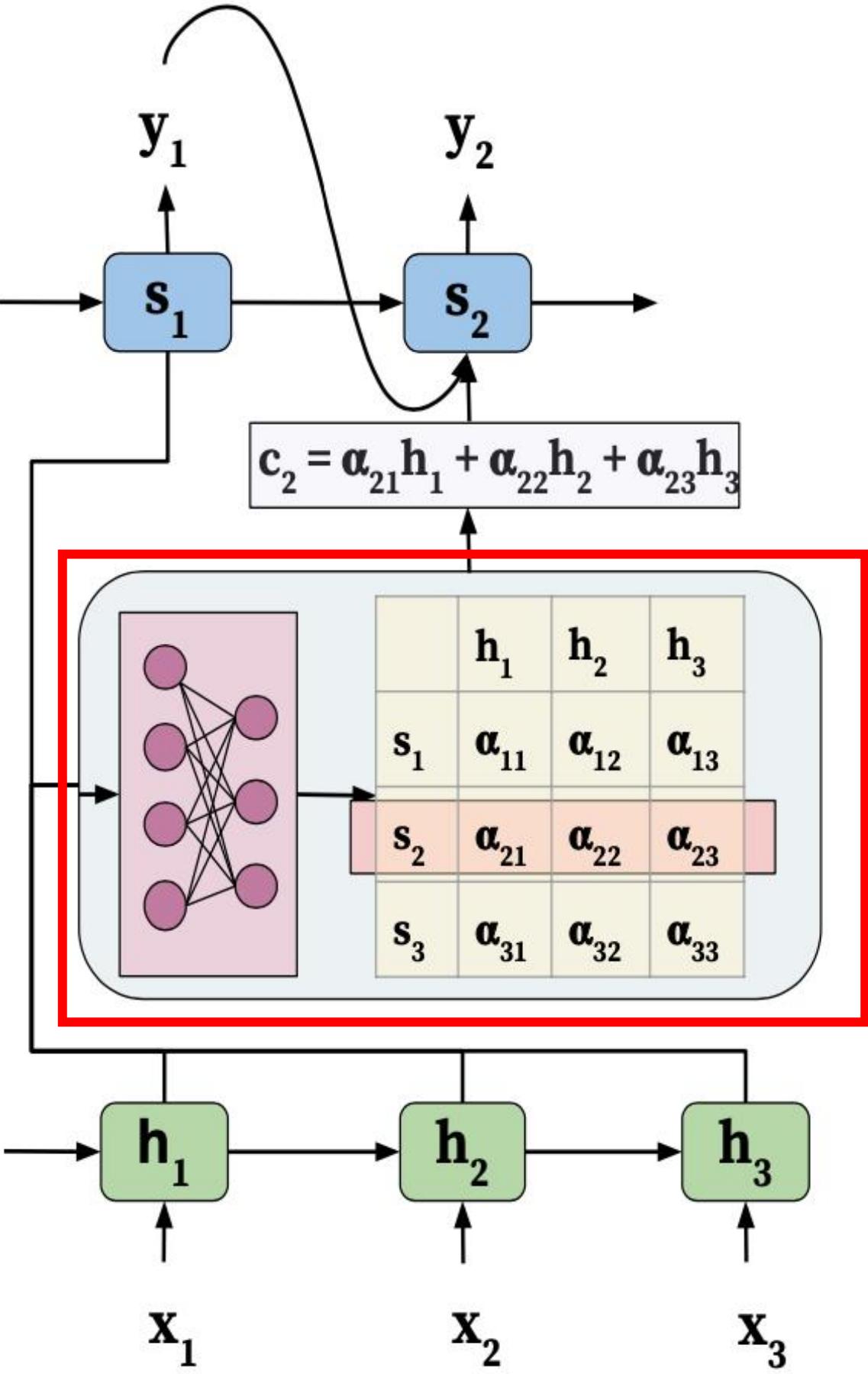


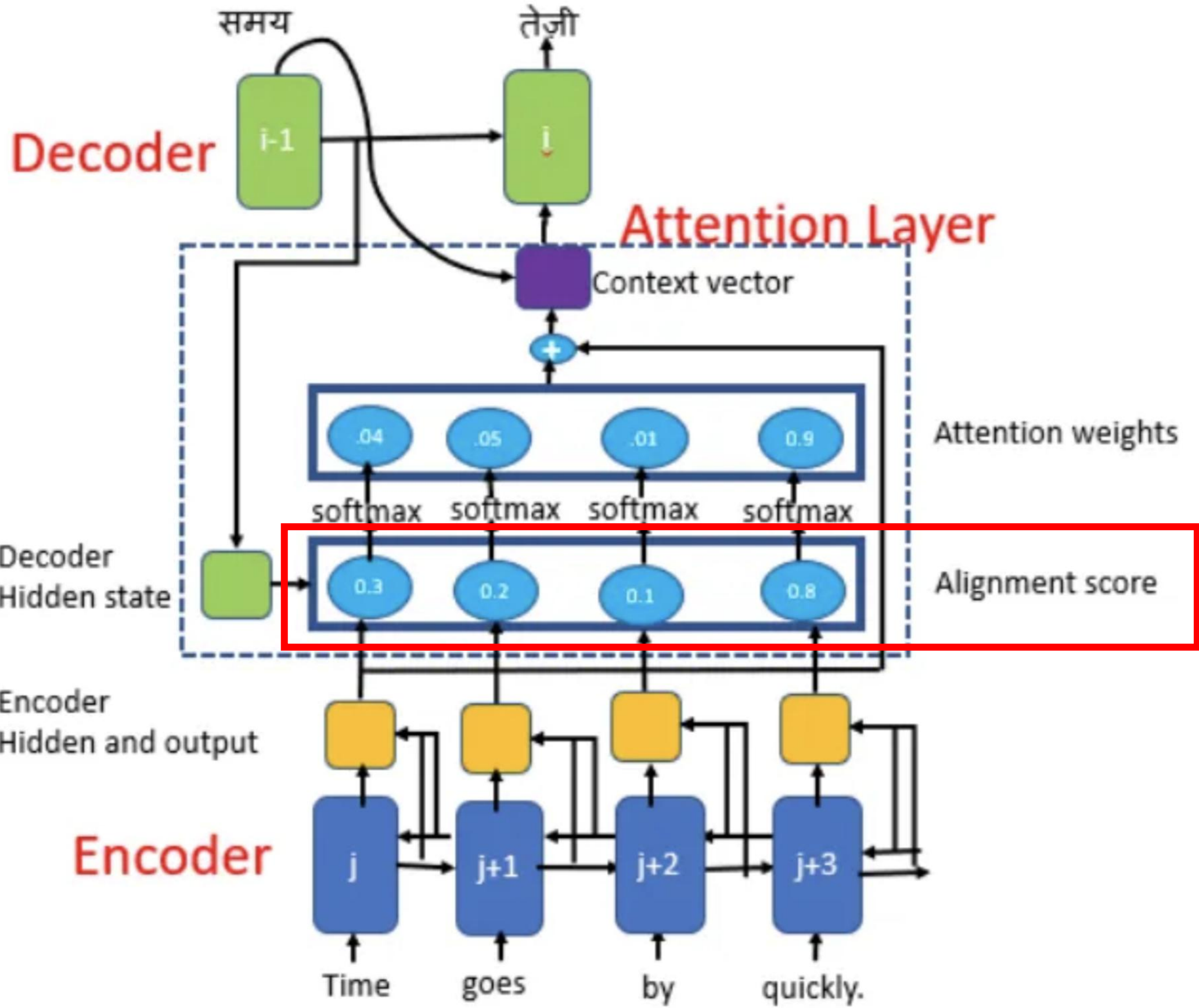
Table 2. Summary of Alignment Functions

$$A(q, K, V) = \sum_i p(a(k_i, q)) * v_i$$

Function	Equation	References
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Table 2. Summary of Alignment Functions



Why Attention?

注意力机制的特点和优势

1. 注意力机制有助于克服循环神经网络 (**RNNs**) 的一些挑战, 例如输入序列长度增加时性能下降和顺序处理输入导致的计算效率低下。
2. 在自然语言处理 (**NLP**)、计算机视觉 (**Computer Vision**)、跨模态任务和推荐系统等多个领域中, 注意力机制已成为多项任务中的最先进模型, 取得了显著的性能提升。
3. 注意力机制不仅可以提高主要任务的性能, 还具有其他优势。它们被广泛用于提高神经网络的可解释性, 帮助解释模型的决策过程, 使得原本被认为是黑盒模型的神经网络变得更易解释。这对于人们对机器学习模型的公平性、可追溯性和透明度的关注具有重要意义。



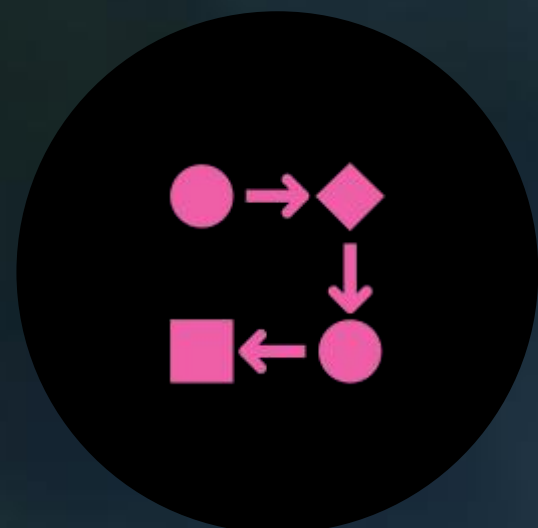
Overcoming challenges with RNNs



解决传统编码器-解码器模型的挑战，避免信息损失和无法建模输入输出对齐的问题。



自动学习注意力权重，捕捉编码器和解码器之间的相关性。



提高模型性能，改善输出质量，并提供更好的解释性。



允许解码器访问整个编码的输入序列，通过注意力权重选择性地关注相关信息。

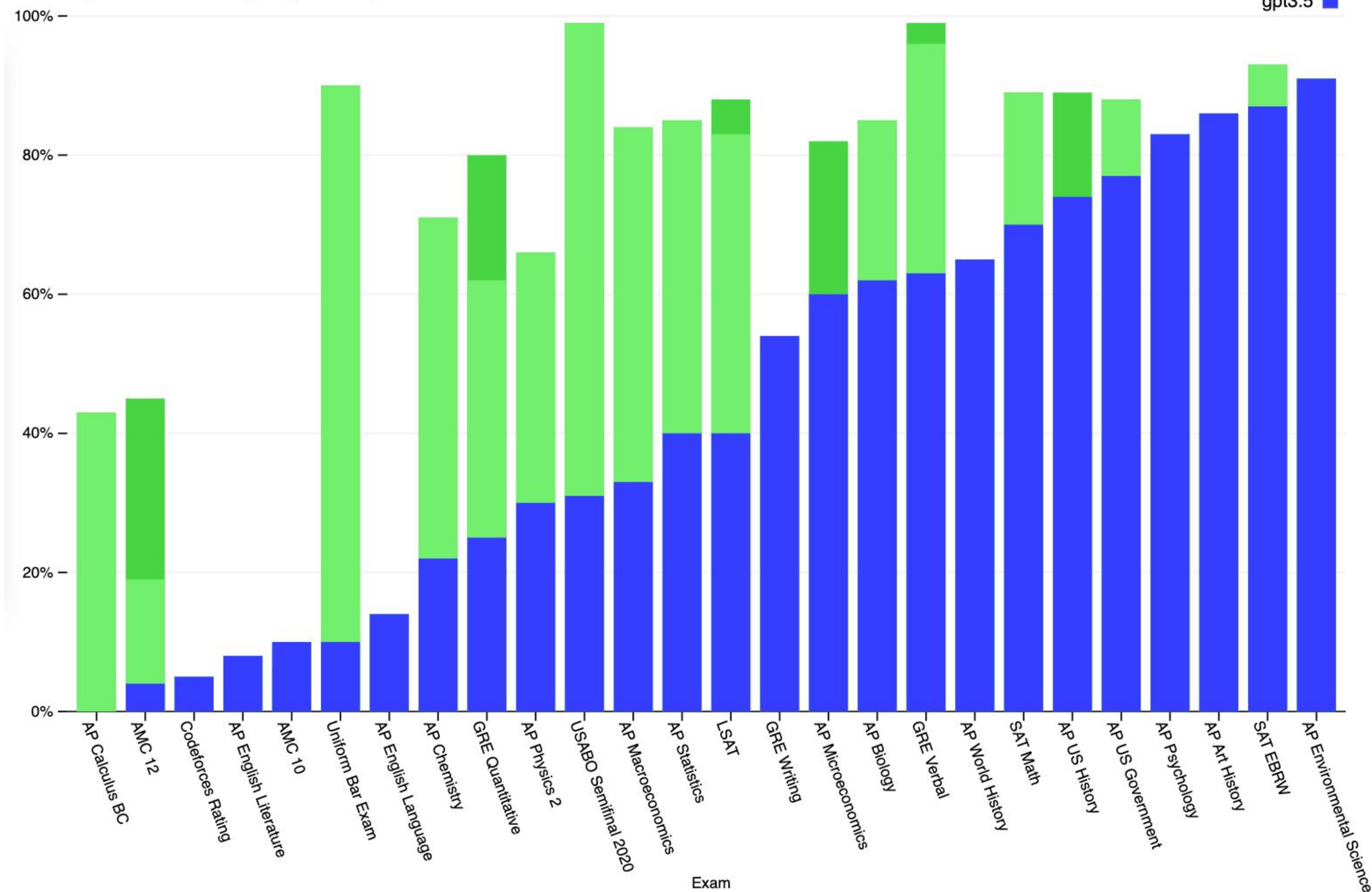


构建上下文向量，使解码器能够全面访问输入序列并重点关注相关部分。

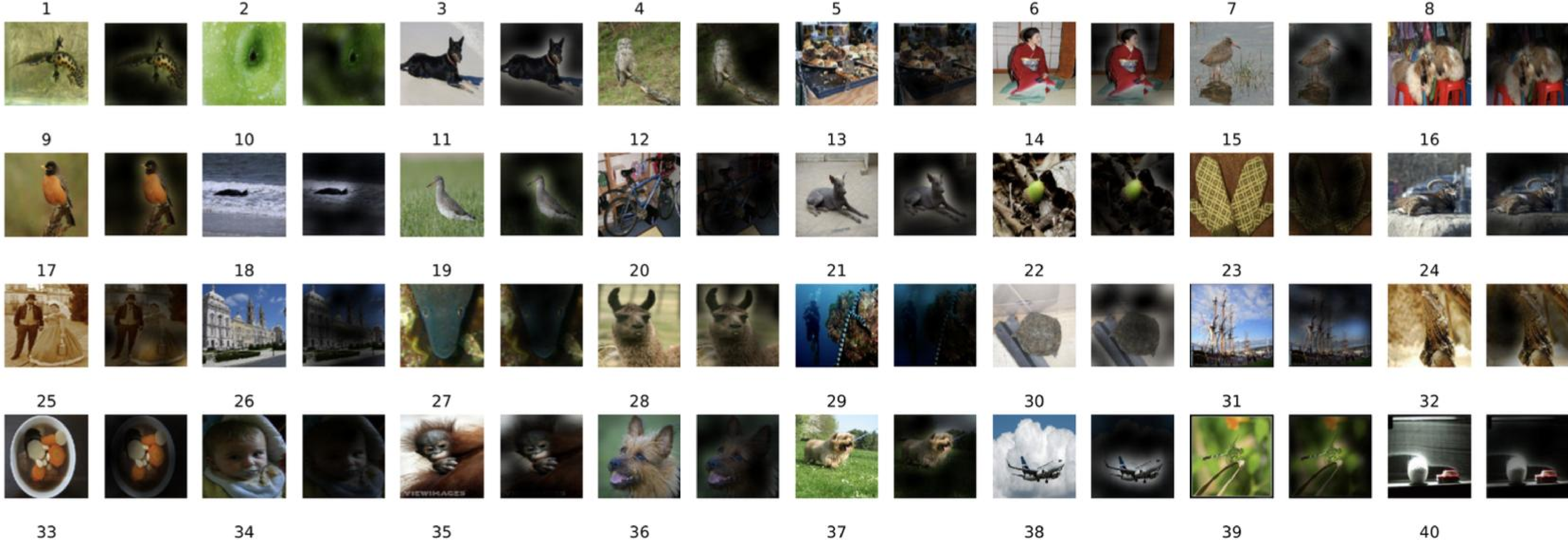
SOTA for Various Tasks

Exam results (ordered by GPT-3.5 performance)

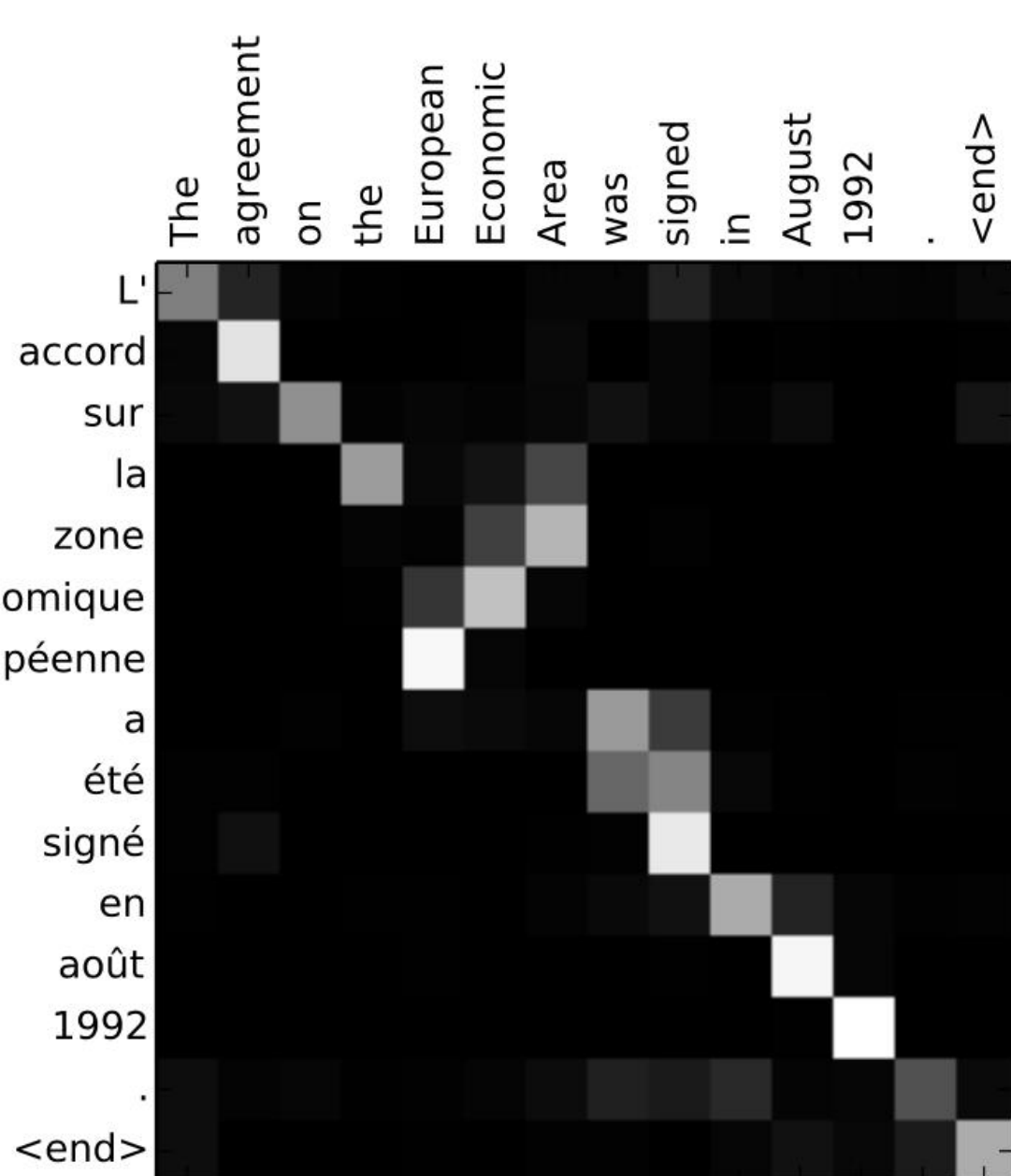
Estimated percentile lower bound (among test takers)



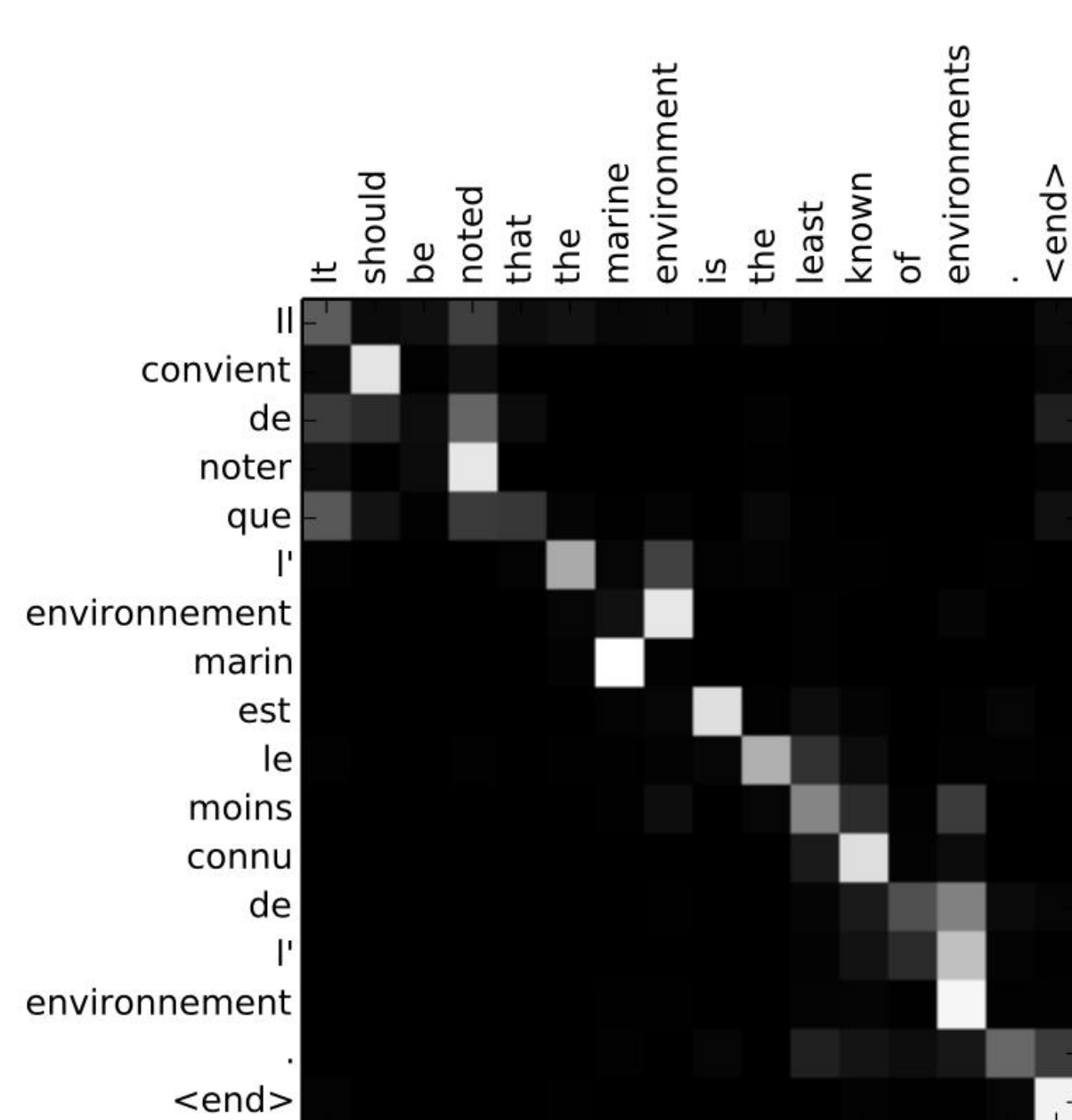
	estimated percentile	estimated percentile	estimated percentile
Uniform Bar Exam (MBE+MEE+MPT) ¹	298/400 ~90th	298/400 ~90th	213/400 ~10th
LSAT	163 ~88th	161 ~83rd	149 ~40th
SAT Evidence-Based Reading & Writing	710/800 ~93rd	710/800 ~93rd	670/800 ~87th
SAT Math	700/800 ~89th	690/800 ~89th	590/800 ~70th
Graduate Record Examination (GRE) Quantitative	163/170 ~80th	157/170 ~62nd	147/170 ~25th
Graduate Record Examination (GRE) Verbal	169/170 ~99th	165/170 ~96th	154/170 ~63rd
Graduate Record Examination (GRE) Writing	4/6 ~54th	4/6 ~54th	4/6 ~54th
USABO Semifinal Exam 2020	87/100 2nd-100th	87/100 95th-100th	43/150 31st-33rd
USNCO Local Section Exam 2022	36/60	38/60	24/60
Medical Knowledge Self-Assessment Program	75%	75%	53%
Codeforces Rating	392 below 5th	392 below 5th	260 below 5th
AP Art History	5 86th-100th	5 86th-100th	5 86th-100th
AP Biology	5 85th-100th	5 85th-100th	4 62nd-85th
AP Calculus BC	4 43rd-59th	4 43rd-59th	1 0th-7th
AP Chemistry	4 71st-88th	4 71st-88th	2 22nd-46th
AP English Language and Composition	2	2	2



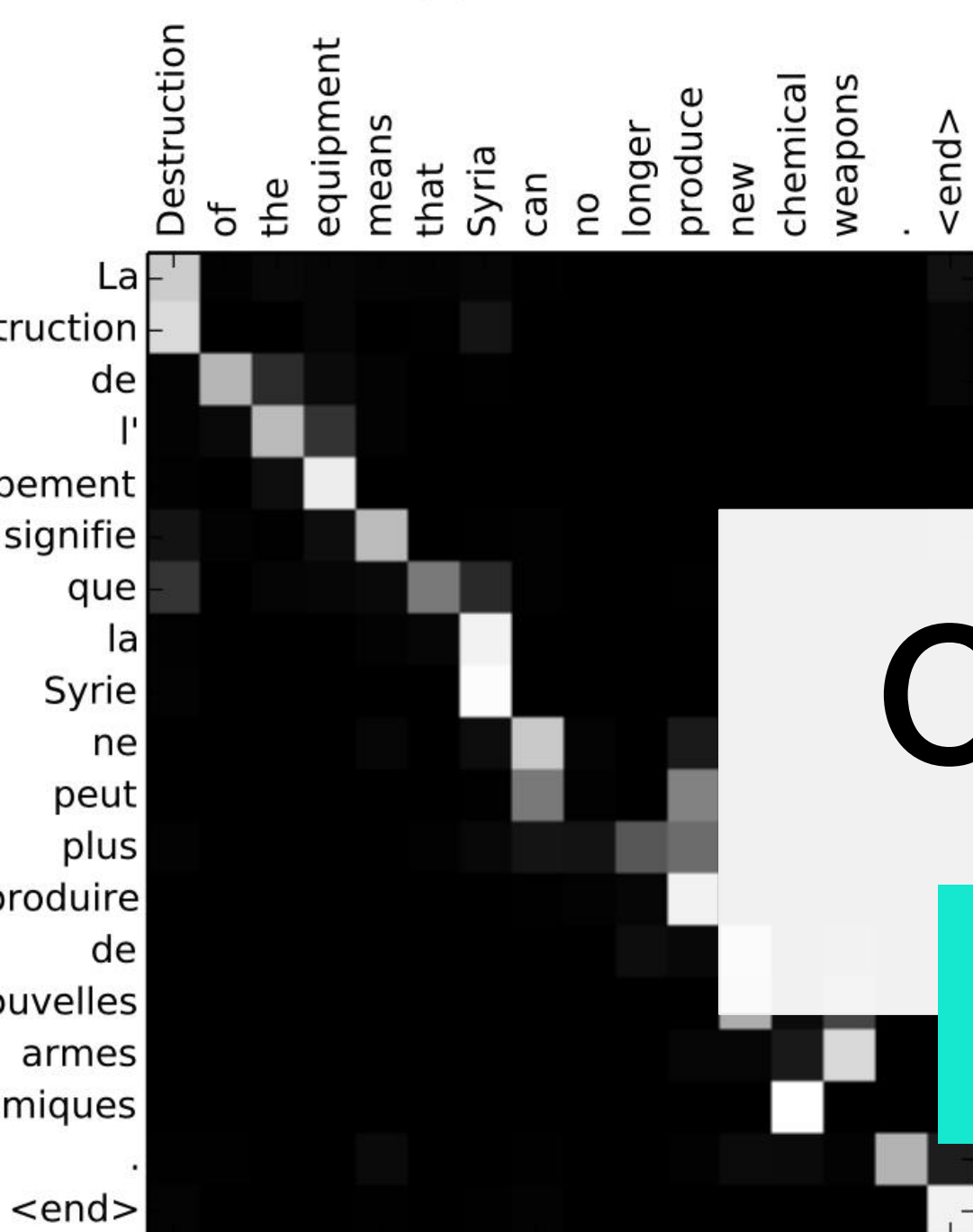
Characterisation & interpretability



(a)



(b)

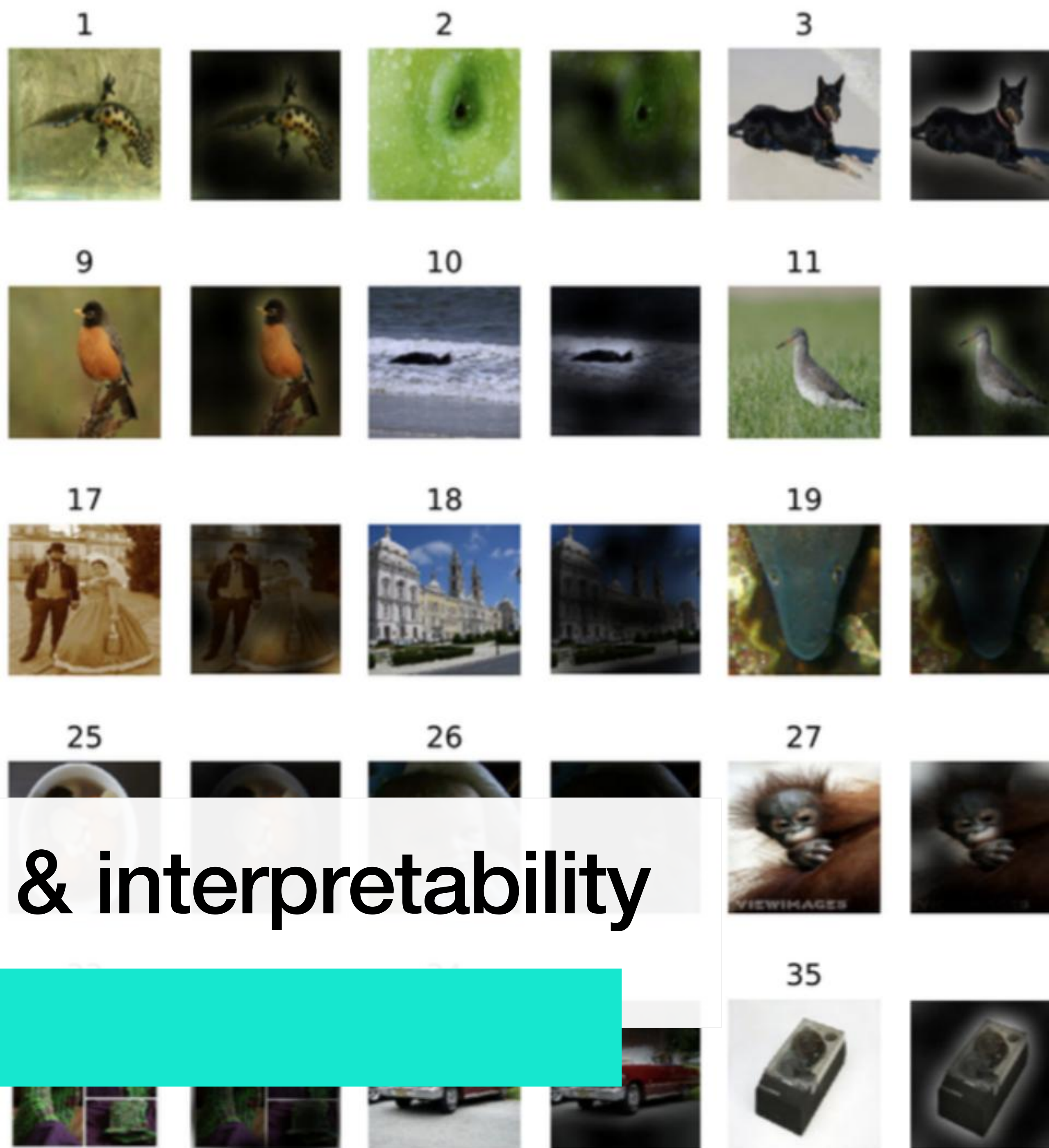


(c)



(d)

Characterisation & interpretability



变革里程碑: **Transformer**的崛起

Search... Expand

Origin paper

Neural Machine Translation by Jointly Learning to Align and Translate

Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio 2014

Sequence to Sequence Learning with Neural Networks

Ilya Sutskever, Oriol Vinyals, Quoc V. Le 2014

Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation

Kyunghyun Cho, Bart van Merriënboer, Çaglar Gülçehre, ... 2014

Attention is All you Need

Ashish Vaswani, Noam M. Shazeer, Niki Parmar, Jakob ... 2017

On the Properties of Neural Machine Translation: Encoder-Decoder Approaches

Kyunghyun Cho, Bart van Merriënboer, Dzmitry Bahdanau, ... 2014

Show, Attend and Tell: Neural Image Caption Generation with Visual Attention

Ke Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron C. ... 2015

On Using Monolingual Corpora in Neural Machine Translation

Çaglar Gülçehre, Orhan Firat, Kelvin Xu, Kyunghyun Cho, Loïc ... 2015

Bleu: a Method for Automatic Evaluation of Machine Translation

K. Papineni, S. Roukos, T. Ward, Wei-Jing Zhu 2002

Long Short-Term Memory

S. Hochreiter, J. Schmidhuber 1997

Neural Machine Translation of Rare Words with Subword Units

Rico Sennrich, B. Haddow, Alexandra Birch 2015

Universal Vector Neural Machine Translation With Effective Attention

Satish Mylapore, Ryan Quincy Paul, Joshua Yi, Robert D. Slater 2020

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

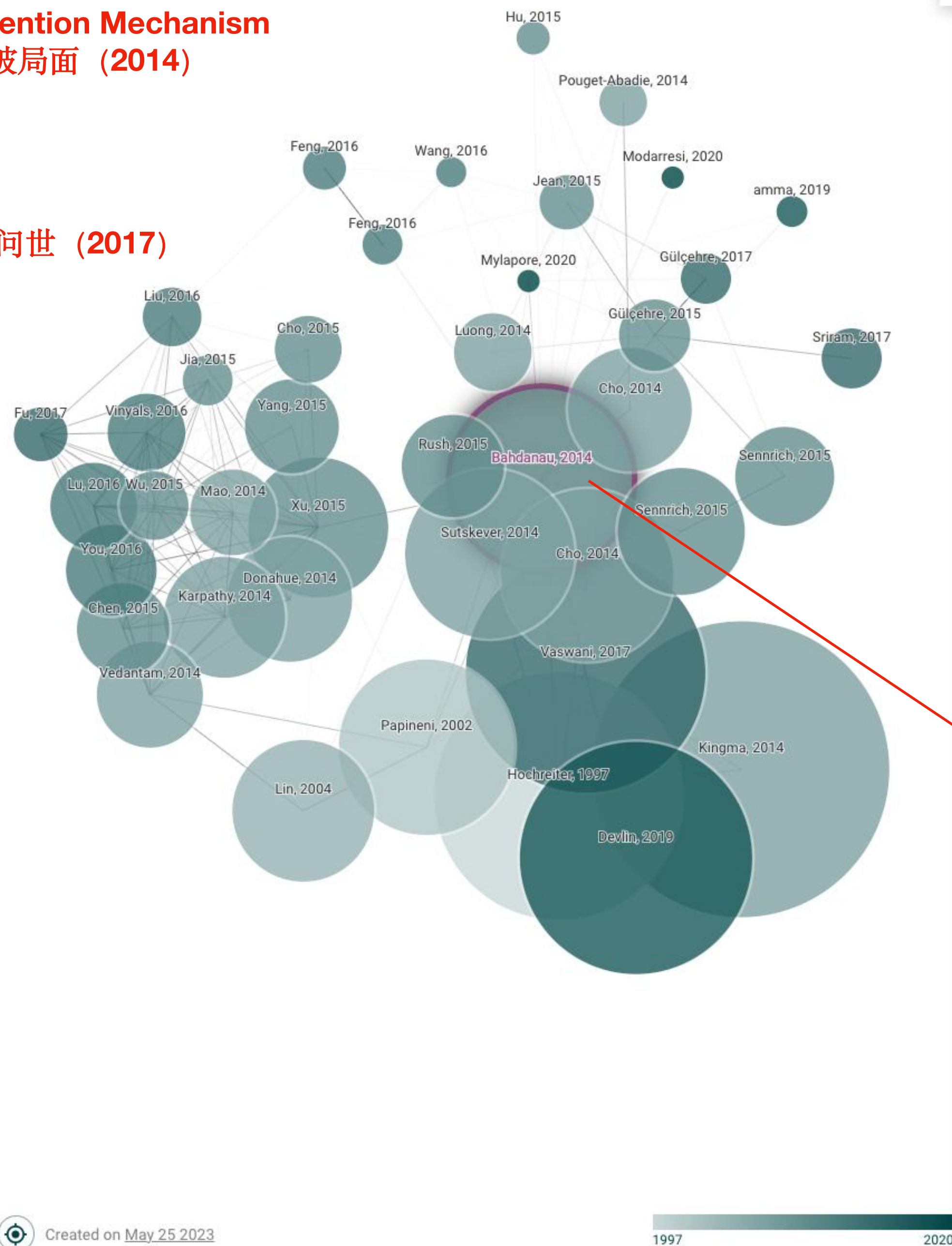
Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova 2019

Deep Visual-Semantic Alignments for Generating Image Descriptions

A. Karpathy, Li Fei-Fei 2014

Attention Mechanism
打破局面 (2014)

Transformer 问世 (2017)



Neural Machine Translation by Jointly Learning to Align and Translate

Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio

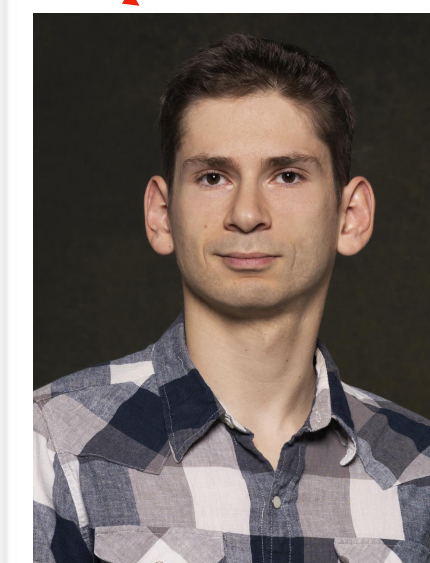
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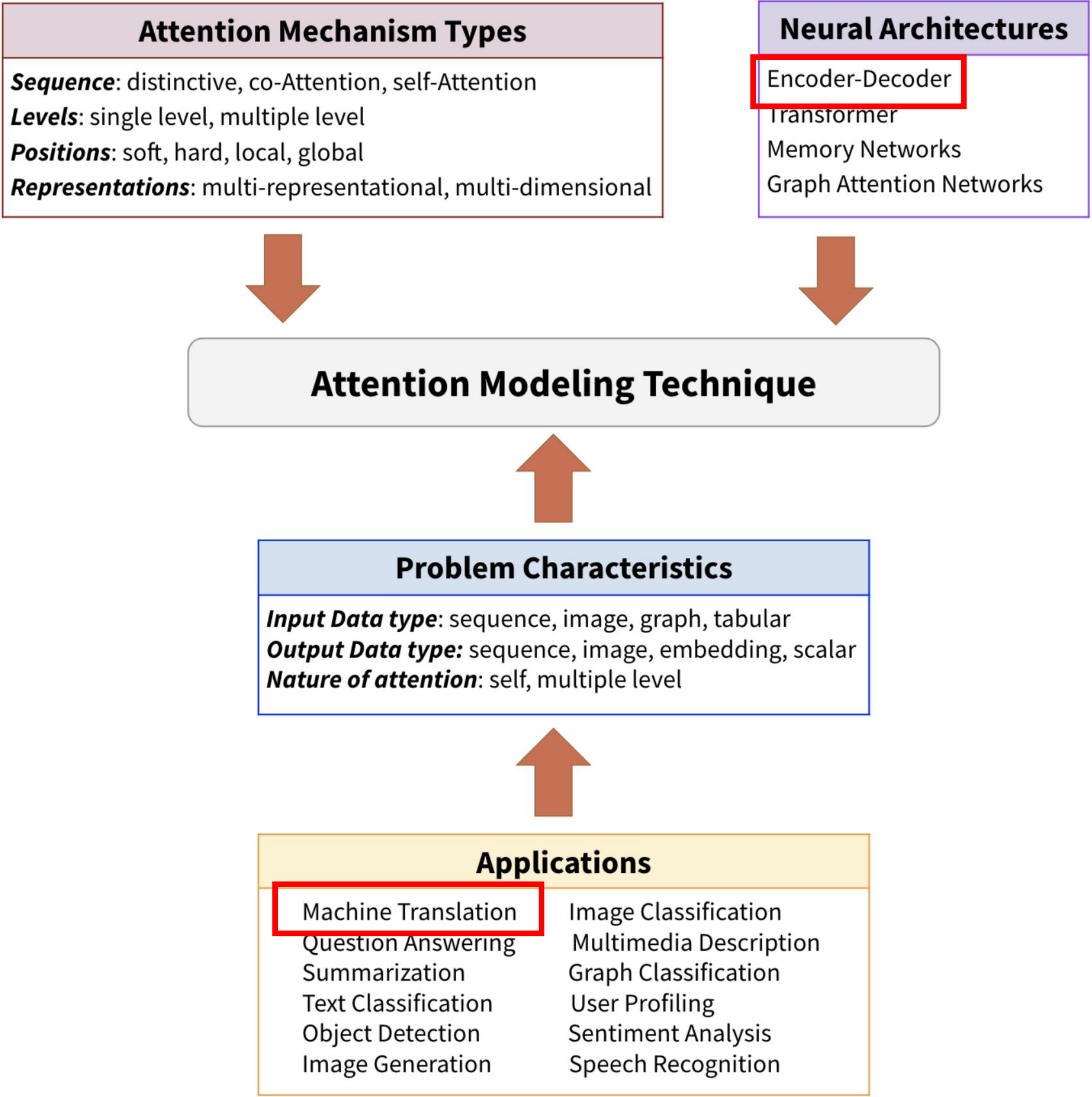
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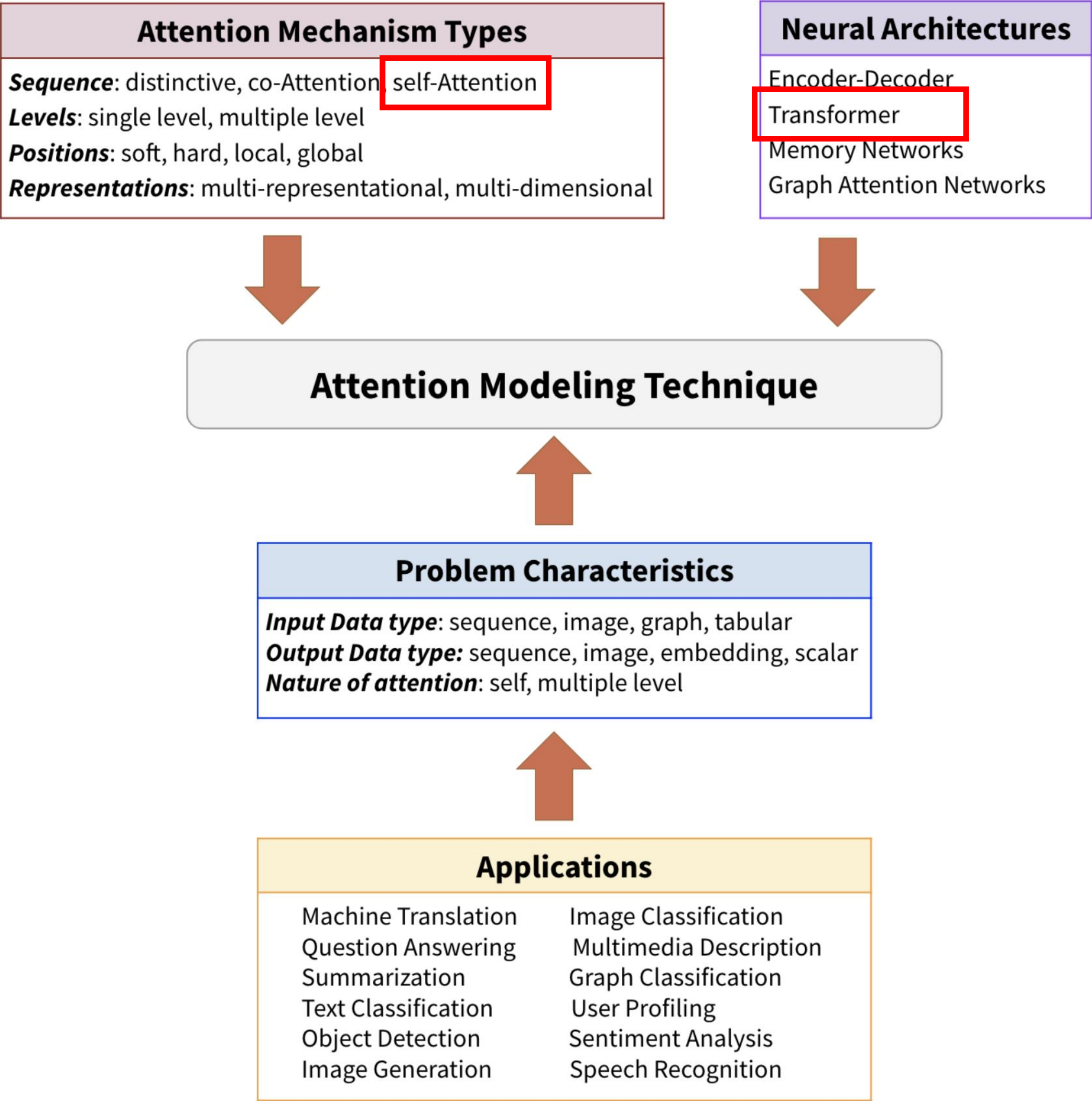
Neural machine translation is a recently proposed approach to machine translation. Unlike the traditional statistical machine translation, the neural machine translation aims at building a single neural network that can be jointly tuned to maximize the translation performance. The models proposed recently for neural machine translation often belong to a family of encoder-decoders and consists of an encoder that encodes a source sentence into a fixed-length vector from which a decoder generates a translation. In this paper, we conjecture that the use of a fixed-length vector is a bottleneck in improving the performance of this basic encoder-decoder architecture, and propose to extend this by allowing a model to automatically (soft-)search for parts of a source sentence that are relevant to predicting a target word, without having to form these parts as a hard segment explicitly. With this new approach, we achieve a translation performance comparable to the existing state-of-the-art phrase-based system on the task of English-to-French translation. Furthermore, qualitative analysis reveals that the (soft-)alignments found by the model agree well with our intuition.



Key Components of Attention Modeling Techniques



Key Components of Attention Modeling Techniques



Attention Is All You Need

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Llion Jones* Google Research llion@google.com	Aidan N. Gomez*[†] University of Toronto aidan@cs.toronto.edu	Lukasz Kaiser* Google Brain lukaszkaizer@google.com	
Illia Polosukhin*[‡] illia.polosukhin@gmail.com			

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

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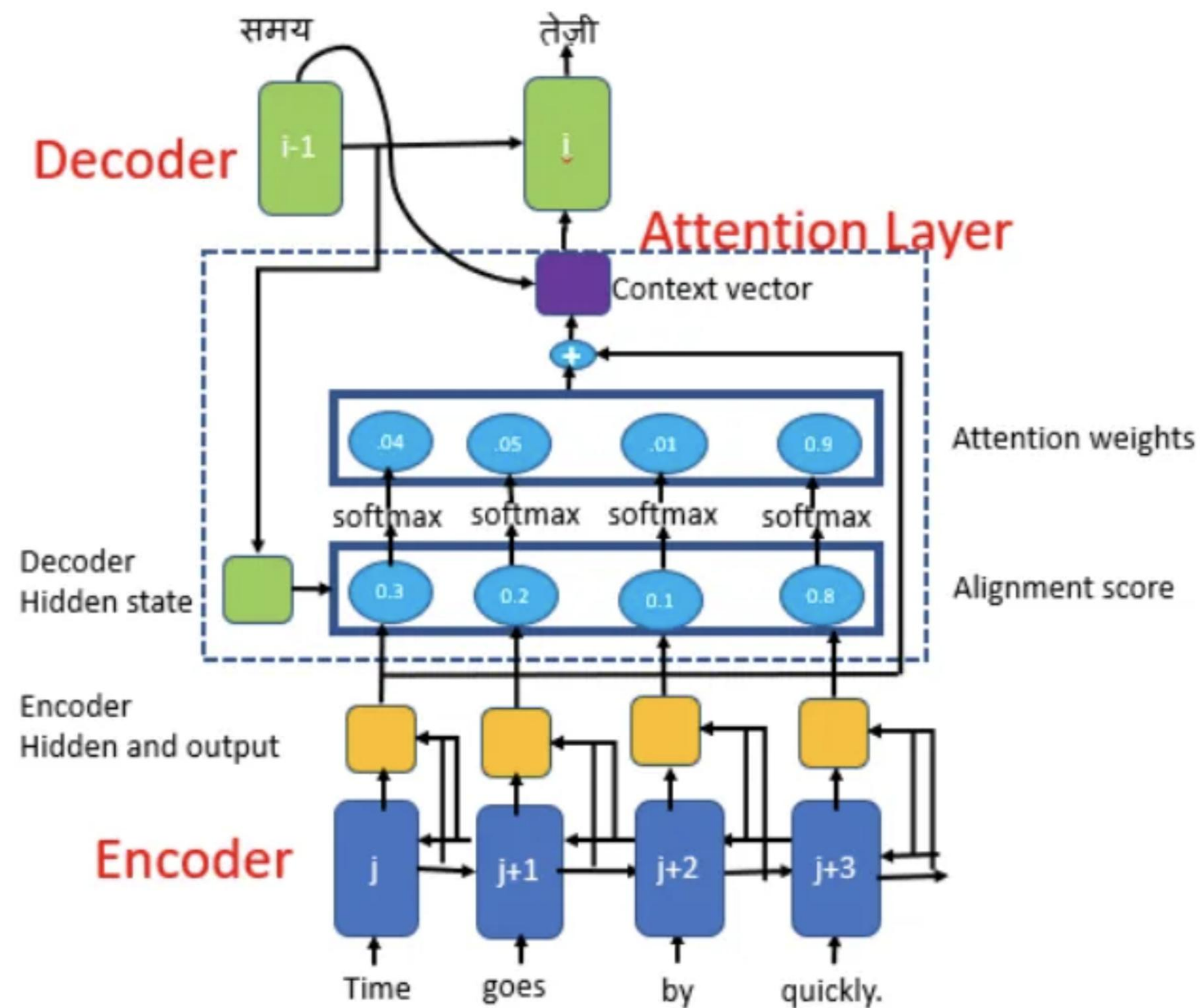
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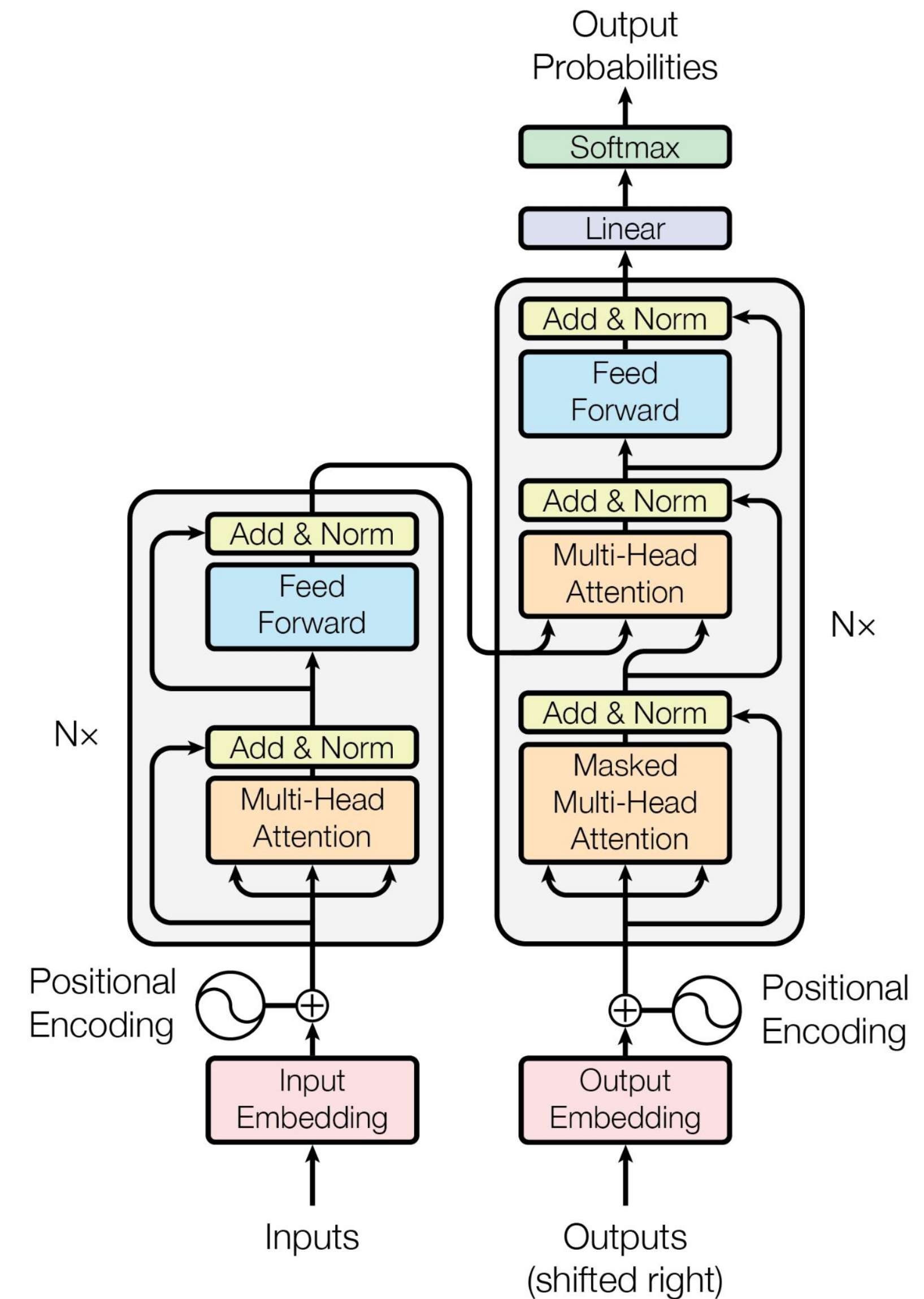
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Transformer : Game Changer



Encoder-decoder architecture with Attention Network



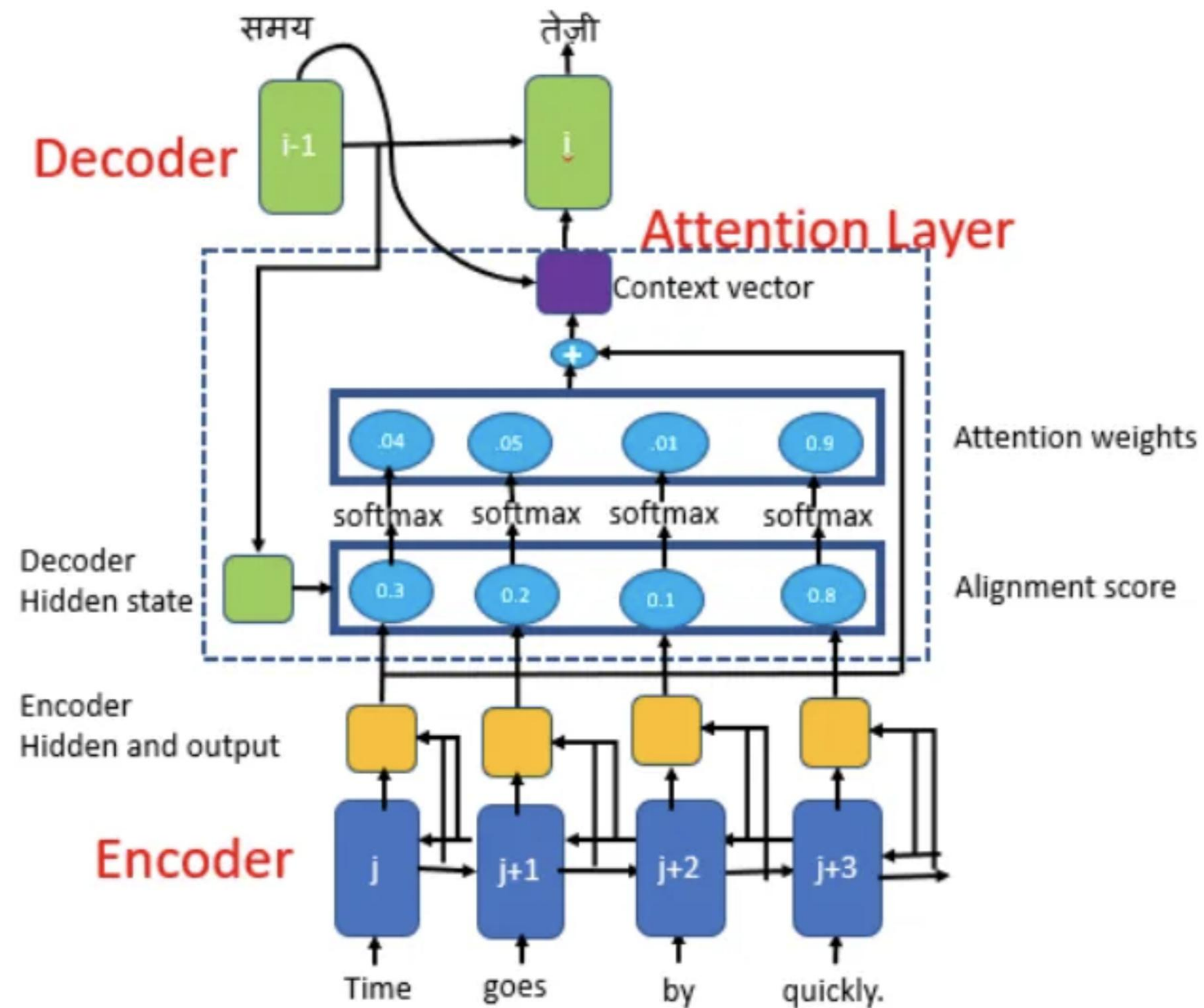
The Transformer - model architecture.

Transformer : Game Changer

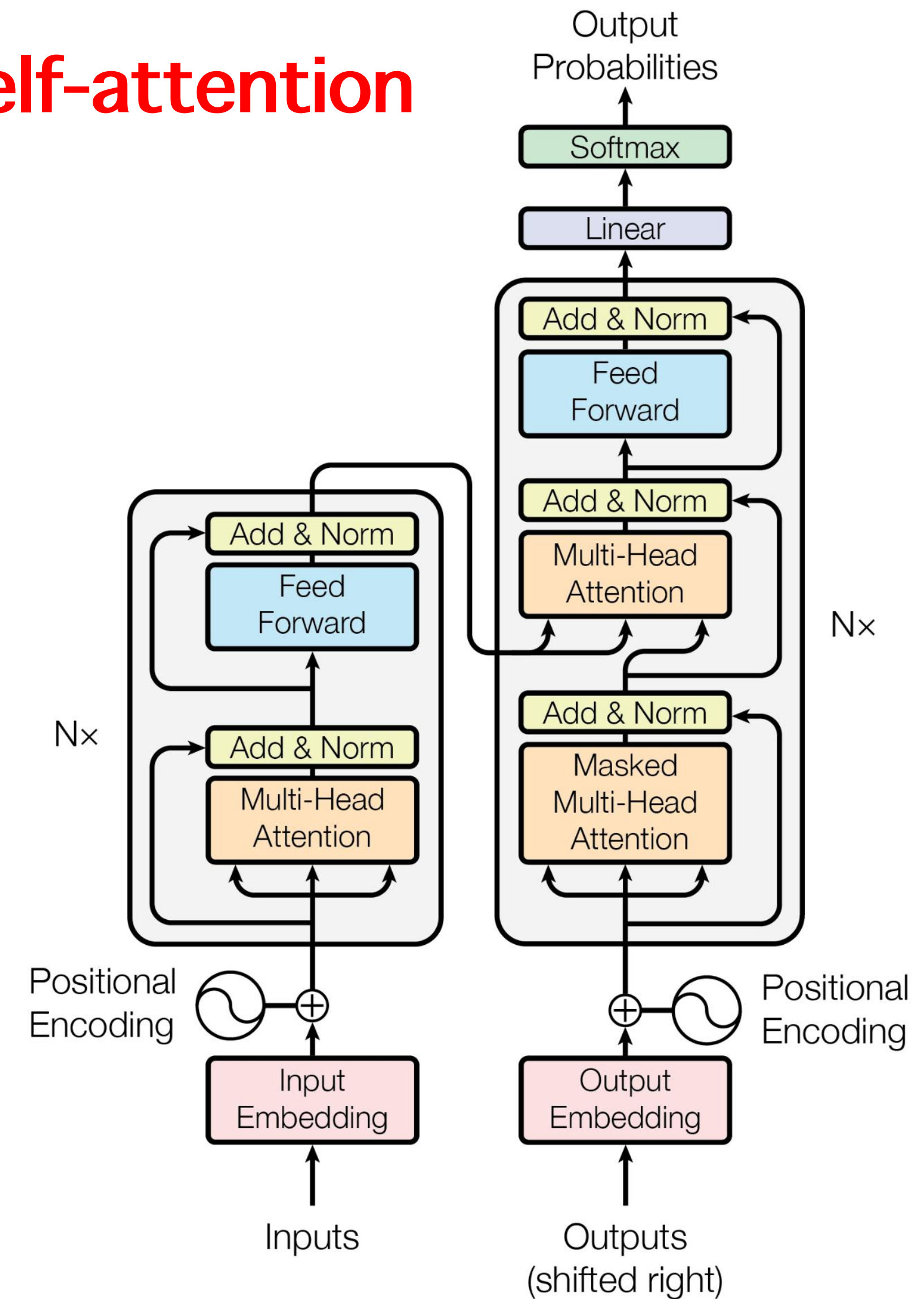
seq-aligned RNNs



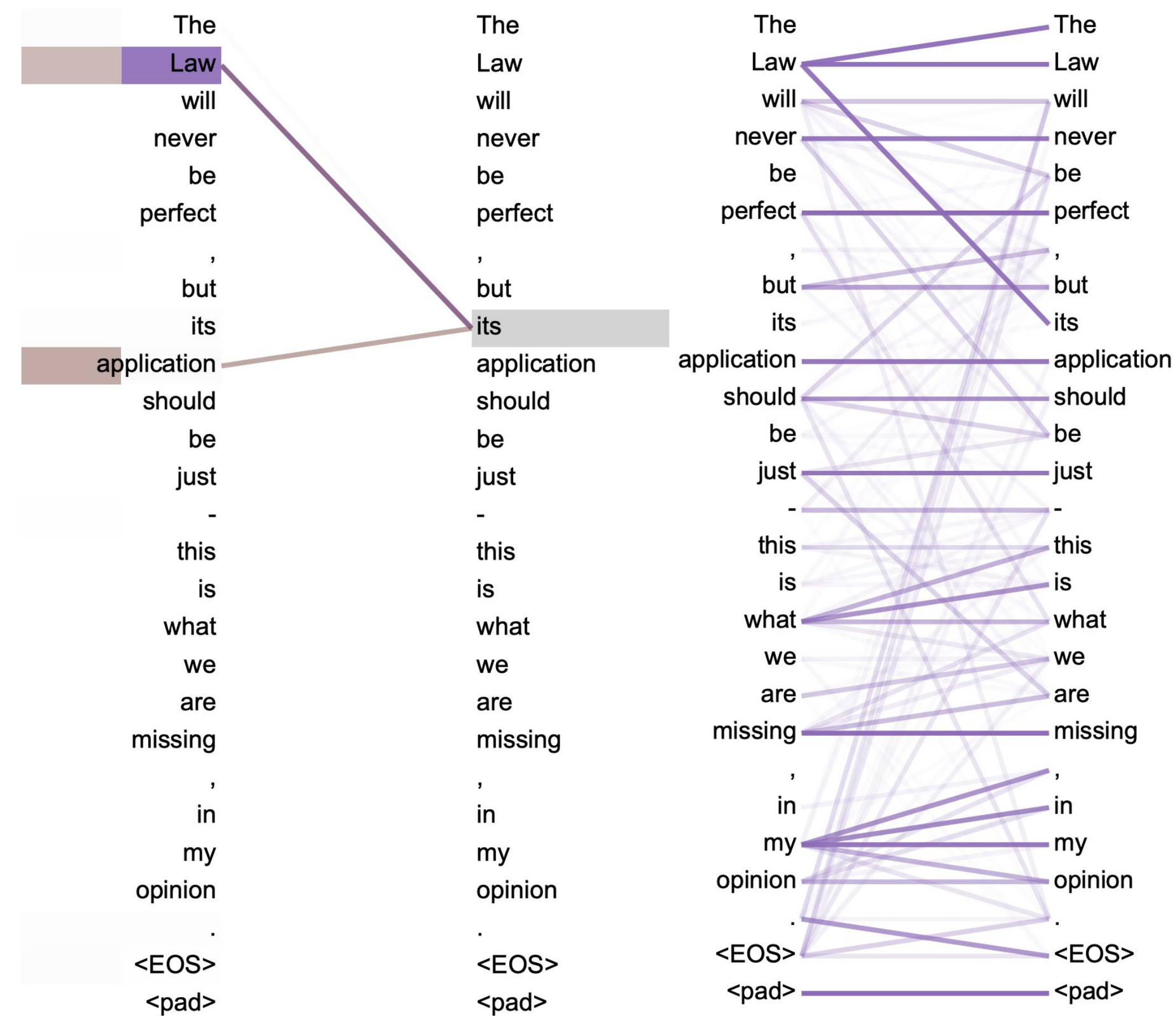
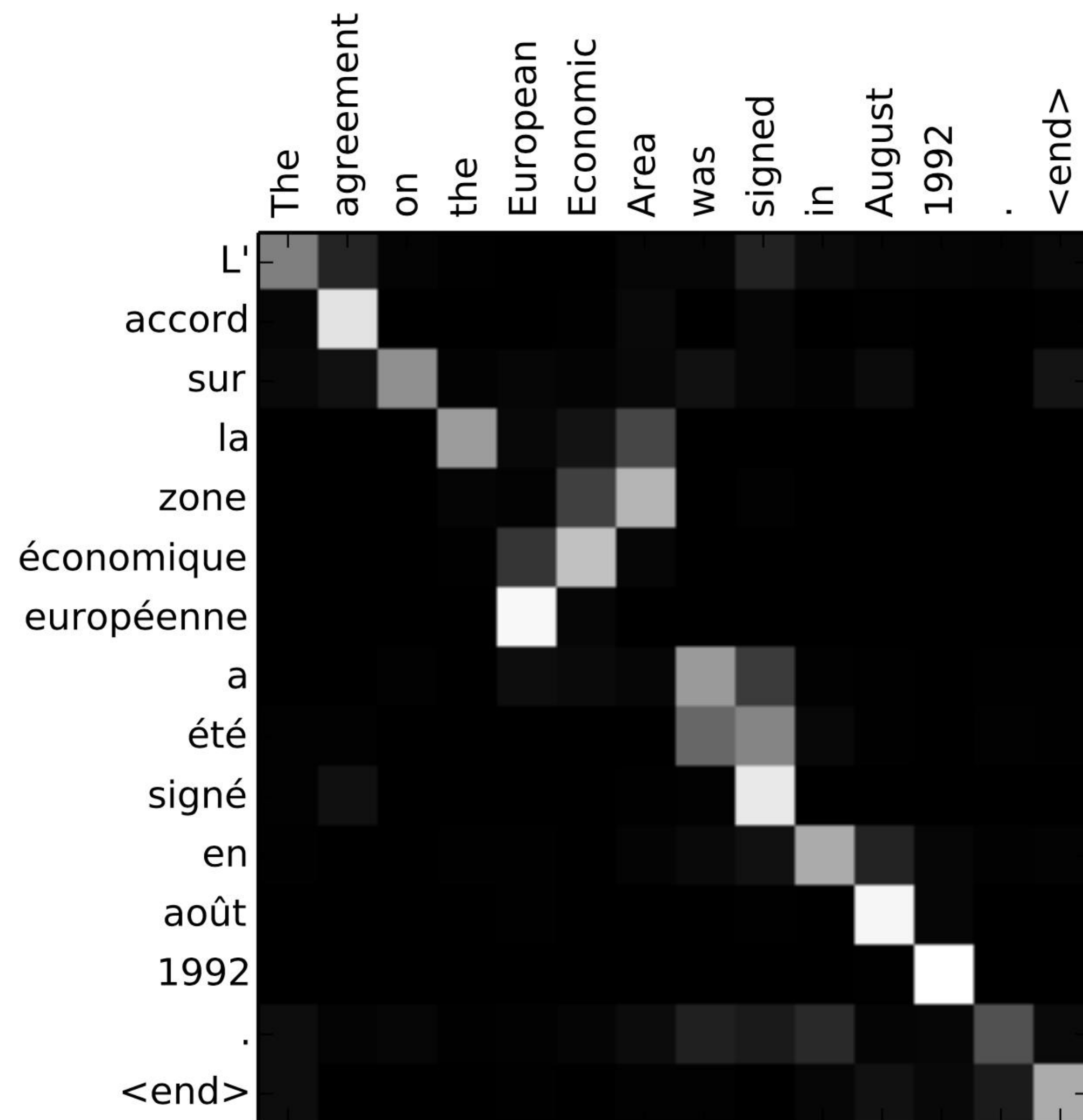
self-attention



Encoder-decoder architecture with Attention Network



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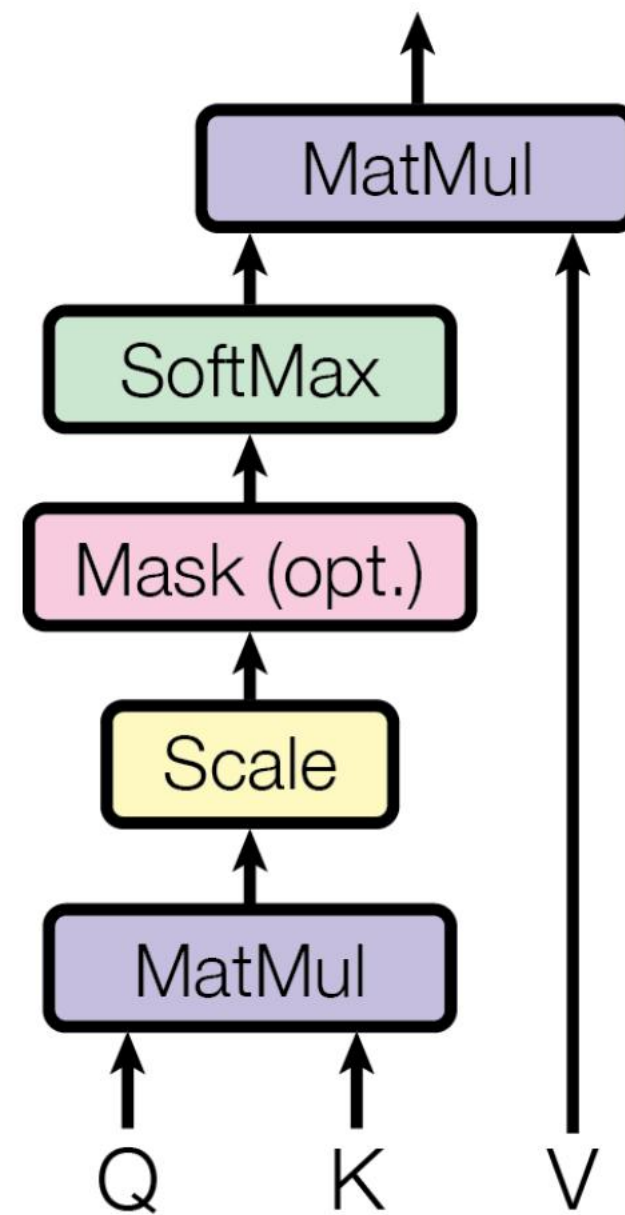
Self-attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Self-attention

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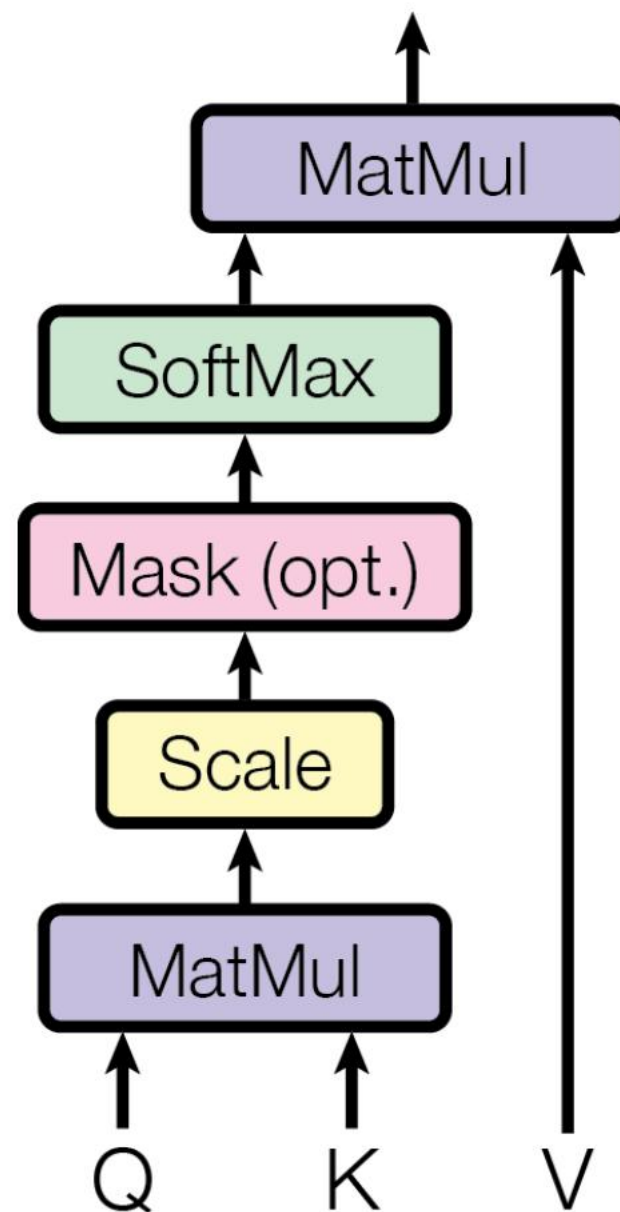
Scaled Dot-Product Attention



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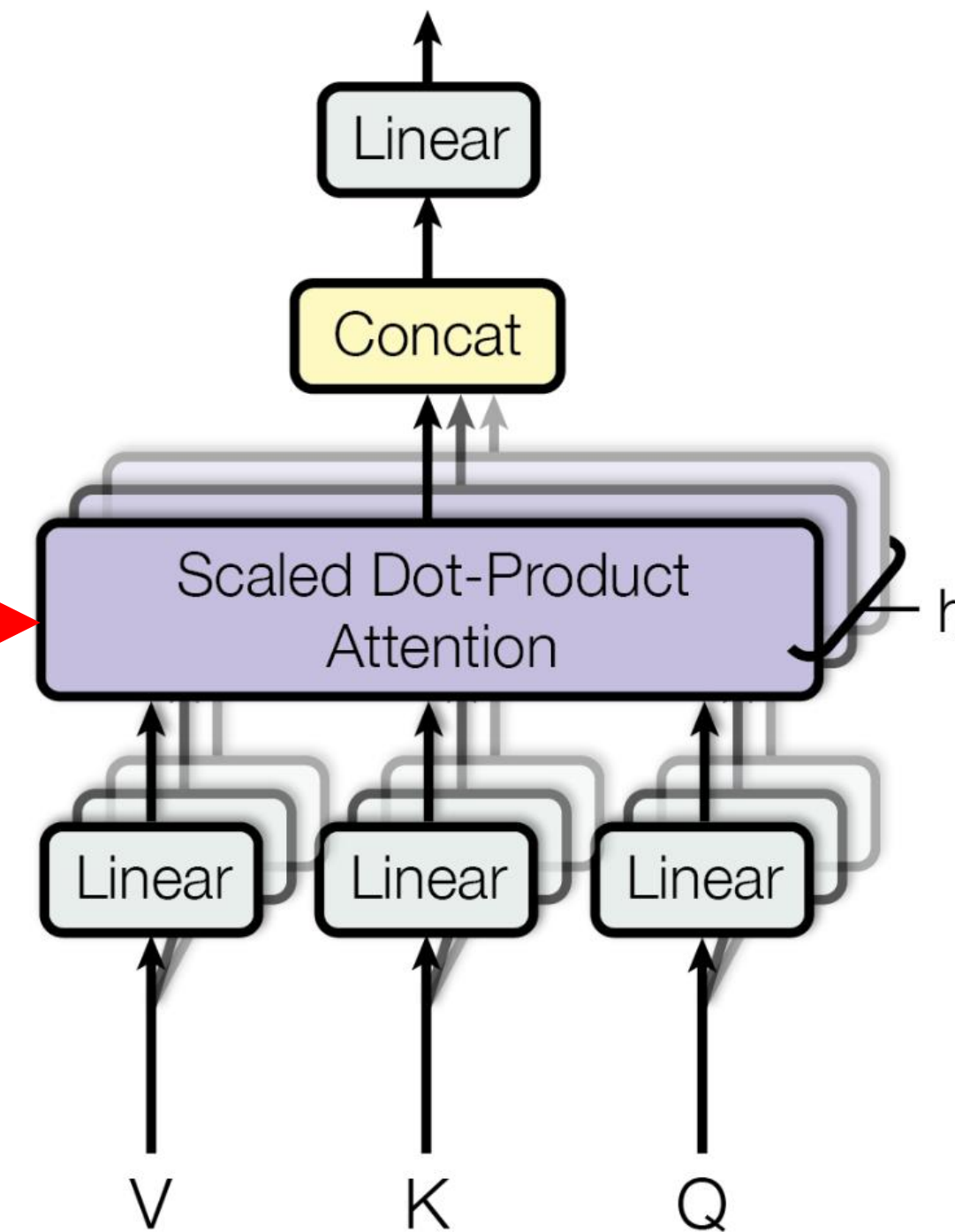
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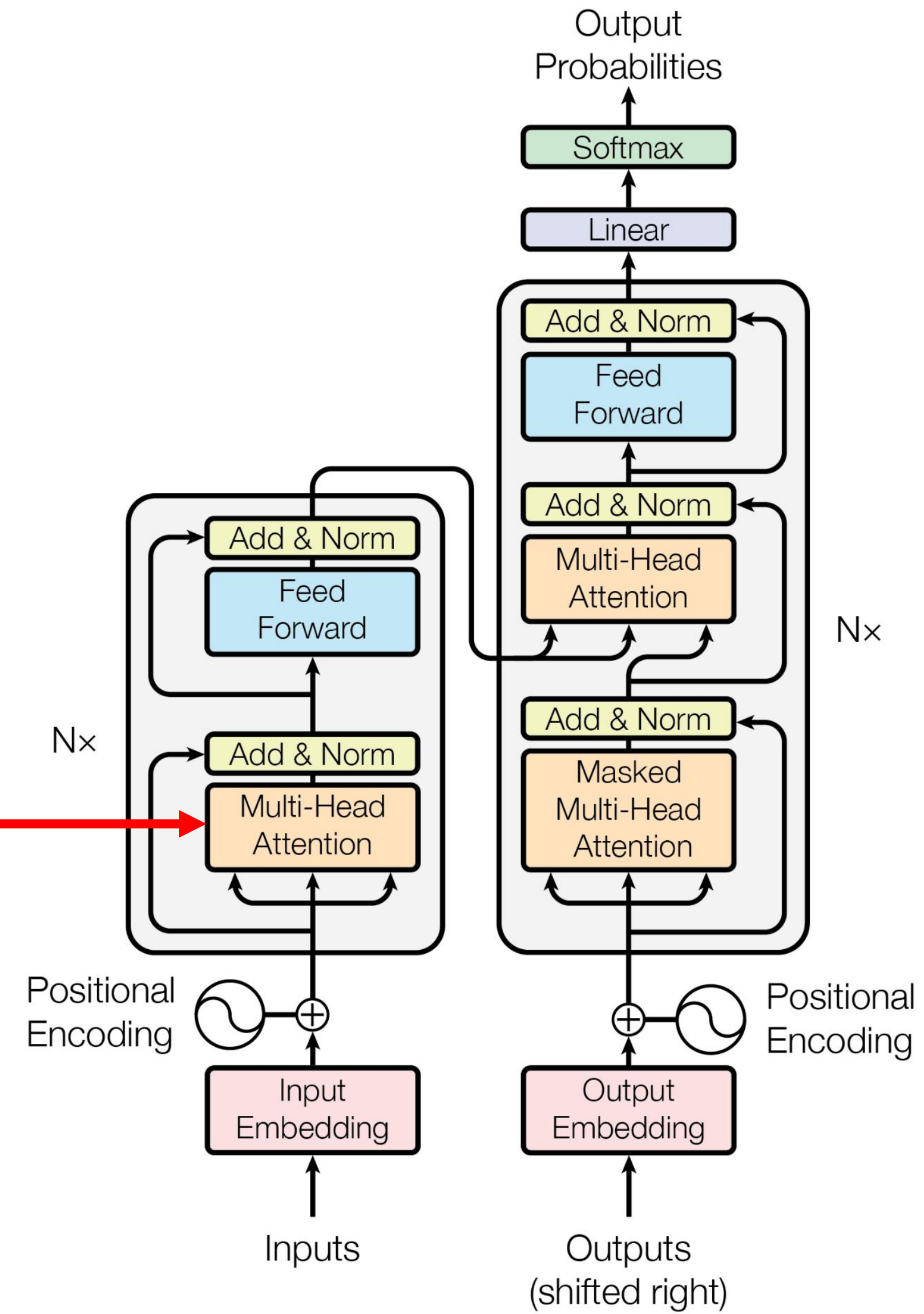
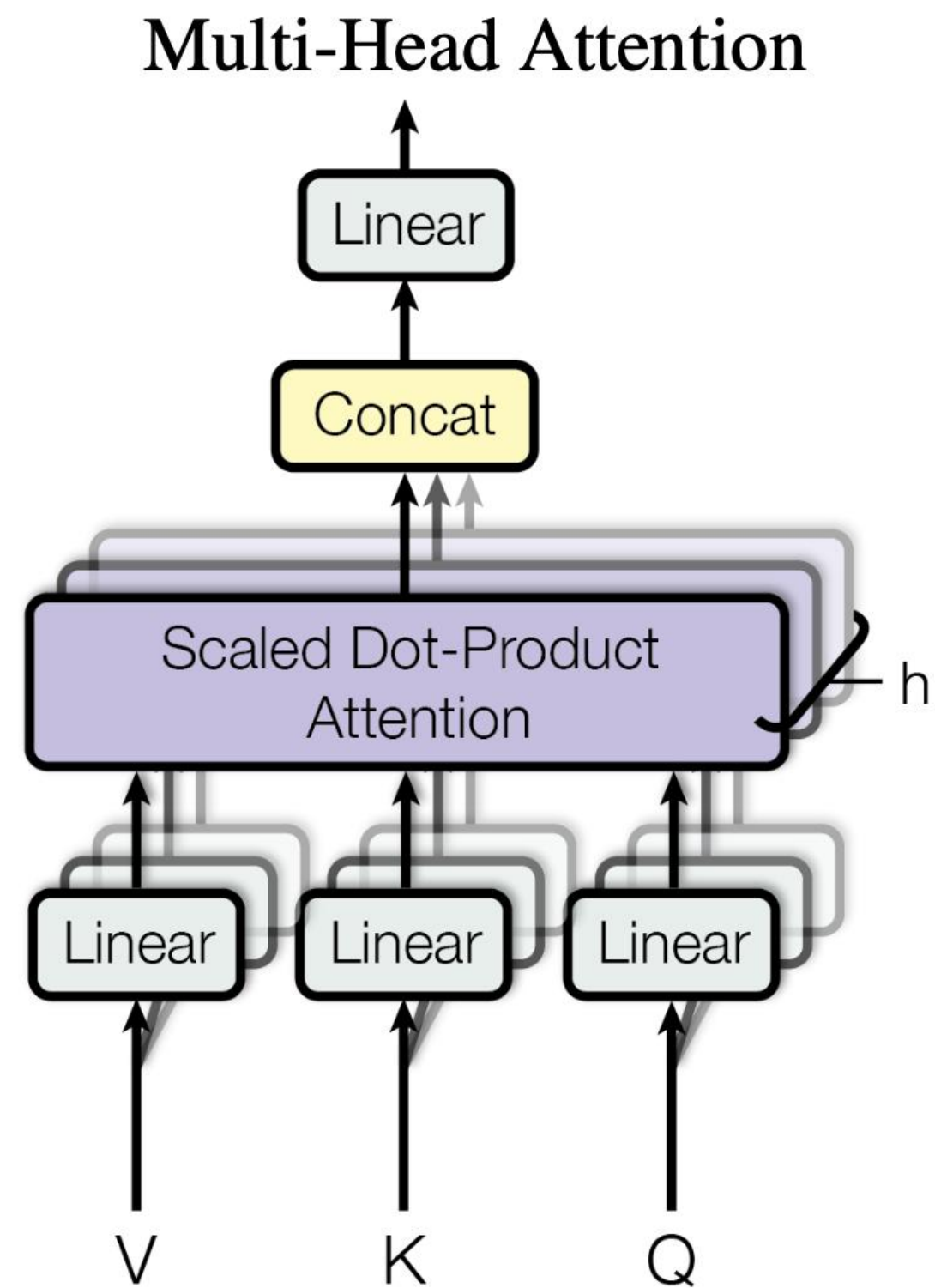
$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

Multi-Head Attention

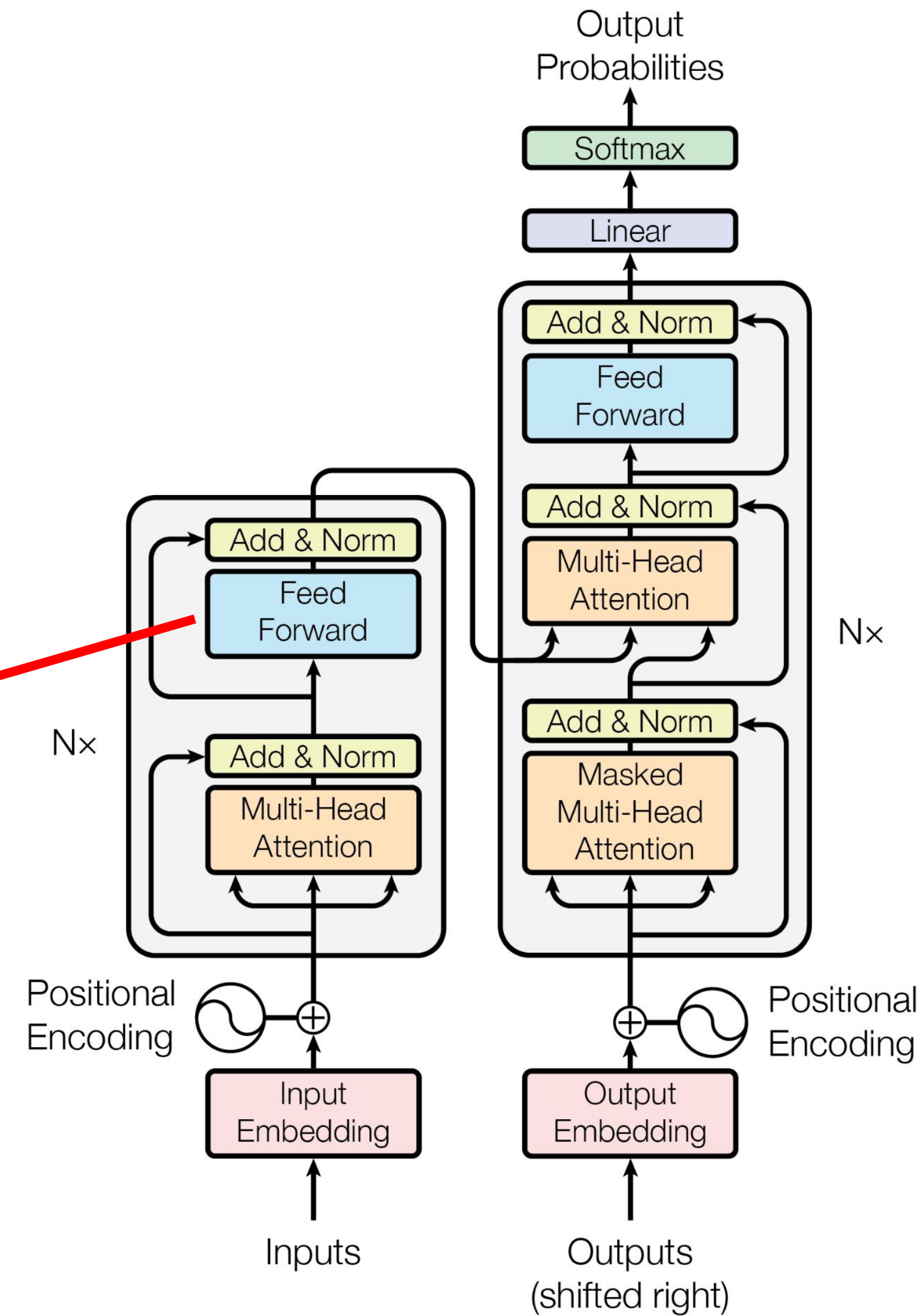


Self-attention

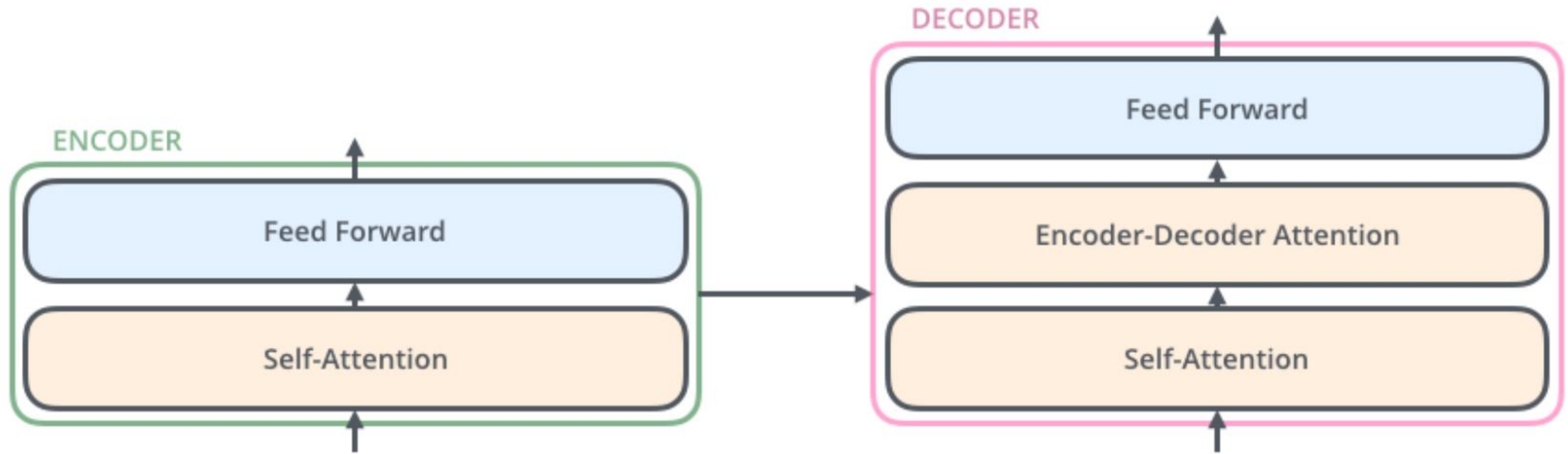


Self-attention

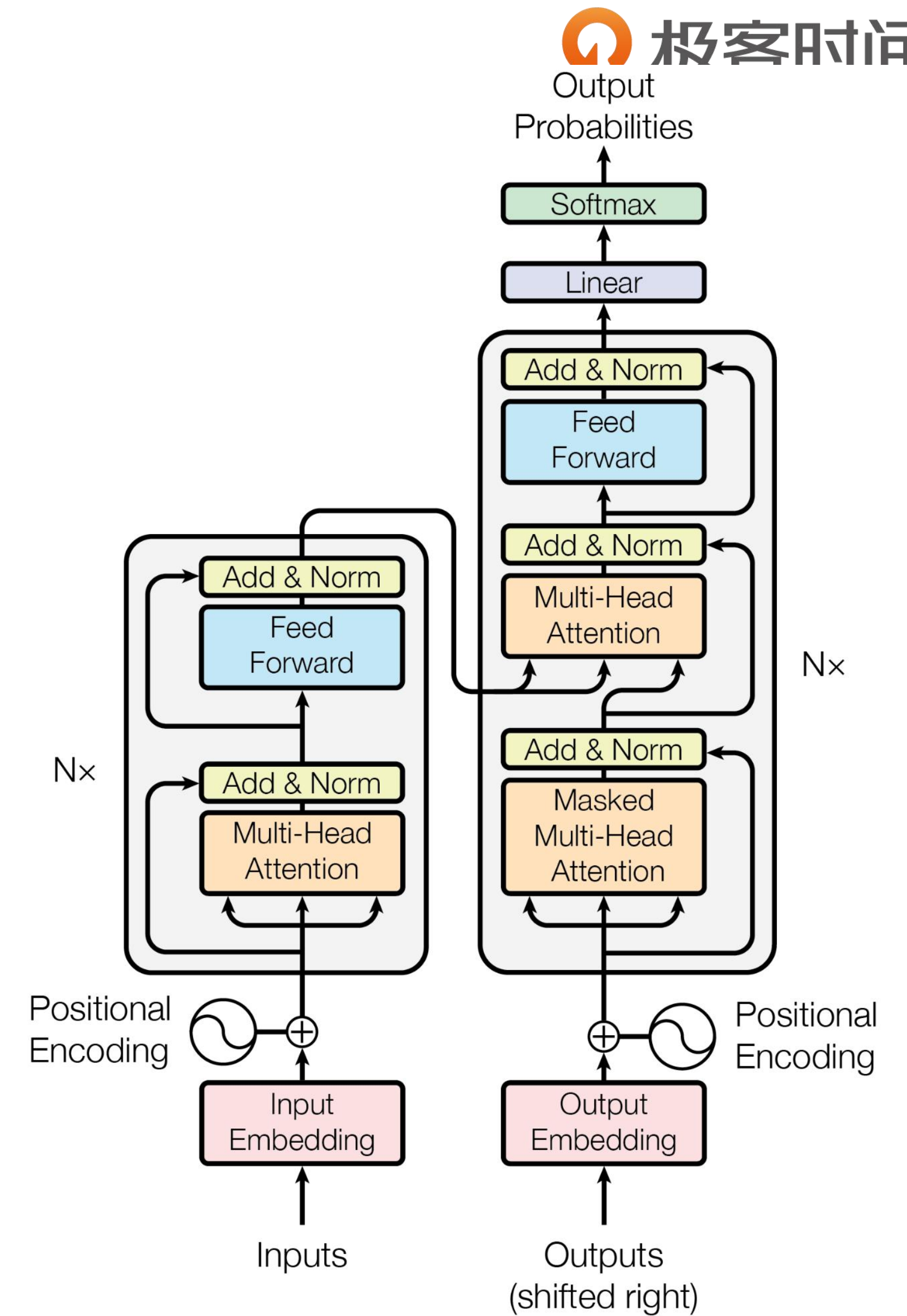
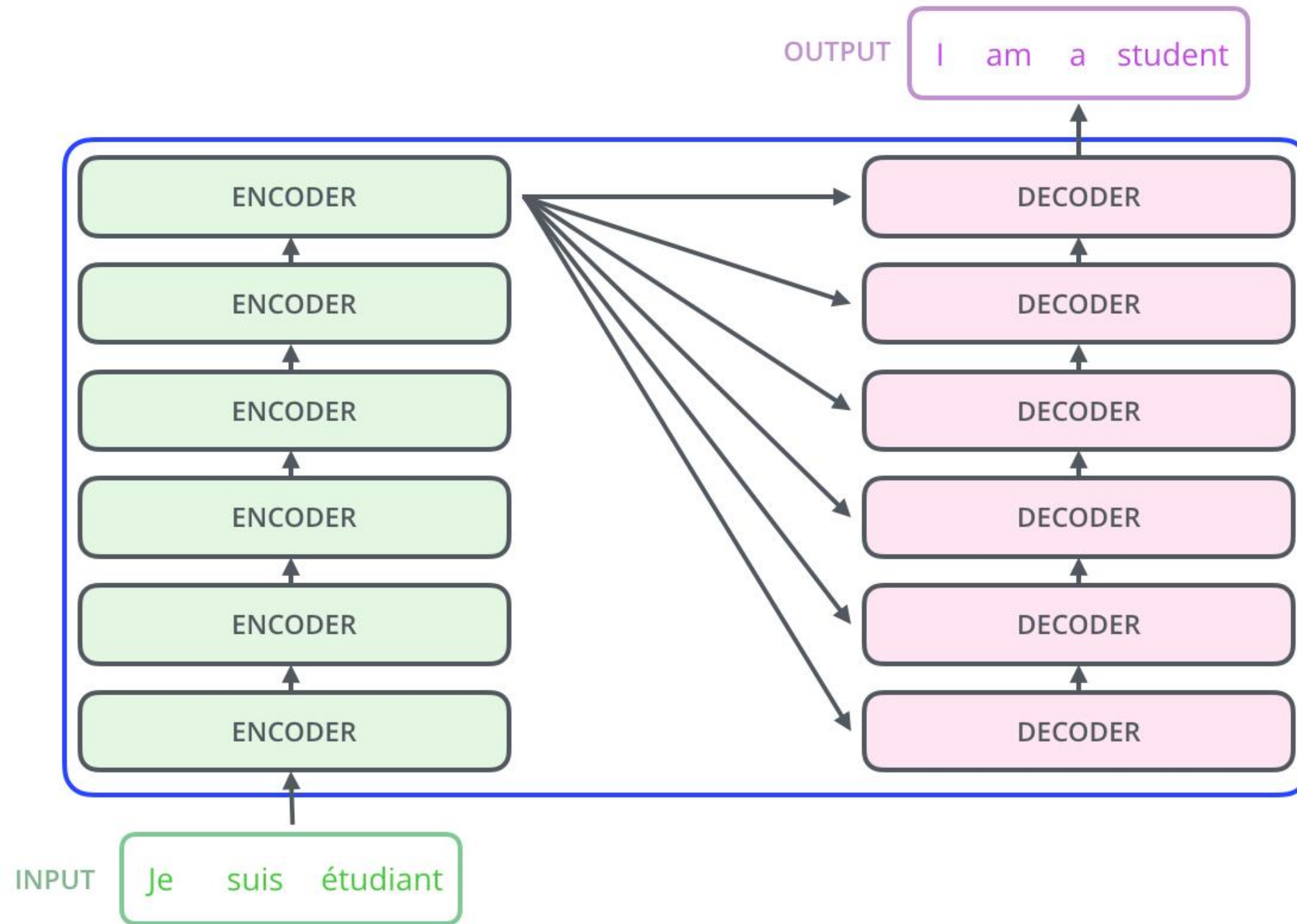
$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$



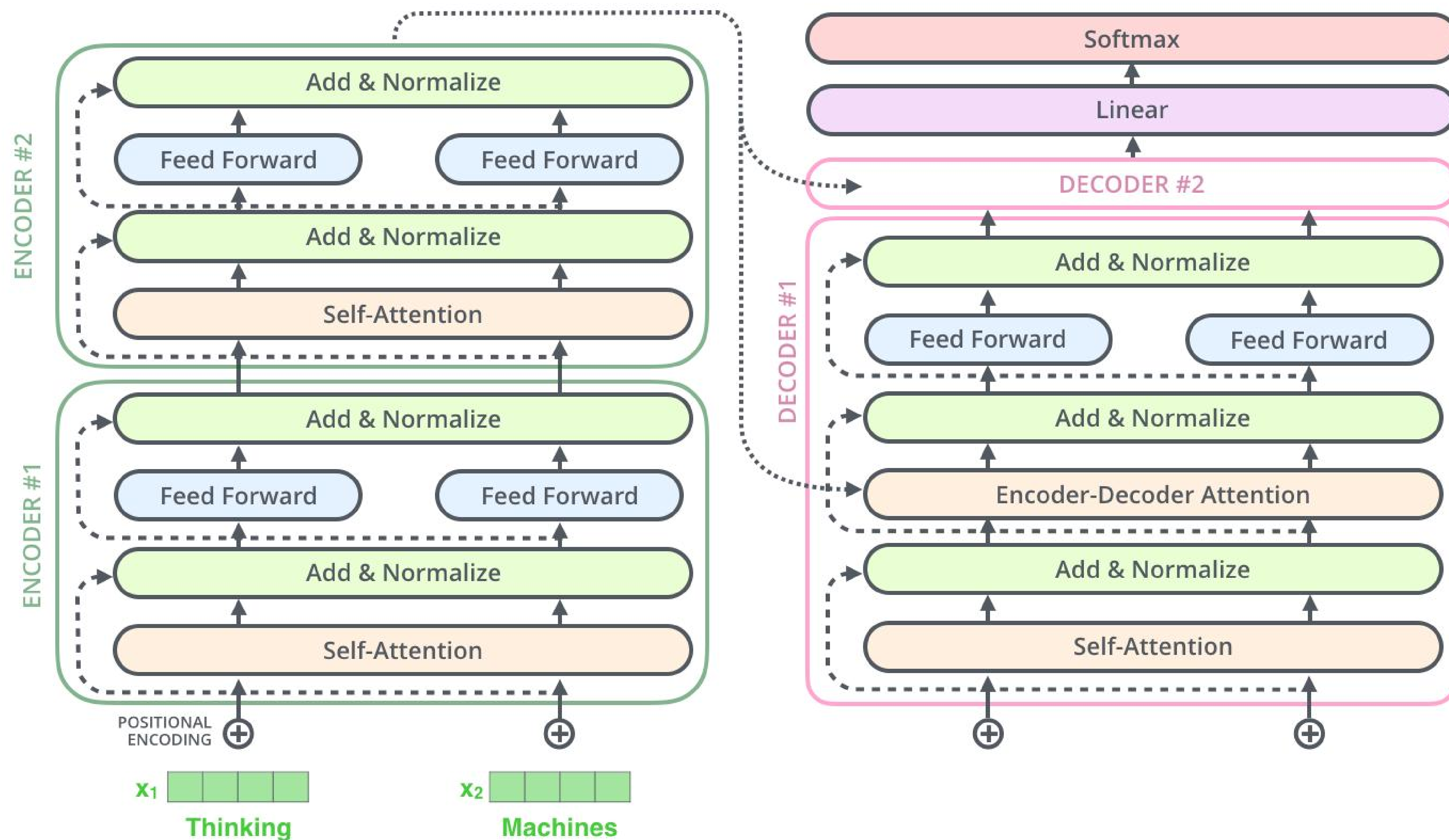
Transformer : Encoder - Decoder



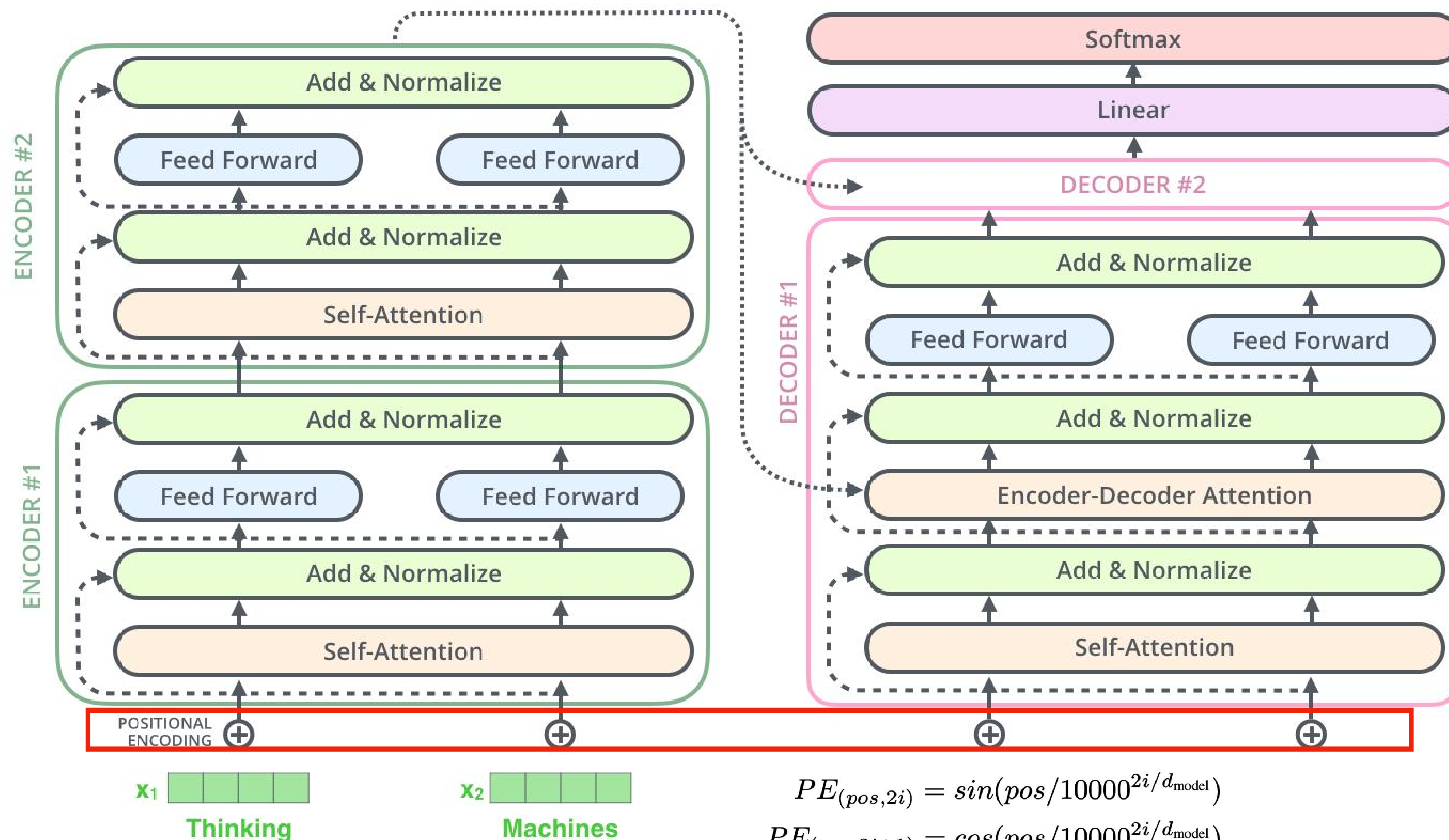
Transformer : A High-Level Look



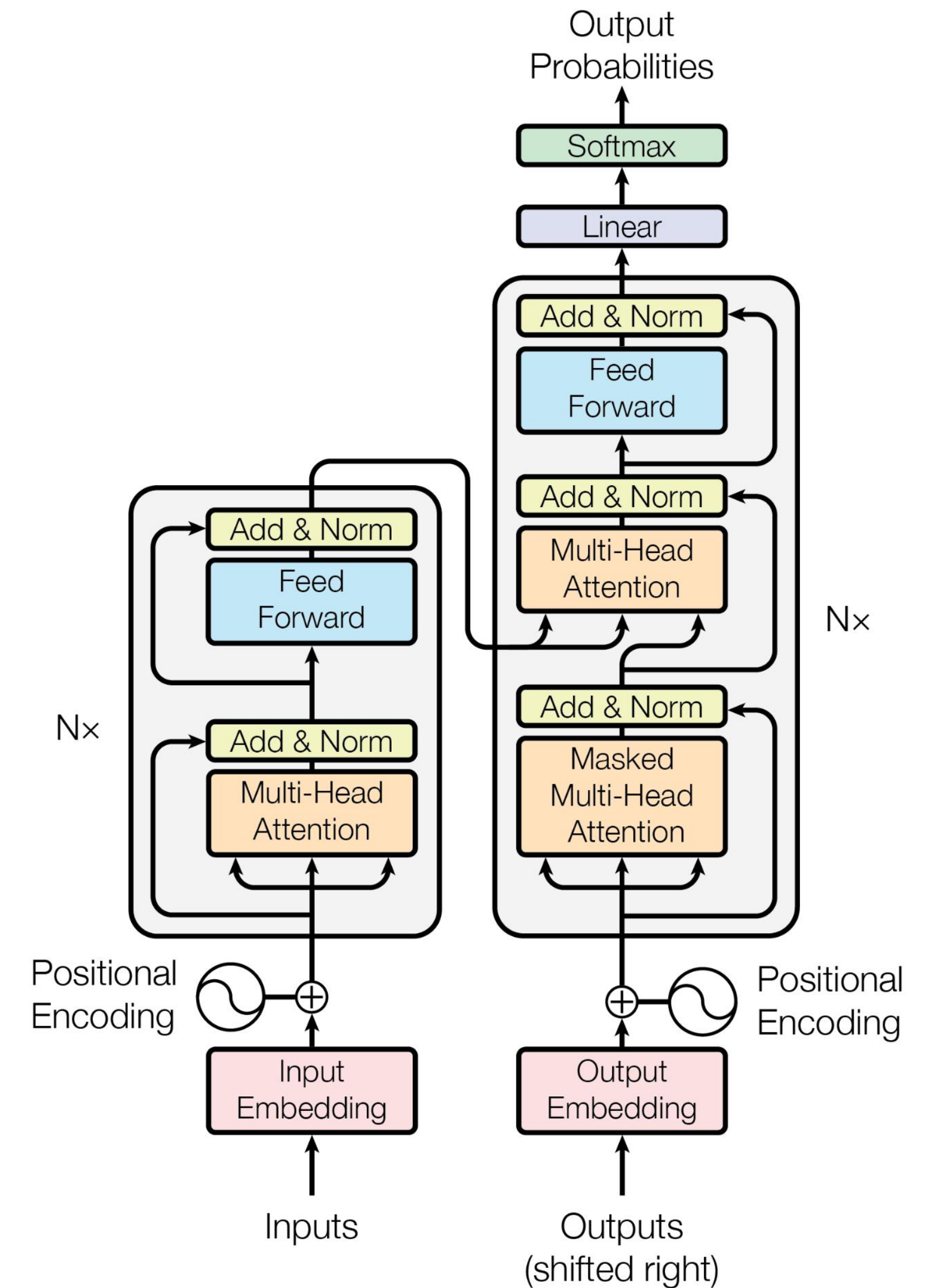
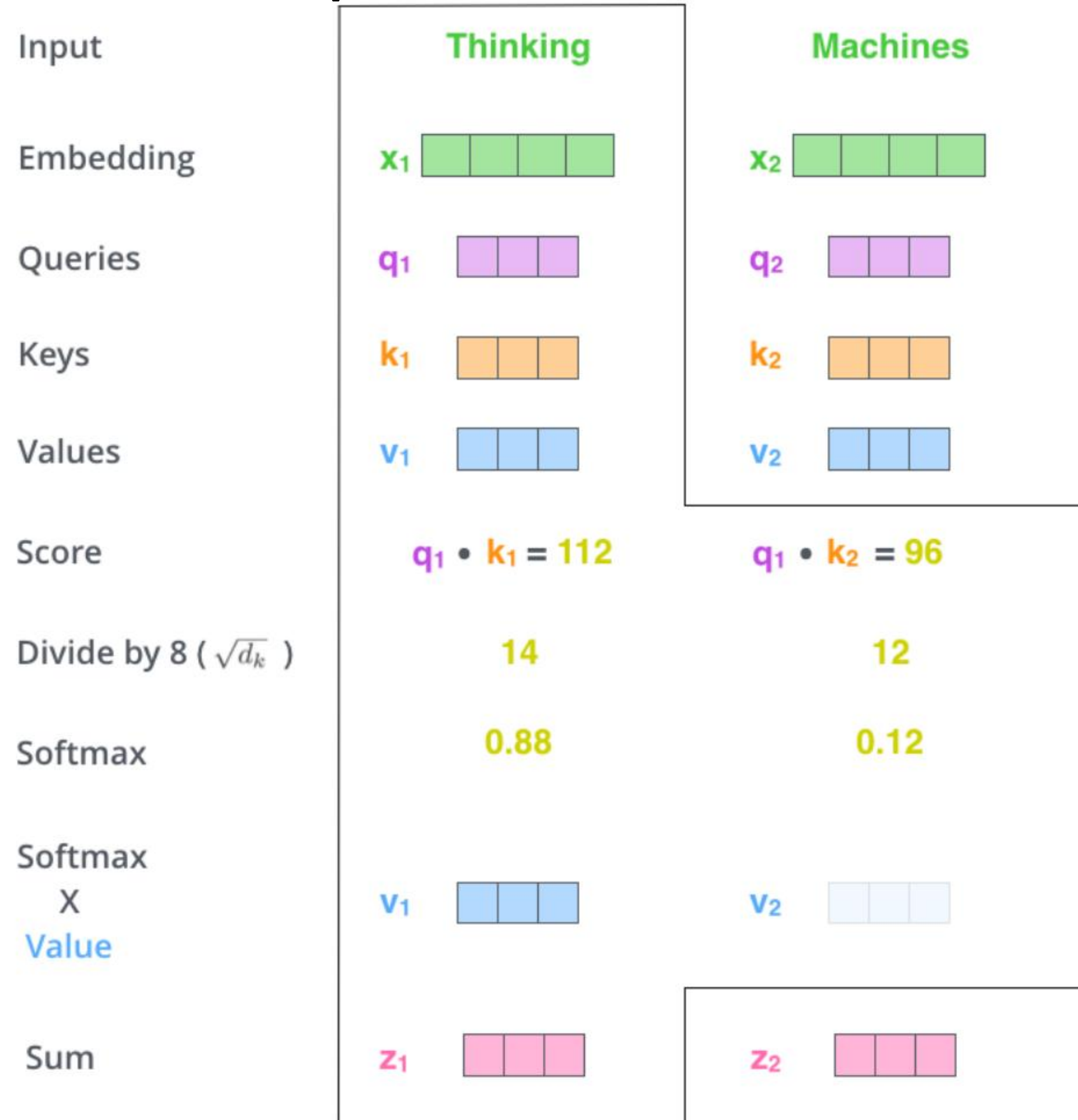
Transformer : A Specific-Level Look



Transformer : A Specific-Level Look



Transformer : A Specific-Level Look



Transformer : Multi-Head Attention Visualizaitons

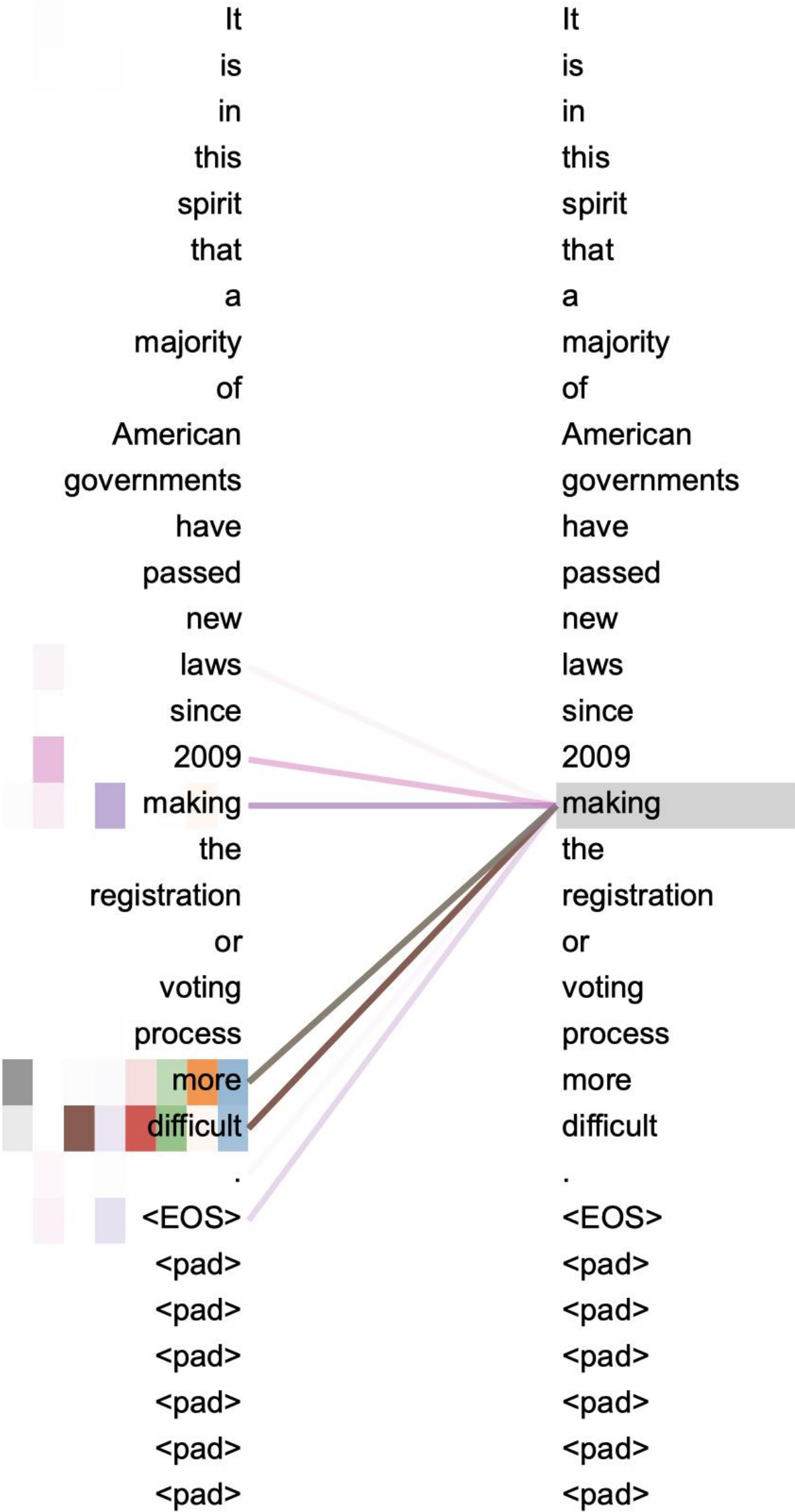


Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	



Why Transformer Important

Attention is all you need:

The first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence- aligned RNNs or convolution

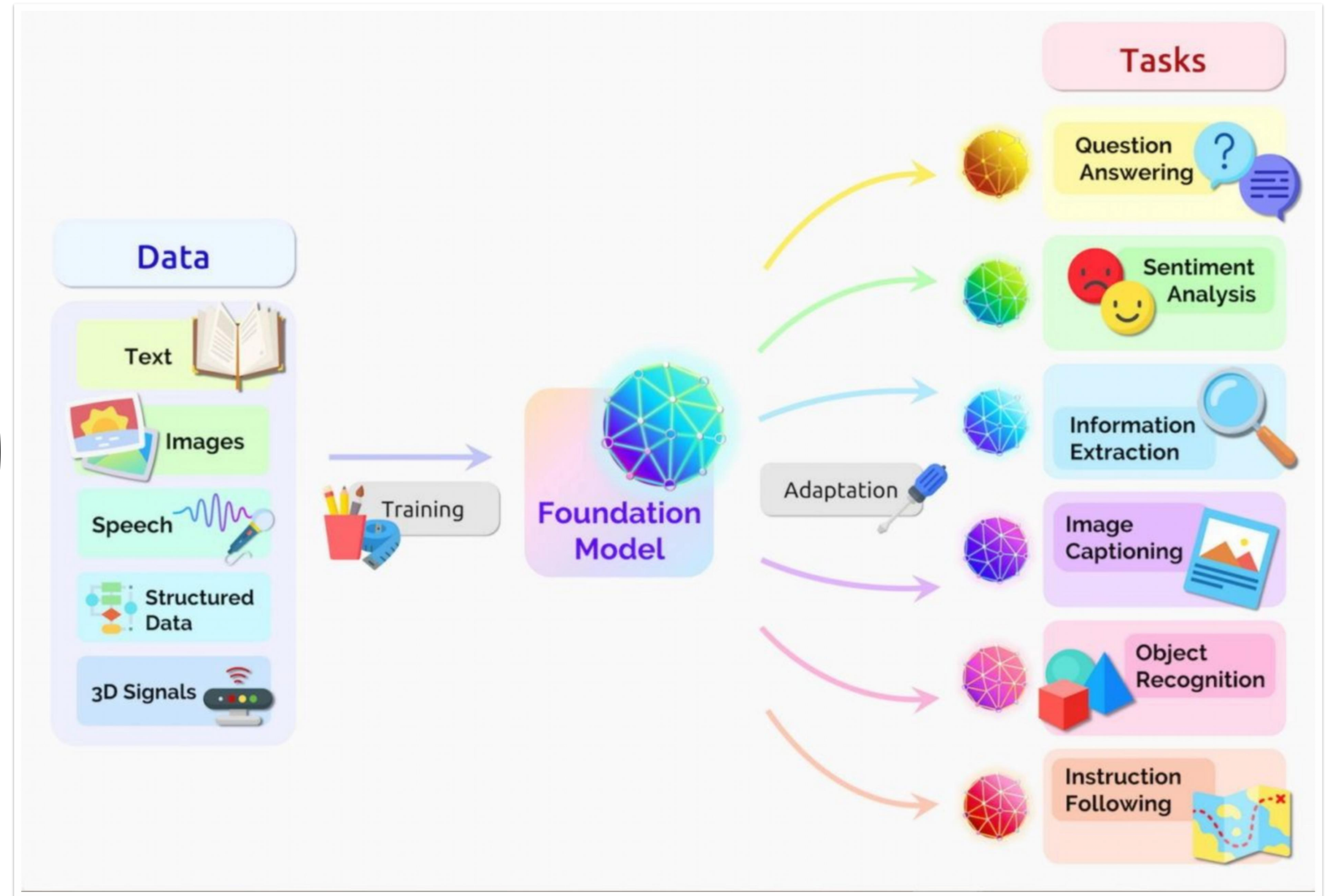
GPU-friendly parallel computation:

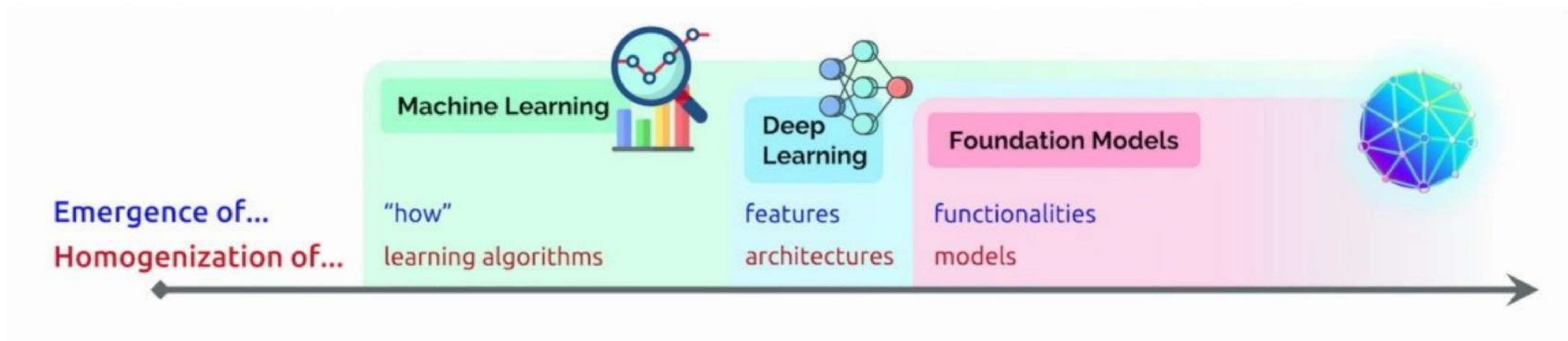
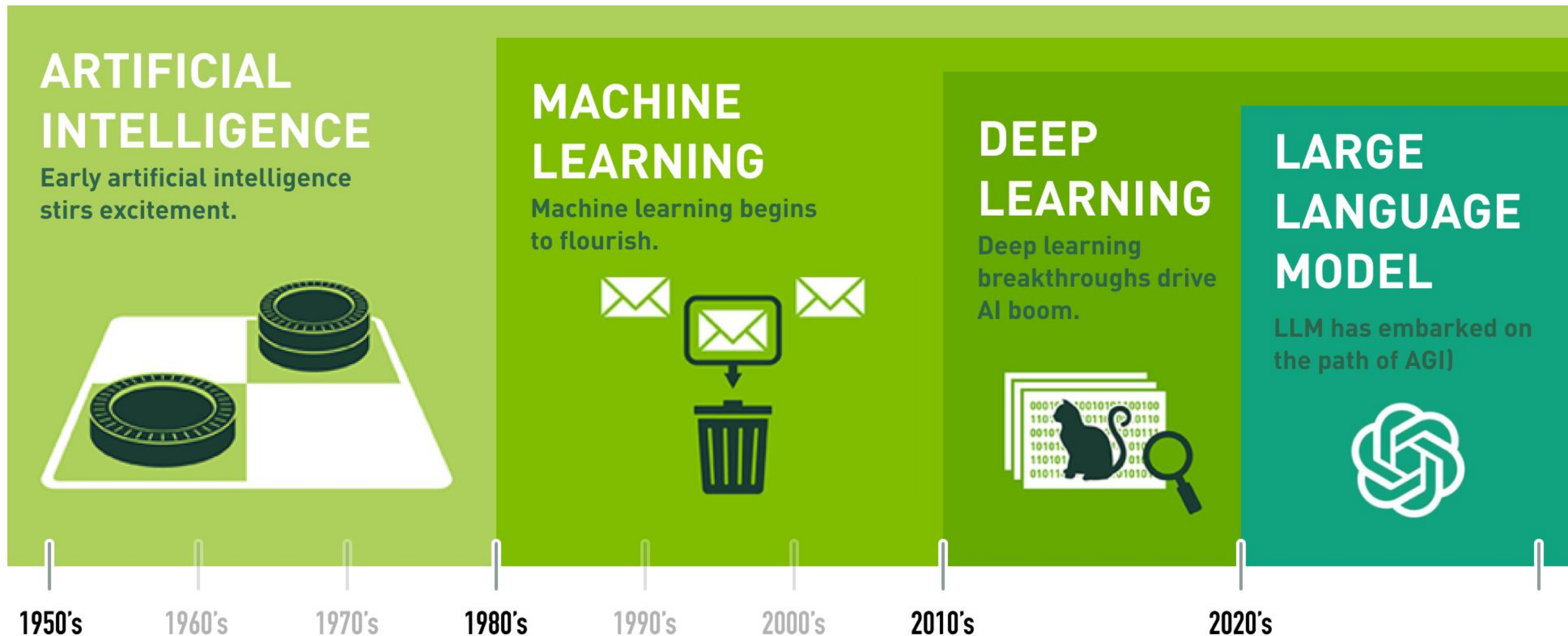
These layers are highly parallelizable, resulting in lower computational costs

Sentence-level representations:

Combined with positional encoding, the Transformer can capture long-range dependencies by considering the relative positions of tokens.

Open A Foundation Model Era





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Ilya Sutskever · 3rd
Co-Founder and Chief Scientist at OpenAI

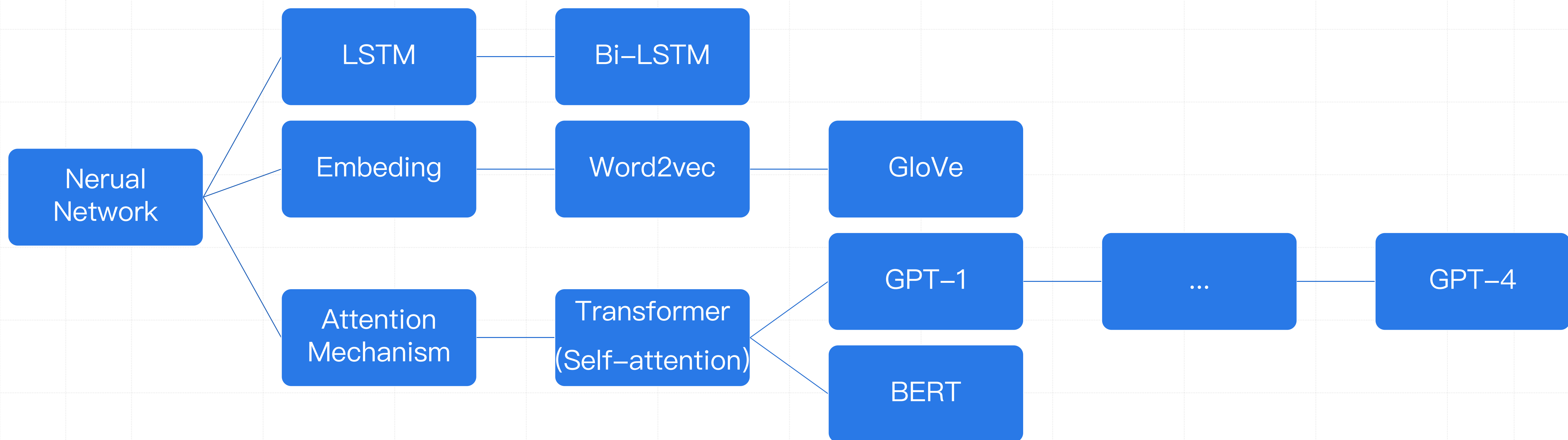


Jacob Devlin · 2nd
Software Engineer at Google

Google 团队
发布 BERT(2019)

OpenAI团队
发布GPT-1(2018)

Attention Mechanism



BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova

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Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are *unidirectional*, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers

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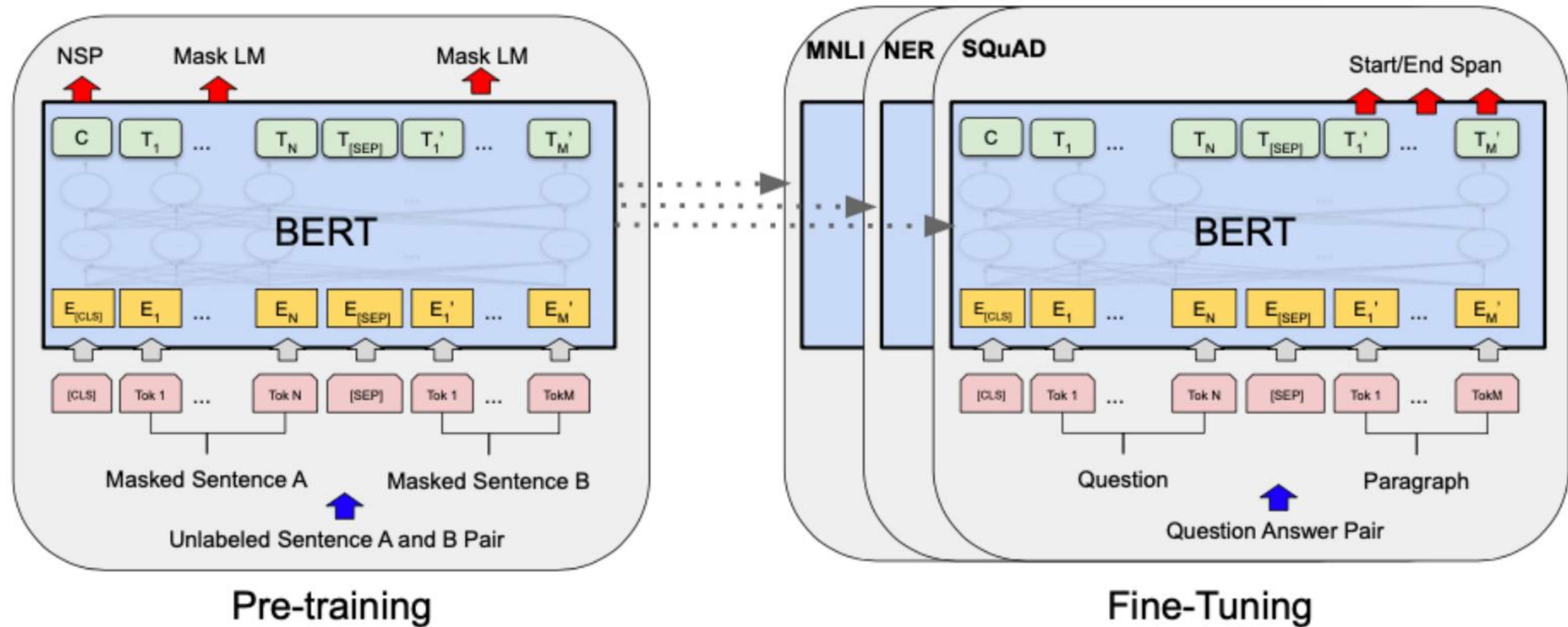
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基准测试	描述
GLUE Benchmark	一个包含九个自然语言理解任务的基准测试，例如：文本蕴含识别、语义角色标注、情感分析等。 BERT 在多项任务中刷新了记录。
SQuAD v1.1	斯坦福问答数据集（ Stanford Question Answering Dataset ），是一个阅读理解型问答任务的基准测试。 BERT 在此任务中超过了人类的表现。
SQuAD v2.0	与 v1.1 版本类似，但增加了一部分问题并没有答案的情况，这对模型的理解能力提出了更高的要求。 BERT 也在此任务中取得了最优成绩。
SWAG	一个多选问答任务，需要根据前文预测下一句话的内容。 BERT 在这个任务中也刷新了最好的成绩。
CoNLL-2003 NER	命名实体识别（ Named Entity Recognition ）的基准测试， BERT 在此任务上表现优异。

BERT : Pre-training + Fine-Tuning Paradigm

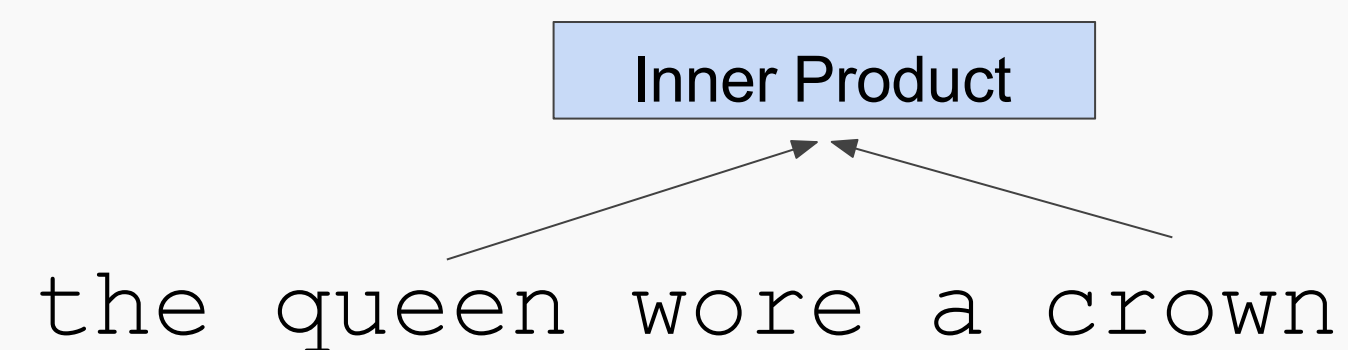
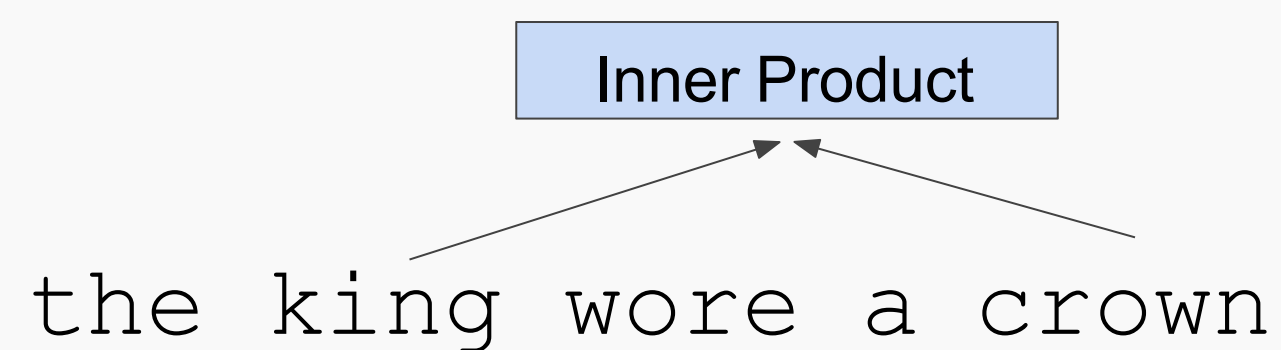


- 1.全方位上下文理解：**与以前的模型（例如**GPT**）相比，**BERT**能够双向理解上下文，即同时考虑一个词的左边和右边的上下文。这种全方位的上下文理解使得**BERT**能够更好地理解语言，特别是在理解词义、消歧等复杂任务上有明显优势。
- 2.预训练+微调（Pre-training + Fine-tuning）的策略：****BERT**模型先在大规模无标签文本数据上进行预训练，学习语言的一般性模式，然后在具体任务的标签数据上进行微调。这种策略让**BERT**能够在少量标签数据上取得很好的效果，大大提高了在各种**NLP**任务上的表现。
- 3.跨任务泛化能力：****BERT**通过微调可以应用到多种**NLP**任务中，包括但不限于文本分类、命名实体识别、问答系统、情感分析等。它的出现极大地简化了复杂的**NLP**任务，使得只需一种模型就能处理多种任务。
- 4.多语言支持：****BERT**提供了多语言版本（**Multilingual BERT**），可以支持多种语言，包括但不限于英语、中文、德语、法语等，使得**NLP**任务能够覆盖更广的语言和区域。
- 5.性能优异：**自**BERT**模型提出以来，它在多项**NLP**基准测试中取得了优异的成绩，甚至超过了人类的表现。它的出现标志着**NLP**领域进入了预训练模型的新时代。
- 6.开源和可接入性：****BERT**模型和预训练权重由**Google**公开发布，让更多的研究者和开发者可以利用**BERT**模型进行相关研究和应用开发，推动了整个**NLP**领域的发展。

- Word embeddings are the basis of deep learning for NLP



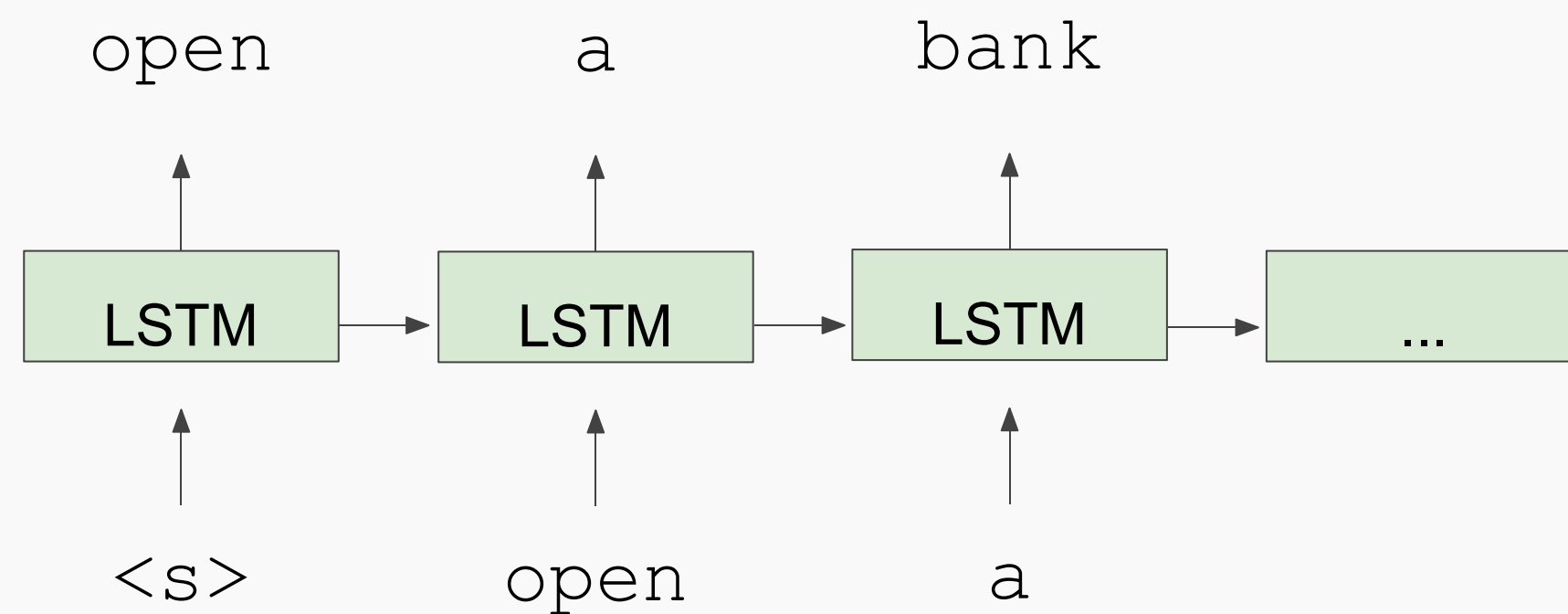
- Word embeddings (`word2vec`, `GloVe`) are often *pre-trained* on text corpus from co-occurrence statistics



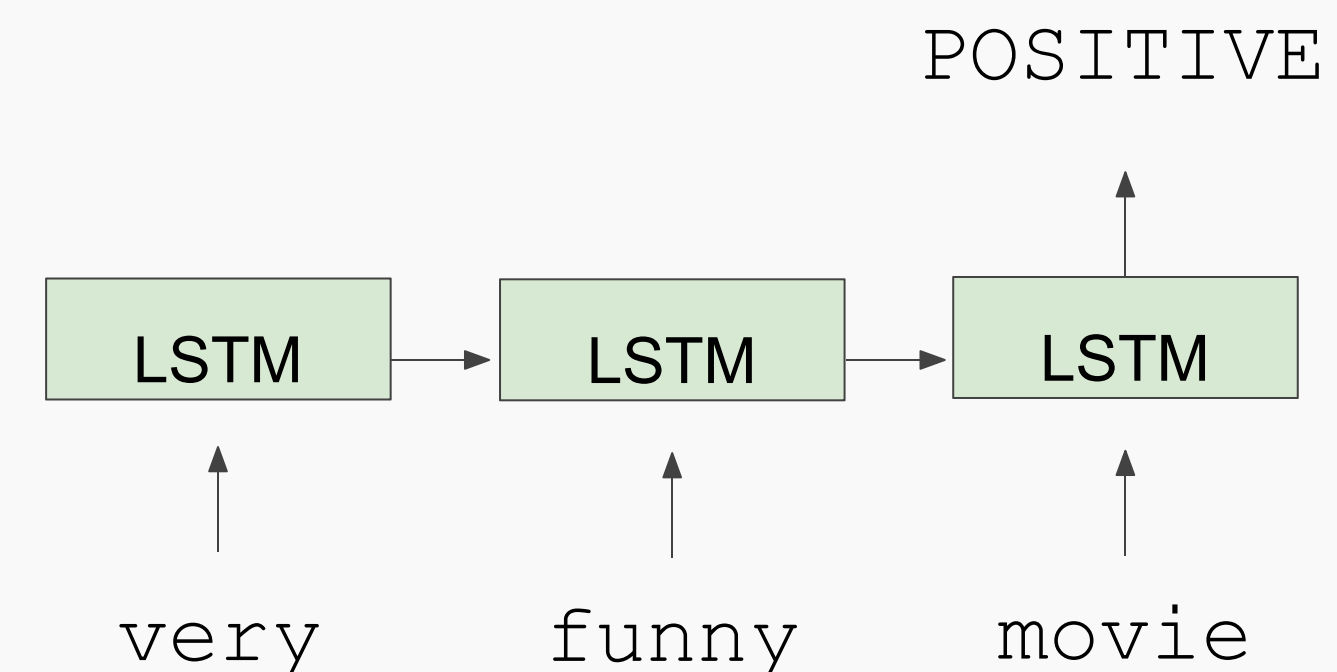
History of Contextual Representations

- *Semi-Supervised Sequence Learning*, Google, 2015

Train LSTM Language Model



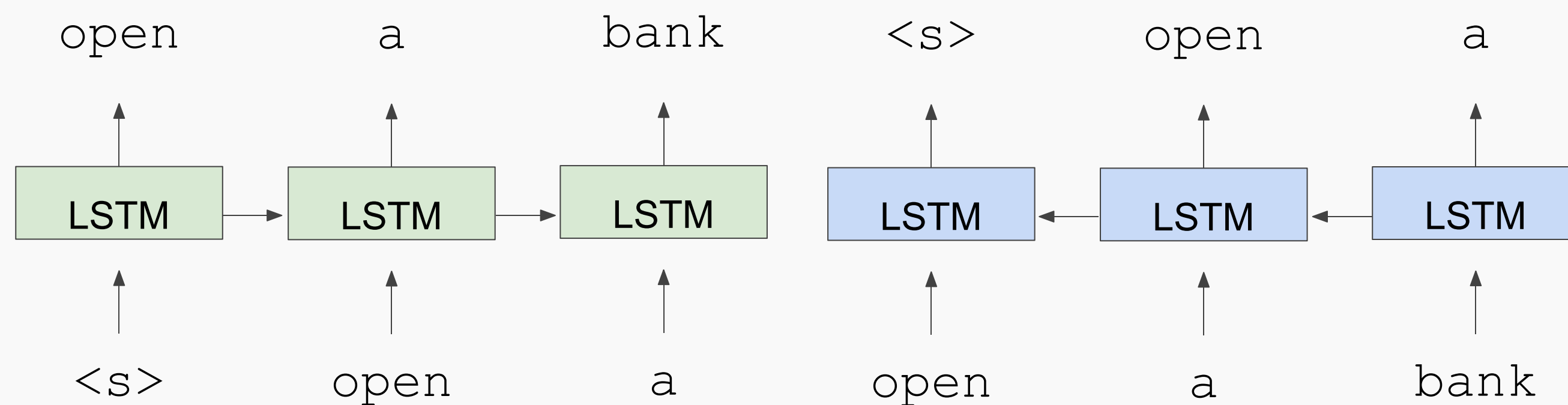
Fine-tune on Classification Task



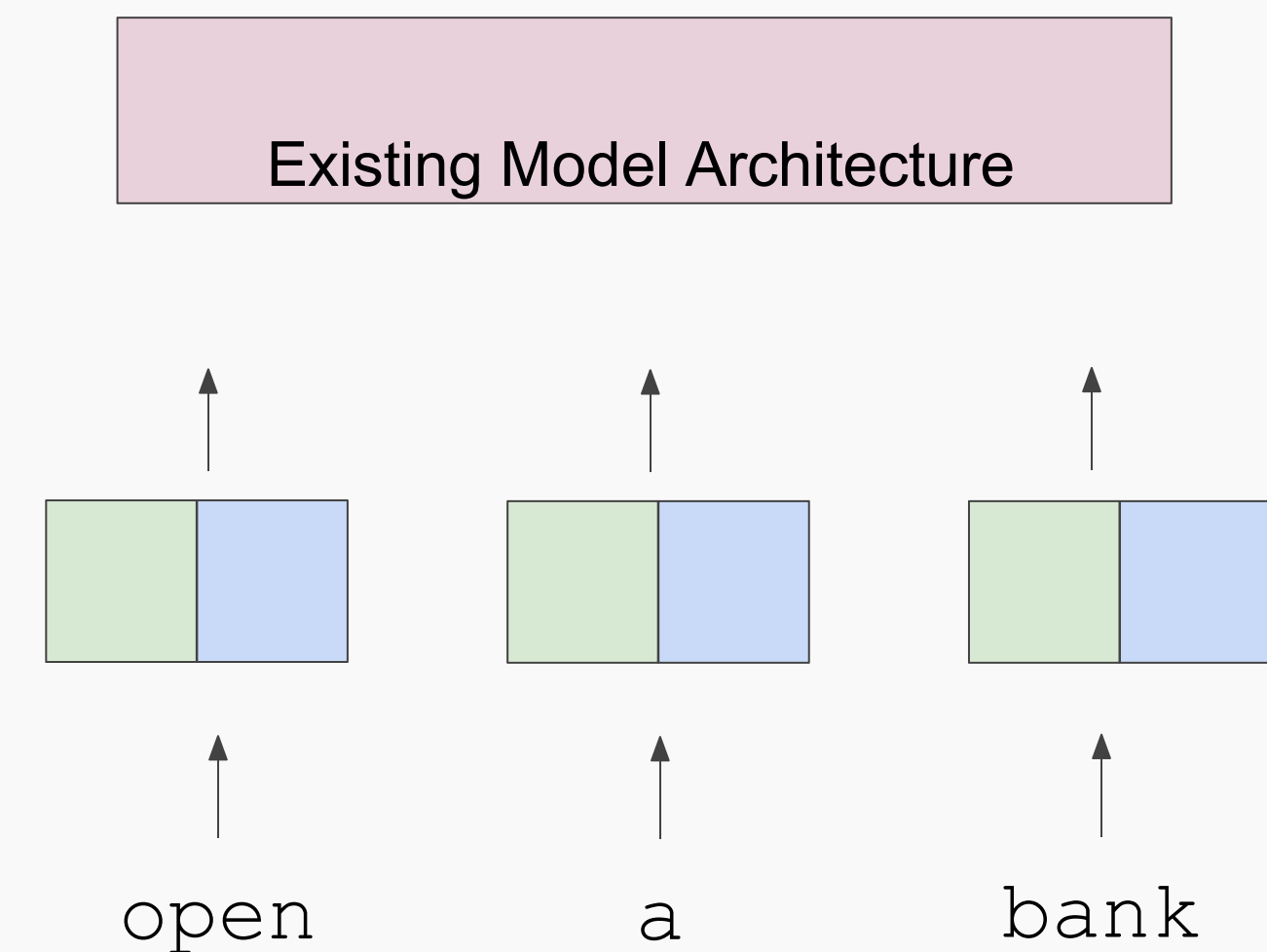
History of Contextual Representations

- *ELMo: Deep Contextual Word Embeddings*, AI2 & University of Washington, 2017

Train Separate Left-to-Right and Right-to-Left LMs

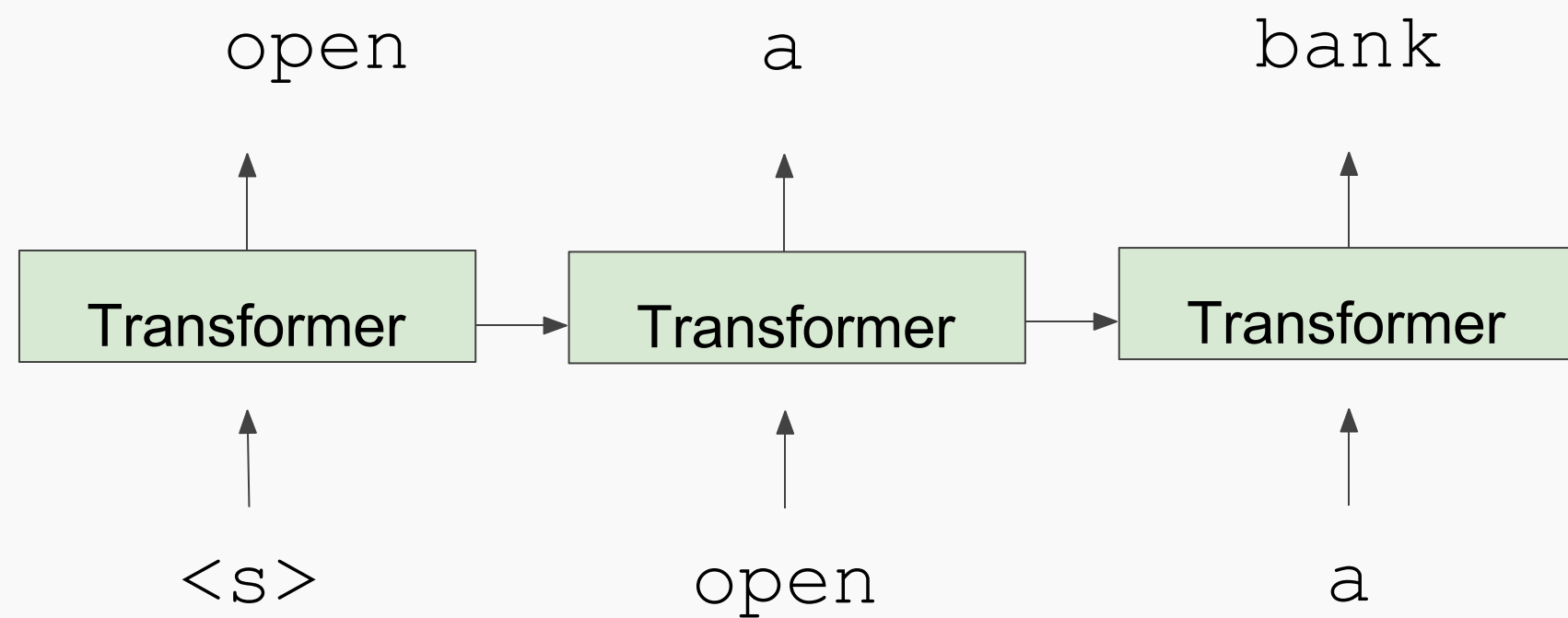


Apply as “Pre-trained Embeddings”

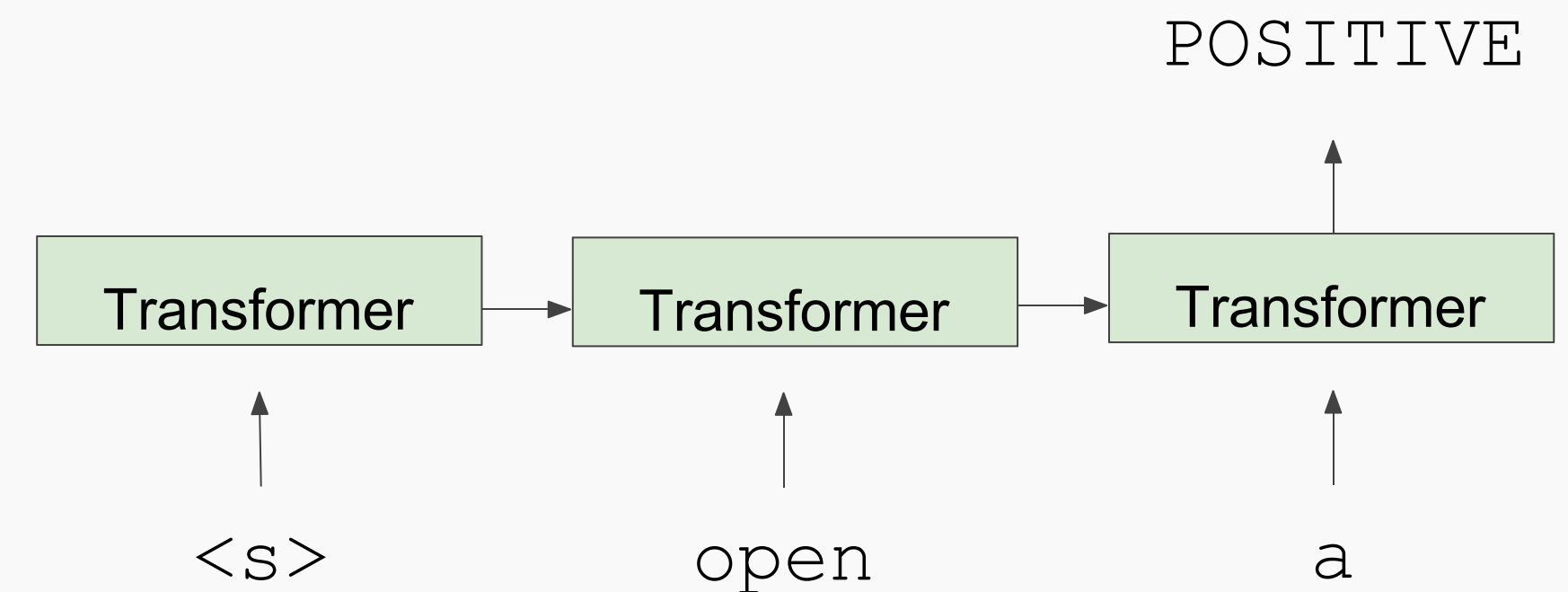


- *GPT-1: Improving Language Understanding by Generative Pre-Training*, OpenAI, 2018

Train Deep (12-layer) Transformer LM



Fine-tune on Classification Task



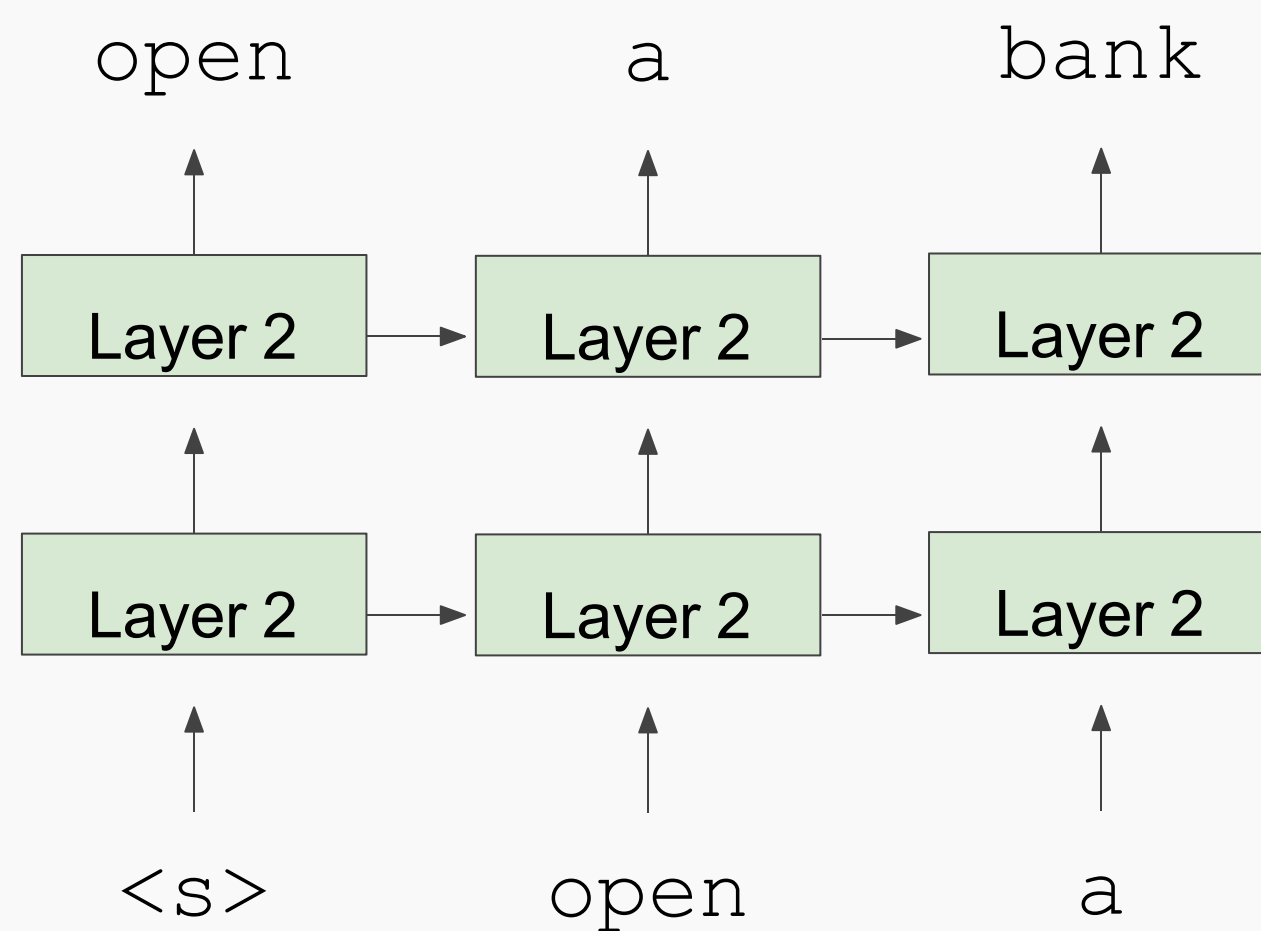
Problem with Previous

Methods

- **Problem:** Language models only use left context *or* right context, but language understanding is *bidirectional*.
- Why are LMs unidirectional?
- Reason 1: Directionality is needed to generate a well-formed probability distribution.
 - We don't care about this.
- Reason 2: Words can “see themselves” in a *bidirectional* encoder.

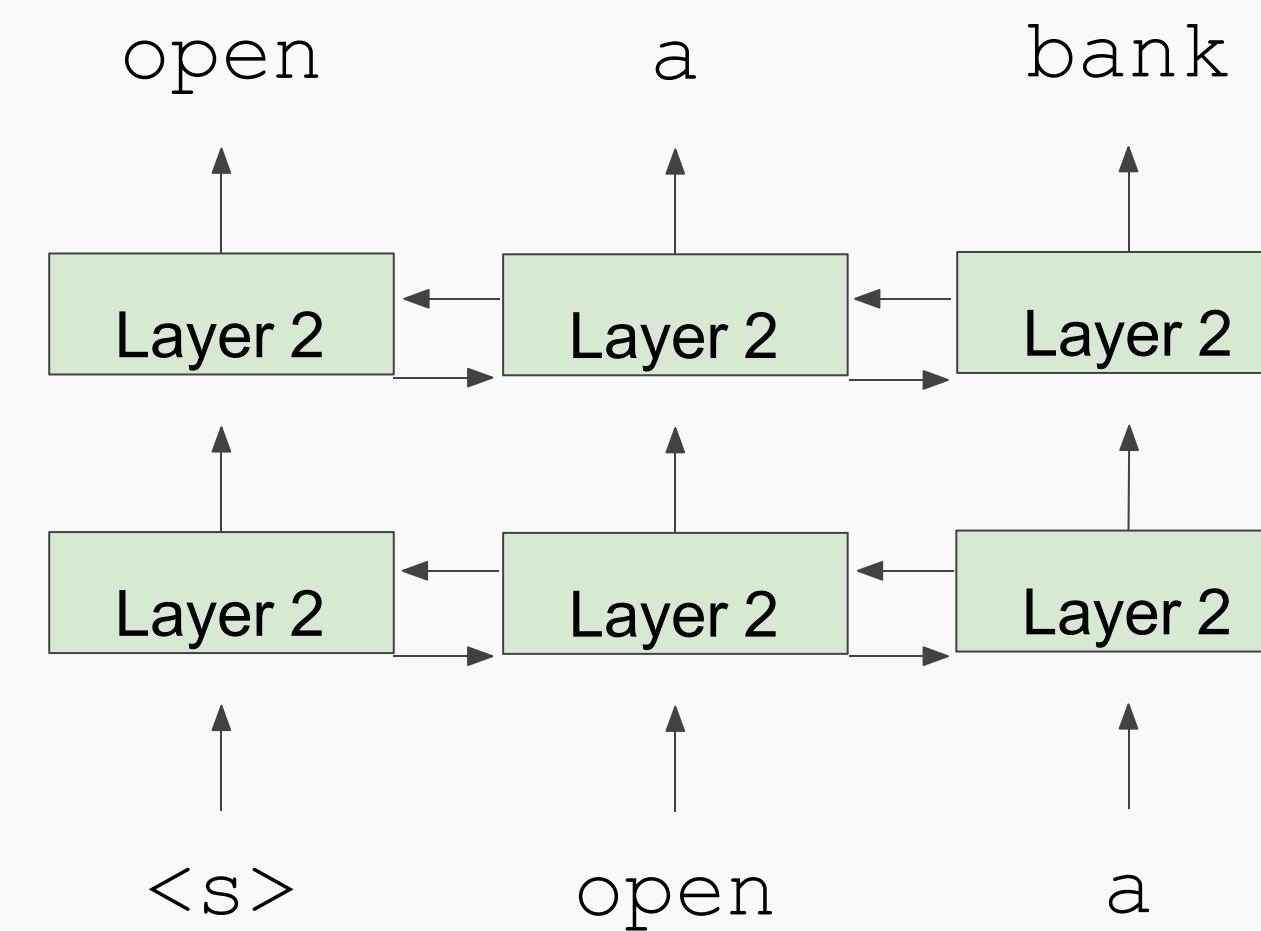
Unidirectional context

Build representation incrementally



Bidirectional context

Words can “see themselves”



- LM ● **Solution:** Mask out $k\%$ of the input words, and then predict the masked words
 - We always use $k = 15\%$

the man went to the [MASK] to buy a [MASK] of milk

store gallon

↑ ↑

- Too little masking: Too expensive to train
- Too much masking: Not enough context

LM

- Problem: Mask token never seen at fine-tuning
- Solution: 15% of the words to predict, but don't replace with [MASK] 100% of the time.

Instead:

- 80% of the time, replace with [MASK]
went to the store → went to the [MASK]
- 10% of the time, replace random word
went to the store → went to the running
- 10% of the time, keep same
went to the store → went to the store

Next Sentence

Prediction

- To learn *relationships* between sentences, predict whether Sentence B is actual sentence that proceeds Sentence A, or a random sentence

Sentence A = The man went to the store.
Sentence B = He bought a gallon of milk.
Label = IsNextSentence

Sentence A = The man went to the store.
Sentence B = Penguins are flightless.
Label = NotNextSentence

BERT vs GPT 差异

特性

训练方式

预测目标

输入处理

适用场景

架构

语言模型

优点

缺点

BERT

自编码 (**Autoencoding**)

给定上下文，预测其中的一个或多个缺失单词

双向，可以同时考虑一个词的左右上下文

适合理解上下文，有助于信息提取、问答系统、情感分析等

基于**Transformer**的编码器

判别式 (**Discriminative**)

对上下文理解能力较强

生成的文本连贯性较弱

GPT

自回归 (**Autoregressive**)

在给定前面的单词时，预测下一个单词

单向 (从左到右或者从右到左)

适合生成式任务，如文章生成、诗歌创作等

基于**Transformer**的解码器

生成式 (**Generative**)

预测的连贯性较强

对上下文理解能力相对较弱

BERT vs GPT
共识

模型架构	Transformer
数据预处理	都需要对数据进行 Tokenization ，一般使用词片方法 (Subword Tokenization)
模型训练	均使用了大量的无标签数据进行预训练
任务迁移	都可以通过 Fine-tuning 方式进行任务迁移
训练目标	都试图通过预训练理解语言的一般模式，如语法、语义、上下文关系等
多语言支持	均支持多语言模型训练

THANKS

 极客时间 | 训练营